TS2000
TS2100
Robot Controller

INSTRUCTION MANUAL

TS2000/TS2100 ROBOT CONTROLLER

INTERFACE MANUAL

Rev.10
April 1, 2020

Notice

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

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The information contained in this manual is subject to change without prior notice to effect improvements.

Preface

This manual describes the type, function and handling method of external cables connecting the TS2000/TS2100 robot controller with an external equipment.

This manual is intended for the system designers and manufacturing engineers.
The TS2000/TS2100 robot controller can work in concert with the external equipment through digital input and output signals that can be programmed by the SCOL language. Also, system input signals that can allows external operation of the controller, system output signals informing an operator of the controller status (PLC processing function) and serial input and output signals that can be connected with the host computer, etc., are provided so that the user can easily construct an FA system.

* Instruction manuals which are referred to from this manual

- Installation & Transport Manual
- Maintenance Manual
- Safety Manual
- User Parameter Manual
- Simple PLC Function Manual
- I/O Common Manual (Type P)

! **CAUTION**	This manual does not contain any detailed descriptions on power and robot connection. For the connection of the power and robot, see the Installation & Transport Manual.

Cautions on Safety

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damages to assets and to assure correct use.
Make sure that you well understand the following details (indications and symbols) before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2).

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.
*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. Details of the actual caution are indicated with pictures or words in or near the symbol.

! **CAUTION**	To perform the work ranging from robot installation to operation with safety, read through and through the Safety Manual provided separately before actually starting the work.

Maintenance and Inspection

To use the robot safety, strictly observe the following matters.

! DANGER	
 Prohibited	• NEVER burn, disassemble or charge the battery. Otherwise, it may explode.
 Mandatory	• Before performing the maintenance and inspection, be sure to turn off the main power switch of the controller. • When disposing of batteries, be sure to follow the user's regulations.

! CAUTION	
 Disassembly Prohibited	• The user should NEVER replace or change parts other than those stipulated in the instruction manual. Otherwise, the performance will deteriorate, resulting in troubles.
 Mandatory	• To replace parts, use the spare parts designated by SHIBAURA MACHINE. • Carry out the maintenance and inspection on a regular basis. Otherwise, the equipment may go wrong or accidents will be caused.

! CAUTION	
	To perform the maintenance and inspection of the robot with safety, read through and through the Maintenance Manual provided separately before actually starting the work.

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1. Type of External Cable

1.1 Layout and Name of Connectors

The TS2000/TS2100 robot controller is connected with the robot and external equipment, using connectors and terminal block provided on the front and rear sides of the controller.

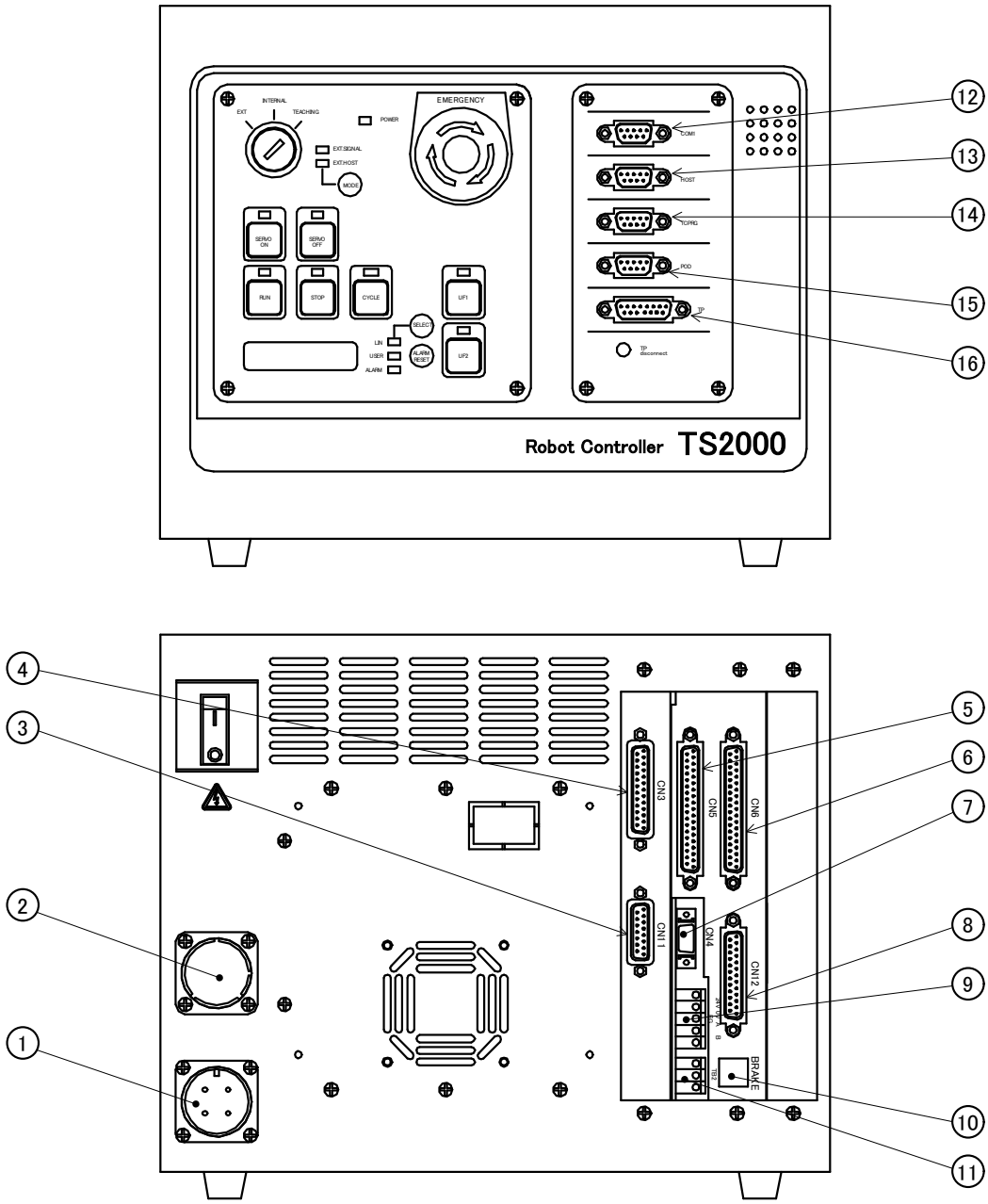


Fig. 1.1 Layout and name of connectors

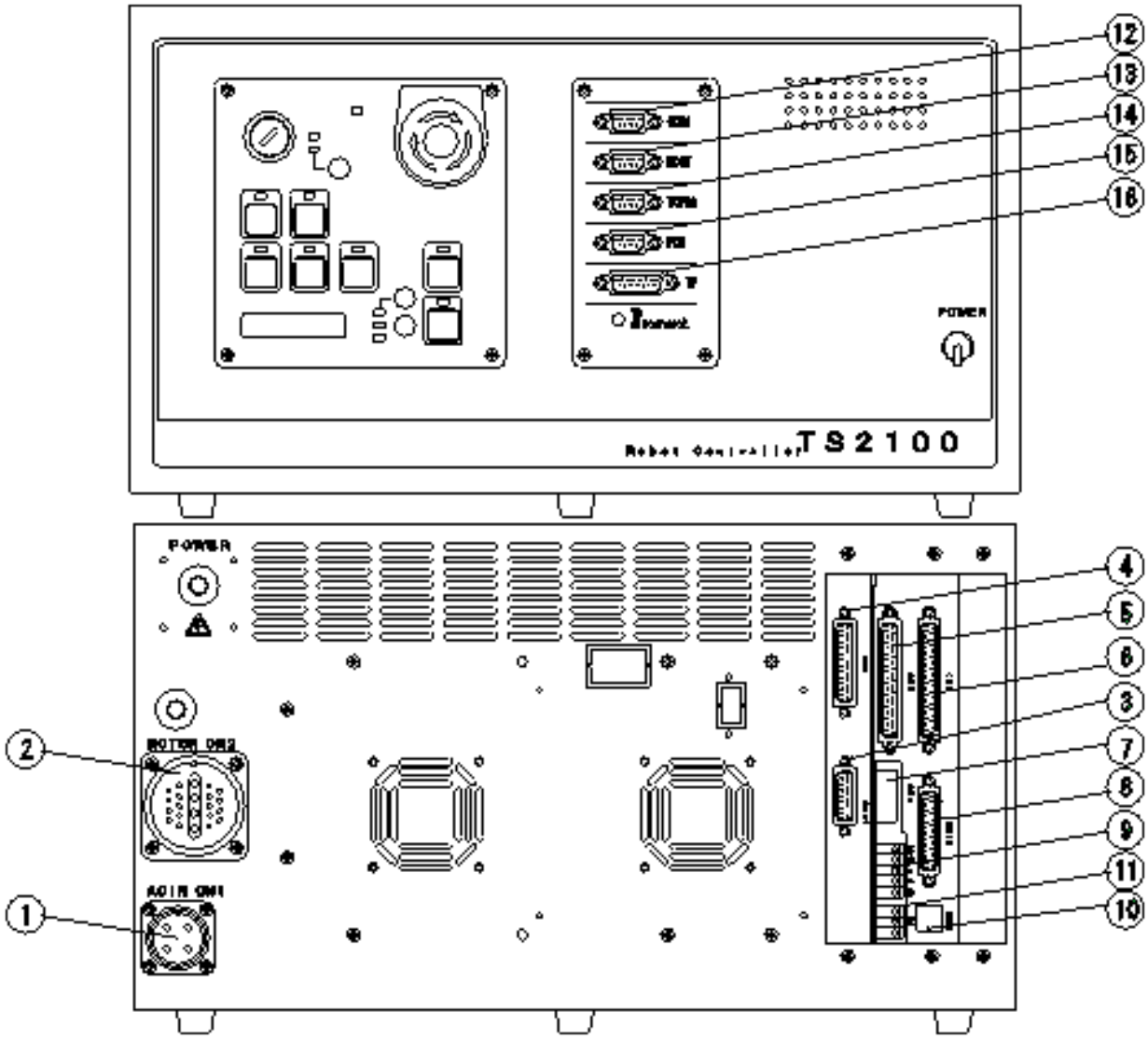


Fig. 1.2 Layout and name of connectors

1.2 TS2000/TS2100 Power Cable "ACIN CN1"

Fig. 1.1/Fig.1.2–[1] (with connector)

TS2000 : The power cable is an interface feeding the main power (2 phases, AC180/250 V, 50/60 Hz) to the TS2000 robot controller.

TS2100 : The power cable is an interface feeding the main power (3 phases, AC 180/250 V, 50/60 Hz) to the TS2100 robot controller.

Connector "ACIN CN1" is used.

TS2100 : The power cable is an interface feeding the main power (3 phases, AC 180/250 V, 50/60 Hz) is an optional specification.

For details, see the Installation & Transport Manual provided separately.

1.3 Robot Control Cables

1.3.1 TS2000/TS2100 Motor Drive Cable "MOTOR CN2"

Fig. 1.1/Fig.1.2–[2] (with cable)

The motor drive cable connects the TS2000/TS2100 robot controller and robot. It feeds the 3-phase AC power to each axis feed motor of the robot.

Connector "MOTOR CN2" is used.

For details, see the Installation & Transport Manual provided separately.

1.3.2 TS2000/TS2100 Encoder Cable "CN3"

Fig. 1.1/Fig.1.2–[4] (with cable)

The encoder cable is an interface which inputs the rotation angle detection encoder signal (axis 1 to axis 4) of each robot axis to the TS2000/TS2100 robot controller. Connector "CN3" is used.

For details, see the Installation & Transport Manual provided separately.

1.3.3 TS2000/TS2100 Encoder Cable "CN11" (Option)

Fig. 1.1/Fig.1.2–[3] (with cable)

The encoder cable is an interface which inputs the rotation angle detection encoder signal (axis 5) of the robot axis to the TS2000/TS2100 robot controller.

Connector "CN11" is used.

1.3.4 Robot Control Signal Cable "CN4"

Fig. 1.1/Fig.1.2–[7] (with cable)
This cable is used to turn on and off the parking brake for securing the robot motor shaft, and input and output the robot control signals such as hand operation.
Among the robot control signal cables, the input/output signal cables for controlling the hand effector such as robot hand can be controlled by the robot language program. Also, they can be turned on and off manually through the teach pendant.
Connector "CN4" is used.
For details, see the Installation & Transport Manual provided separately.
For the robot language, see the Robot Language Manual. For the operation of hand input and output signals from the teach pendant, see the Operation Manual.

1.3.5 Robot Control Signal Cable "BRAKE"

Fig. 1.1/Fig.1.2–[10] (Cable is optional.)
Separate from the cable "CN4" described in Para. 1.3.4 above, this cable is used to turn on and off the parking brake for securing the robot motor shaft. Connector "BRAKE" is used.
When this signal cable is used, the robot side is also addressed optionally.

1.4 Digital Output Power Select Cable "TB2" (with jumper)

Fig. 1.1/Fig.1.2–[11]
This is the terminal block for selecting the power (P24 V) for the digital output (32 numbers) of the TS2000/TS2100 robot controller. When the external power supply (P24 V) is used, the power is supplied from this terminal block.

1.5 External I/O Signal Cables

1.5.1 External Input Signal Cable "CN5"

Fig. 1.1/Fig.1.2–[5] (with dummy connector)
This cable is used to input the digital signal from the external equipment to the TS2000/TS2100 robot controller.

The external input signal comes in twenty-four (24) digital input signals that can be programmed at the user's discretion, using the SCOL language and ten (10) external operation input signals that allow operation of the controller from the external equipment. This signal, together with the external output signal in Para. 1.5.2, allows the robot controller to work in concert with the external equipment.
Connector "CN5" is used.

1.5.2 External Output Signal Cable "CN6"

Fig. 1.1/Fig.1.2–[6] (with dummy connector)
The external output signal cable is an interface which outputs the digital signal from the TS2000/TS2100 robot controller to the external equipment.
The external output signal comes in sixteen (16) digital output signals that can be programmed at the user's discretion, using the SCOL language and twelve (12) external operation output signals that output the controller status to the external equipment. This signal, together with the external input signal in Para. 1.5.1, allows the robot controller to work in concert with the external equipment.
Additionally, four (4) signals of external servo ON, external servo OFF and external emergency stop 1 & 2 are included in this cable.
Connector "CN6" is used.

1.5.3 External I/O Signal Cable "CN12"

Fig. 1.1/Fig.1.2–[8]
The external input/output signal cable is an interface which inputs the digital signal from the external equipment to the TS2000/TS2100 robot controller, and outputs the digital signal from the TS2000/TS2100 robot controller to the external equipment.
The external input signal comes in eight (8) digital input signals that can be programmed at the user's discretion, using the SCOL language.
The external output signal comes in eight (8) digital output signals that can be programmed at the user's discretion, using the SCOL language.
These signals, together with the external input signal in Para. 1.5.1 and external output signal in Para. 1.5.2, allow the robot controller to work in concert with the external equipment.
Connector "CN12" is used.

1.6 Serial I/O Signal Cable

1.6.1 Serial I/O Signal "COM1"

Fig. 1.1/Fig.1.2–[12]
Of the serial input and output signals of four (4) channels equipped on the TS2000/TS2100 robot controller, the D-SUB 9-pin connector located on the first line of the front connector unit in Fig. 1.1 is COM1.
COM1 is exclusively used for the RS232C and allows data communication with an image processing equipment or other FA equipment that can connect an RS232C interface. Connector "COM1" is used.

1.6.2 Serial I/O Signal "HOST"

Fig. 1.1/Fig.1.2–[13]
Of the serial input and output signals of four (4) channels equipped on the TS2000/TS2100 robot controller, the D-SUB 9-pin connector located on the second line from the top of the front connector unit in Fig. 1.1 is HOST.
HOST is exclusively used for the RS232C and allows transfer and saving of various parameters and updating of the system when connected with the host computer. Connector "HOST" is used.

1.6.3 TCPRGOS "TCPRG"

Fig. 1.1/Fig.1.2–[14]
The TS2000/TS2100 robot controller uses exclusive sequence control tool "TCPRGOS" (option) to transfer, save and monitor sequence programs in the built-in programmable controller (PLC).
To connect with this sequence control tool, the D-SUB 9-pin connector located on the third line from the top of the front connector unit in Fig. 1.1 is TCPRG. Connector "TCPRG" is used.
For the use of TCPRGOS, see the Simple PLC Function Manual (option) provided separately.

1.6.4 Serial I/O Signal "POD"

Fig. 1.1/Fig.1.2–[15]
Of the serial input and output signals of four (4) channels equipped on the TS2000/TS2100 robot controller, the D-SUB 9-pin connector located on the fourth line from the top of the front connector unit in Fig. 1.1 is POD.
POD is exclusively used for the RS232C and allows communication with the touch panel controller where an RS232C interface can be connected. Connector "POD" is used.

1.7 Teach Pendant Cable "TP"

Fig. 1.1/Fig.1.2–[16] (with dummy connector)
This is an interface connecting the TS2000/TS2100 robot controller and teach pendant (TP1000). The TP1000 is an option. By connecting the TP cable, creation of motion programs, manual robot guidance, etc., are possible through the teach pendant. Connector "TP" is used. The TP cable is secured to the teach pendant and cannot be disconnected from the teach pendant. The standard cable length is 5 m.

1.8 Remote I/O Cable "EXT–I/O"

Fig. 1.1/Fig.1.2–[9]
This is an RS485 communication terminal connecting the optional remote I/O module function (TR48DIOCN/TR48DIOC module, etc.) of the TS2000/TS2100 robot controller. To connect, the terminal block on the rear side of the controller is used.

2. Connecting Power Cable

To connect the power cable, use the attached connector (JL04V–2E18–10PE–B; made by Japan Aviation Electronics Industry).

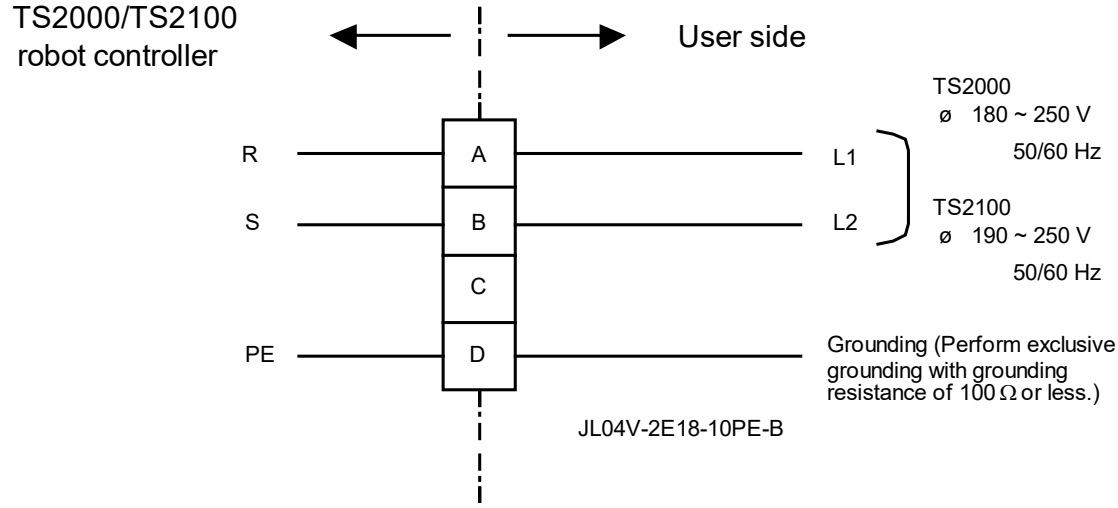


Fig. 2.1 Connection of power cable

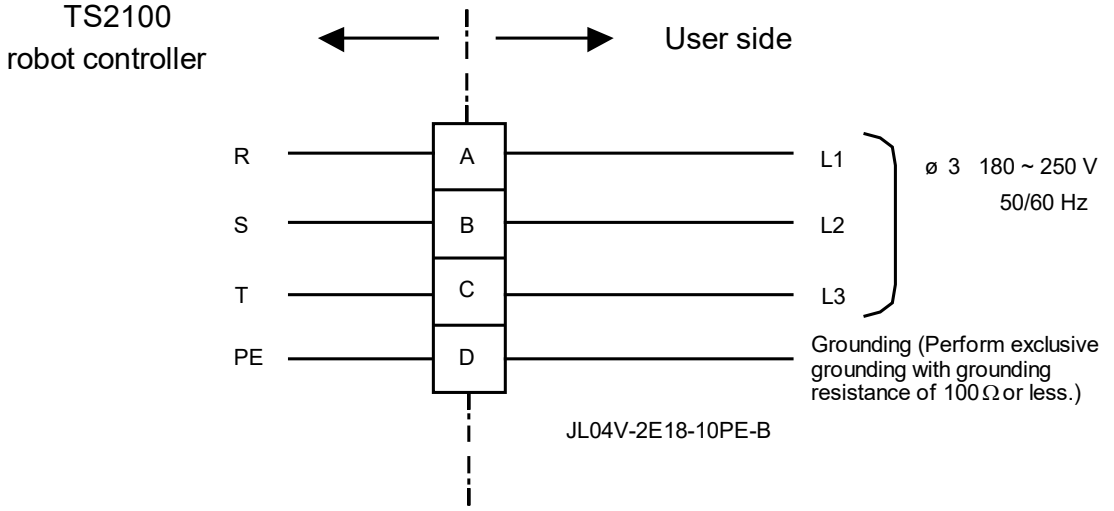


Fig. 2.2 Connection of power cable

For details of the power cable connection, see the Installation & Transport Manual provided separately.

3. Connecting Robot Control Cable

3.1 Connecting Motor Drive Cable

To connect the motors, use the attached cables.

The attached standard cables do not include the wiring for the optional axis 5.

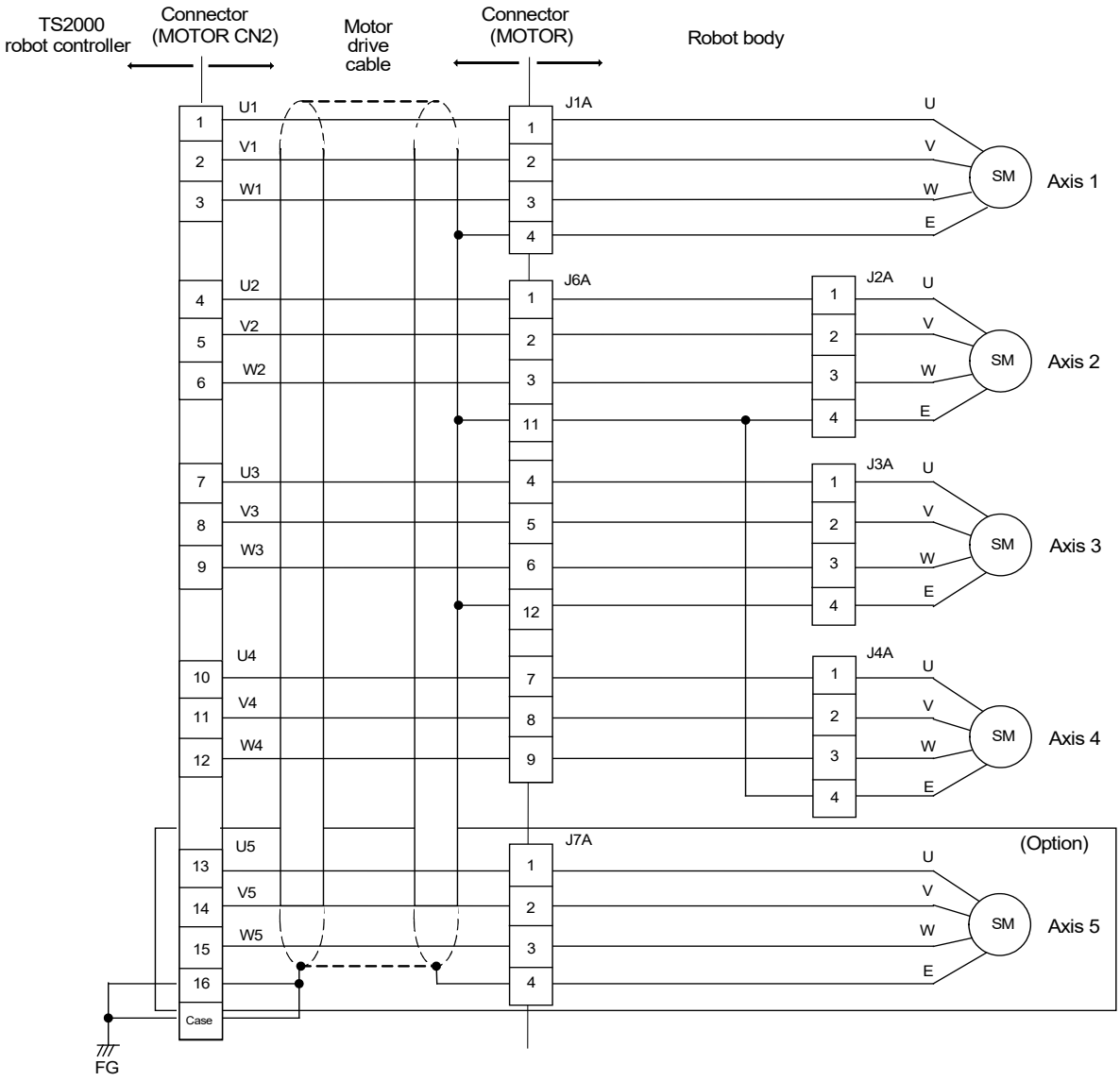


Fig. 3.1 Connection of motor drive cables

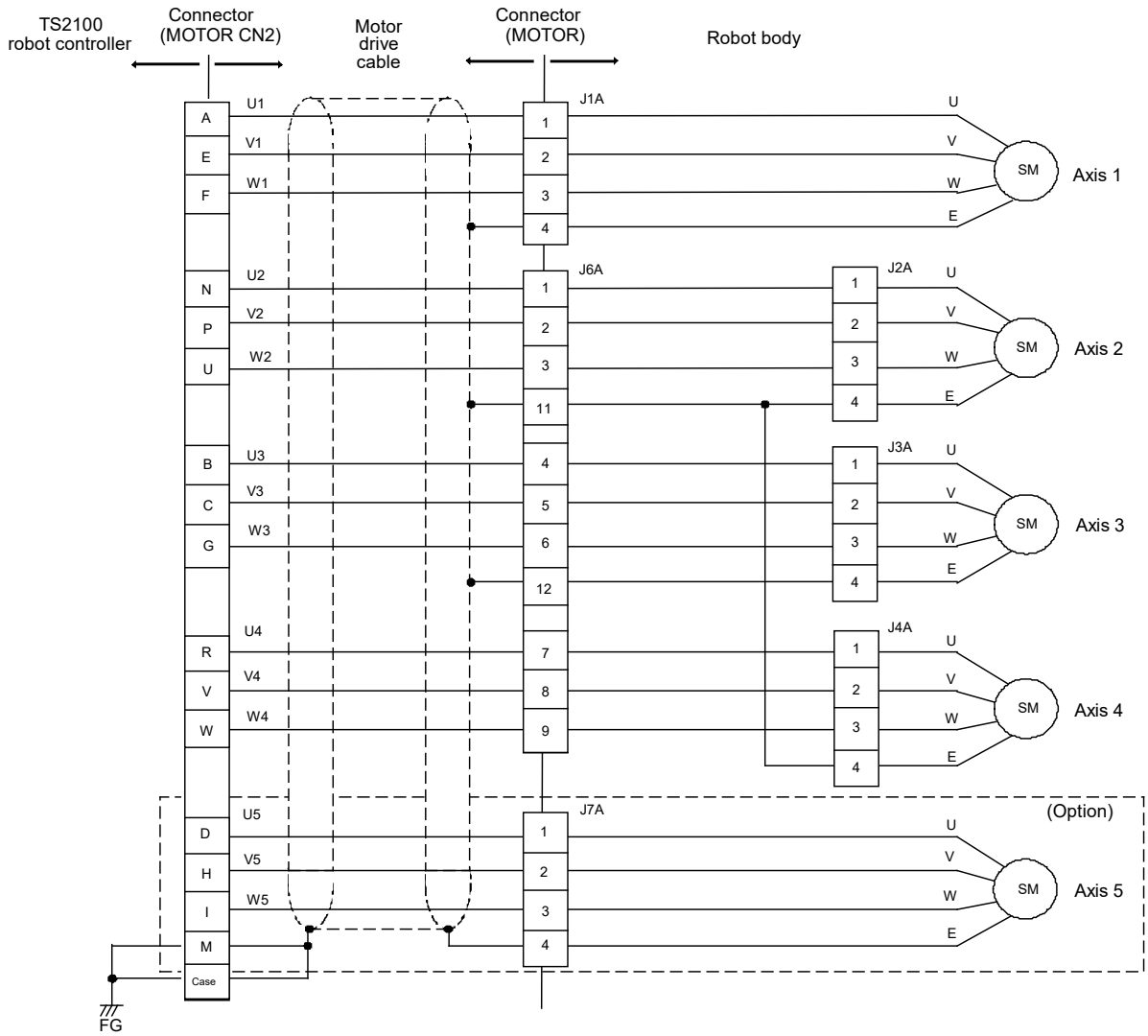


Fig. 3.2 Connection of motor drive cables

For the motor drive cable connection, see the Installation & Transport Manual.

3.2 Connecting Encoder Cable

3.2.1 Connecting Encoder Cable

To connect the encoders to the TS2000/TS2100 digital servo printed board (X8HW,X8HS), use the attached cables.

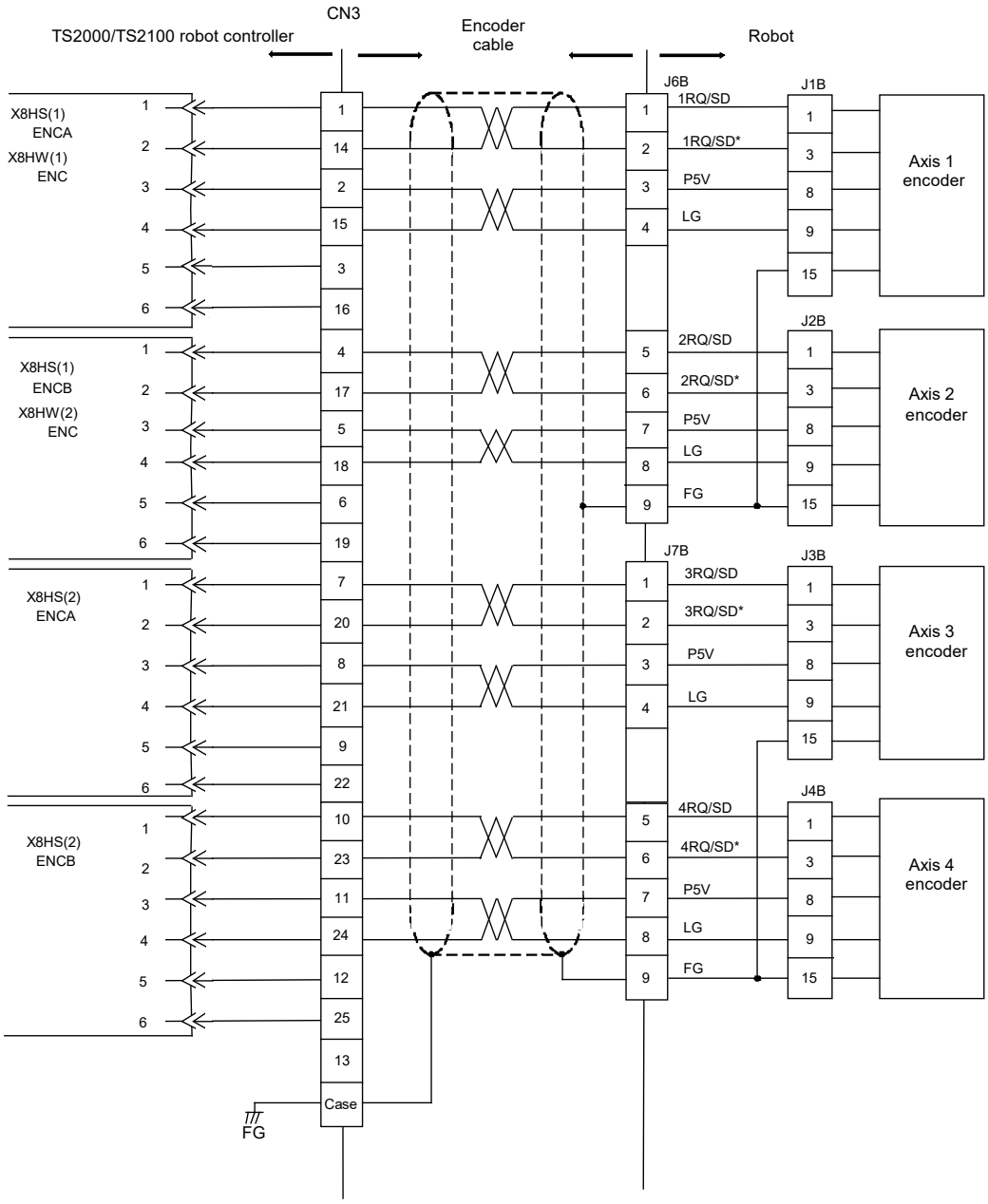


Fig. 3.2 Connection of encoder cables

For the encoder cable connection, see the Installation & Transport Manual.

3.2.2 Connecting Encoder Cable (Option)

When the axis 5 (option of TS2000/TS2100) is used, use the attached cable to connect the encoder.

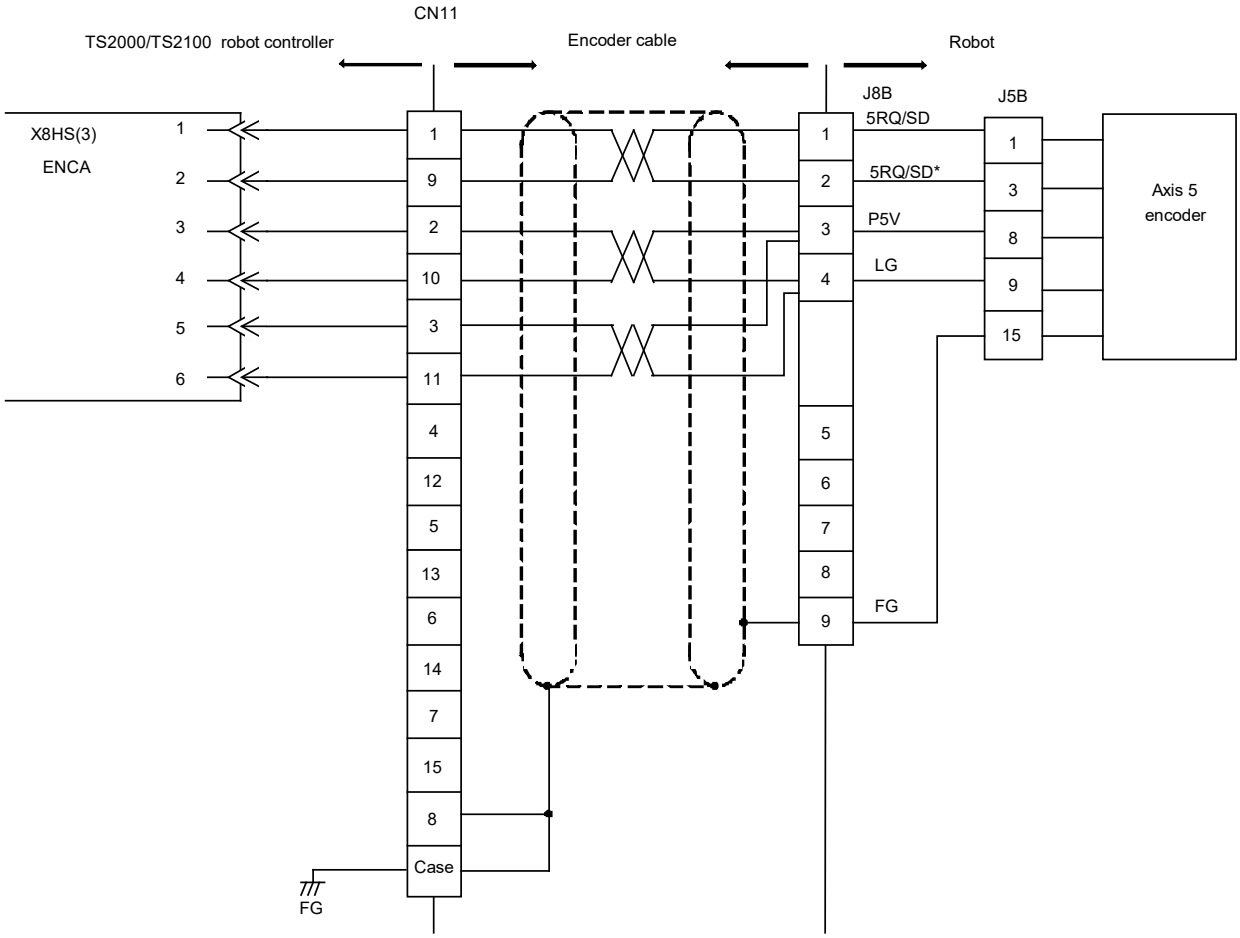


Fig. 3.3 Connection of encoder cable

For the encoder cable connection, see the Installation & Transport Manual.

3.3 Connecting Robot Control Signal Cable

To connect the robot control signal cable, use the attached cable.
The I/O common comes in the two (2) types; Type N [X8HN (output sink type) is selected for the I/O printed board and the polarity is the same as in the SR7000 robot] and Type P [X8HI (output source type) is selected for the I/O printed board). After confirming the type of your controller, connect the robot control signal cable.

[Type N] (When X8HN printed board is used)

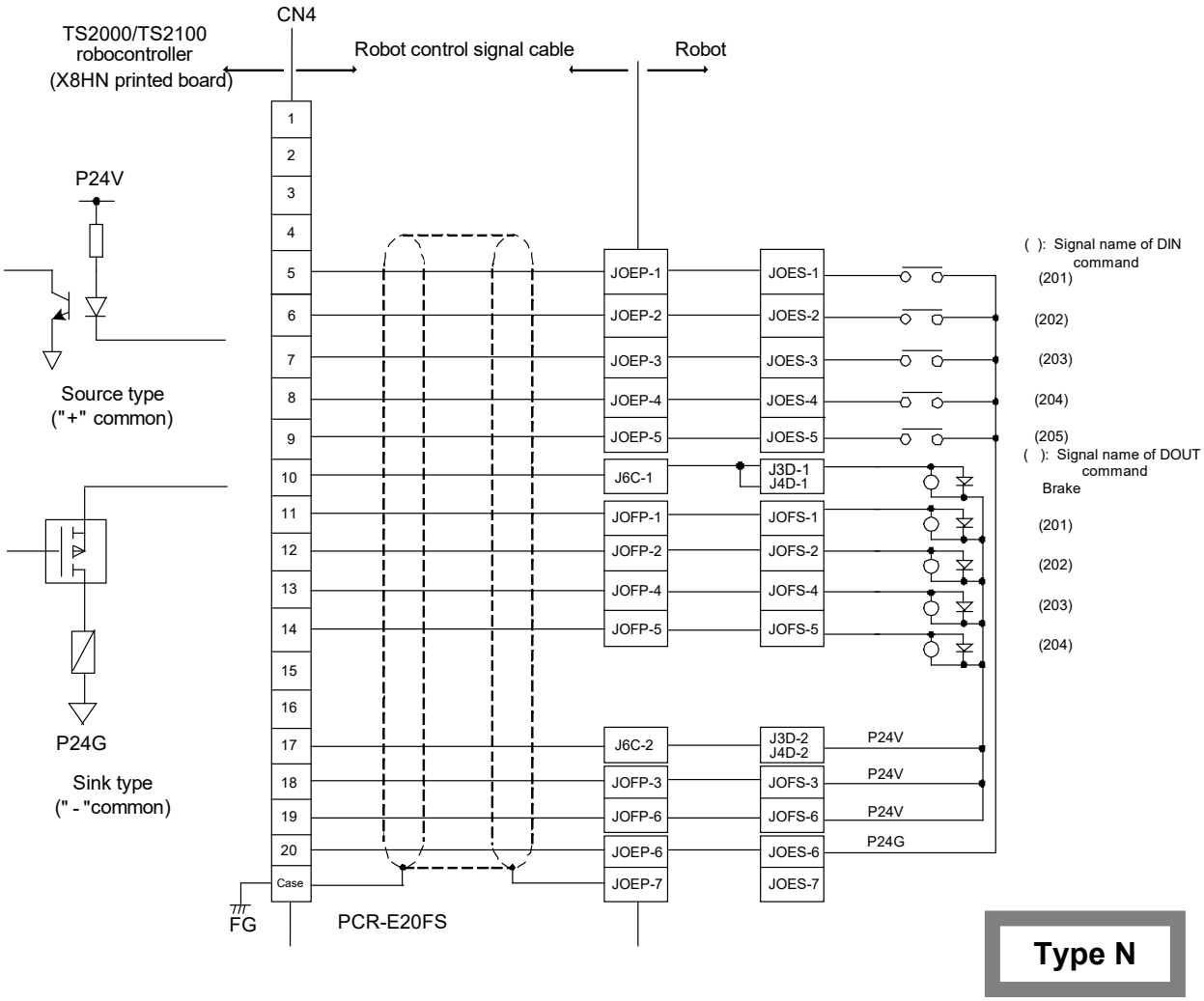


Fig. 3.4 Connection of robot control signal cable (Type N)

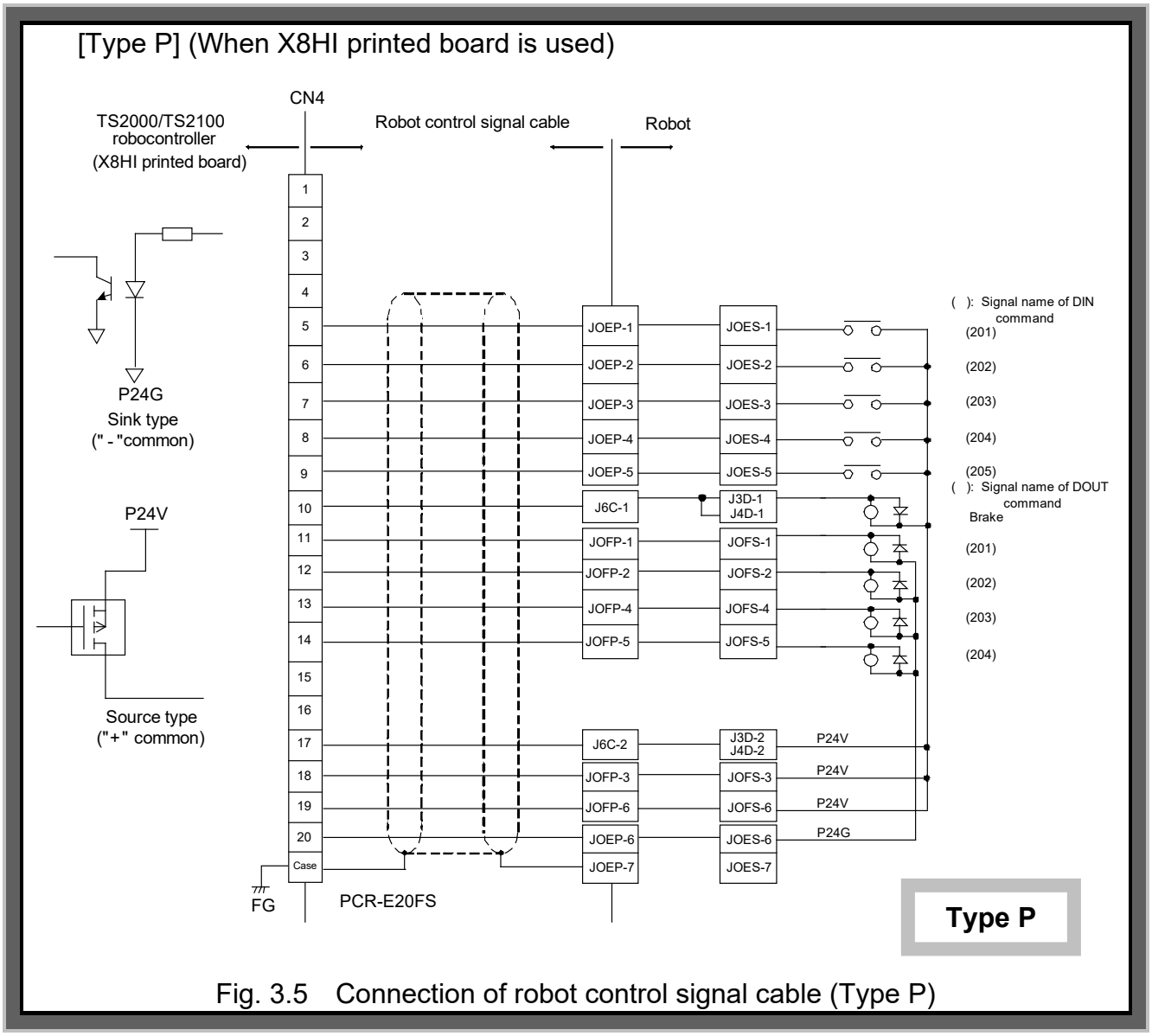


Fig. 3.5 Connection of robot control signal cable (Type P)

The robot control signal controls ON/OFF of the brake for securing the motor shaft, and the end effector such as hand operation.

The TS2000/TS2100 controller is provided with five (5) hand input signals and four (4) hand output signals to control the end effector.

The specifications of the hand input signal are same as those of the digital input signal. The output type, output circuit structure and electric rating of the hand output signal differ from those of the digital output signal, as described below.

All of the hand output signals are turned off when the controller power has turned on or turned off. When designing the hand effector, therefore, take careful precautions not to drop a part at power failure, etc.
The robot control signal can be controlled from the robot language program. In the robot language, the robot control signal is specified by the signal name assigned to each signal.
For the robot language, see the Robot Language Manual.
Additionally, the hand input and output signals can be controlled by the PLC built in the TS2000/TS2100. For details, see the Simple PLC Function Manual.

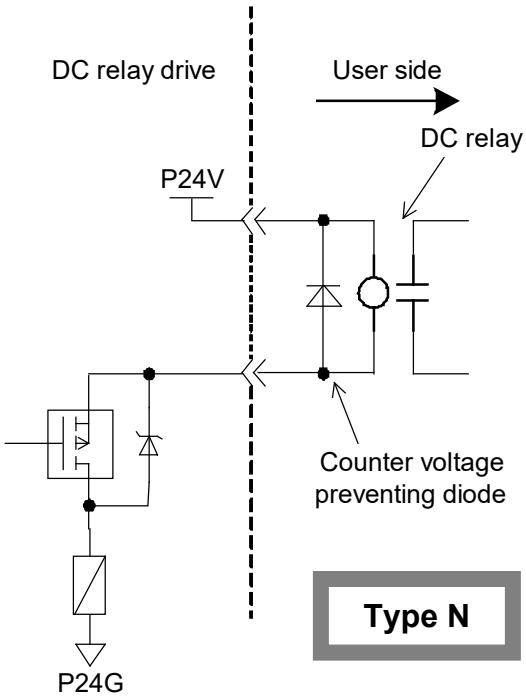
It is also possible to control the double solenoid device by combining two (2) hand output signals. When this happens, two (2) output signals bearing consecutive signal names are used to serve as a double solenoid. For the automatic operation, program using the robot language so that the two (2) output signals forming the double solenoid can be made exclusive. While the controller power is turned off or turned on, both output signals turn off.

The hand output signal can be turned on and off manually through the teach pendant on condition that each hand output signal which turns on and off is defined in the user parameter (USER. PAR) beforehand.
For the setting procedures, see the User Parameter Manual.

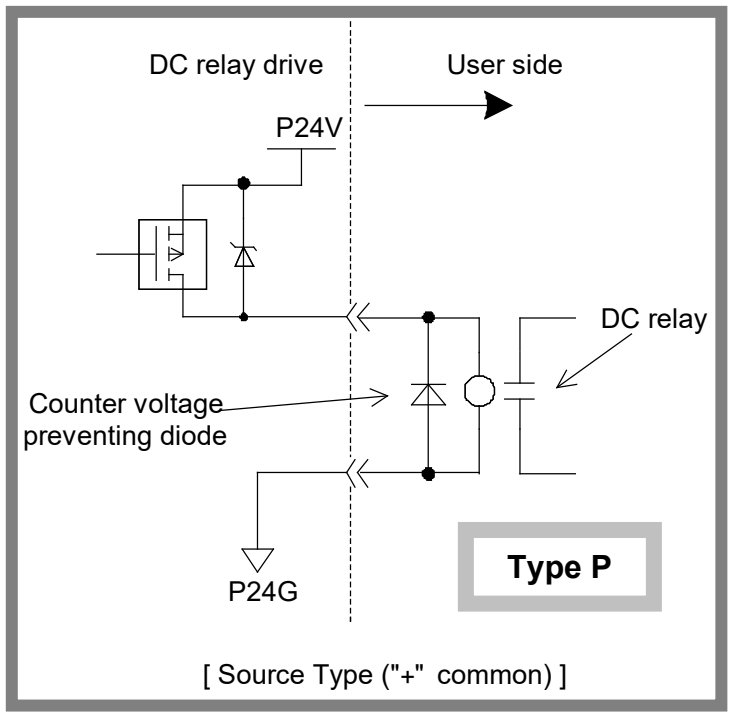
Note: Once the double solenoid is defined, using the user parameter, two (2) output signals are output exclusively by the hand output signal operation through the teach pendant. For the automatic operation, however, program using the robot language so that the two (2) output signals forming the double solenoid can be made exclusive. Also, even if the double solenoid is defined in advance, relevant two (2) signals turn off when the controller power is turned on.
In the commands of OPEN1, OPEN2, CLOSE1, CLOSE2, OPENI1, OPENI2, CLOSEI1 and CLOSEI2, each set signals of (201, 202) and (203, 204) are output exclusively.
The hand output signal cannot be reset by the RESET SIG operation or RESET DOUT command.

The specifications of the hand output signal are as follows:

- Output type : FET output
- Electric rating : Rated voltage DC24 V, rated current 1 A (max.)
- Output circuit structure



[Sink type ("-" common)]



[Source Type ("+" common)]

! CAUTION

If the current which exceeds the rated output current is supplied, the output device may be damaged or the printed board may be burnt. To avoid this, be sure to use within the rated output current.

The current of a total of four (4) hand output signals should be 1 A or less.

NEVER use the hand input/output function of the controller together with the external power supply to prevent a failure.

For further information on the robot control signal cable connection, see the Installation & Transport Manual.

4. Connecting External I/O Signal Cable

4.1 Connecting External Input Signal Cable

To connect the external input signal cable, use the attached connector [XM2D–3701 (socket type connector), XM2S–3711 (connector cover)].
The input common comes in the two (2) types; Type N [X8HN (output sink type) is selected for the I/O printed board and the polarity is the same as in the SR7000 robot] and Type P [X8HI (output source type) is selected for the I/O printed board).
After confirming the type of your controller, connect the external input signal cables.

[Type N] (When X8HN printed board is used)

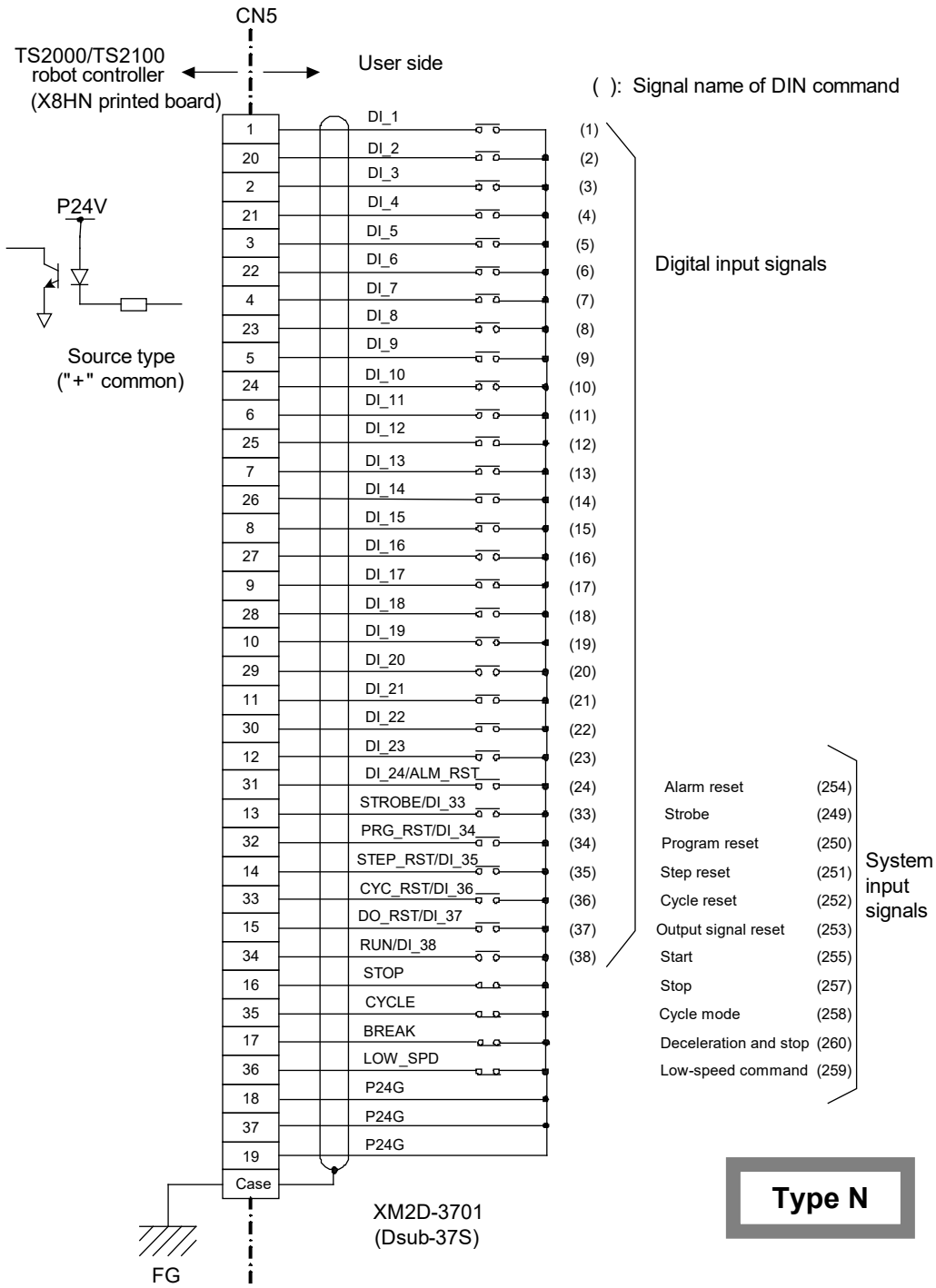


Fig. 4.1 Connection of external input signal cables (Type N)

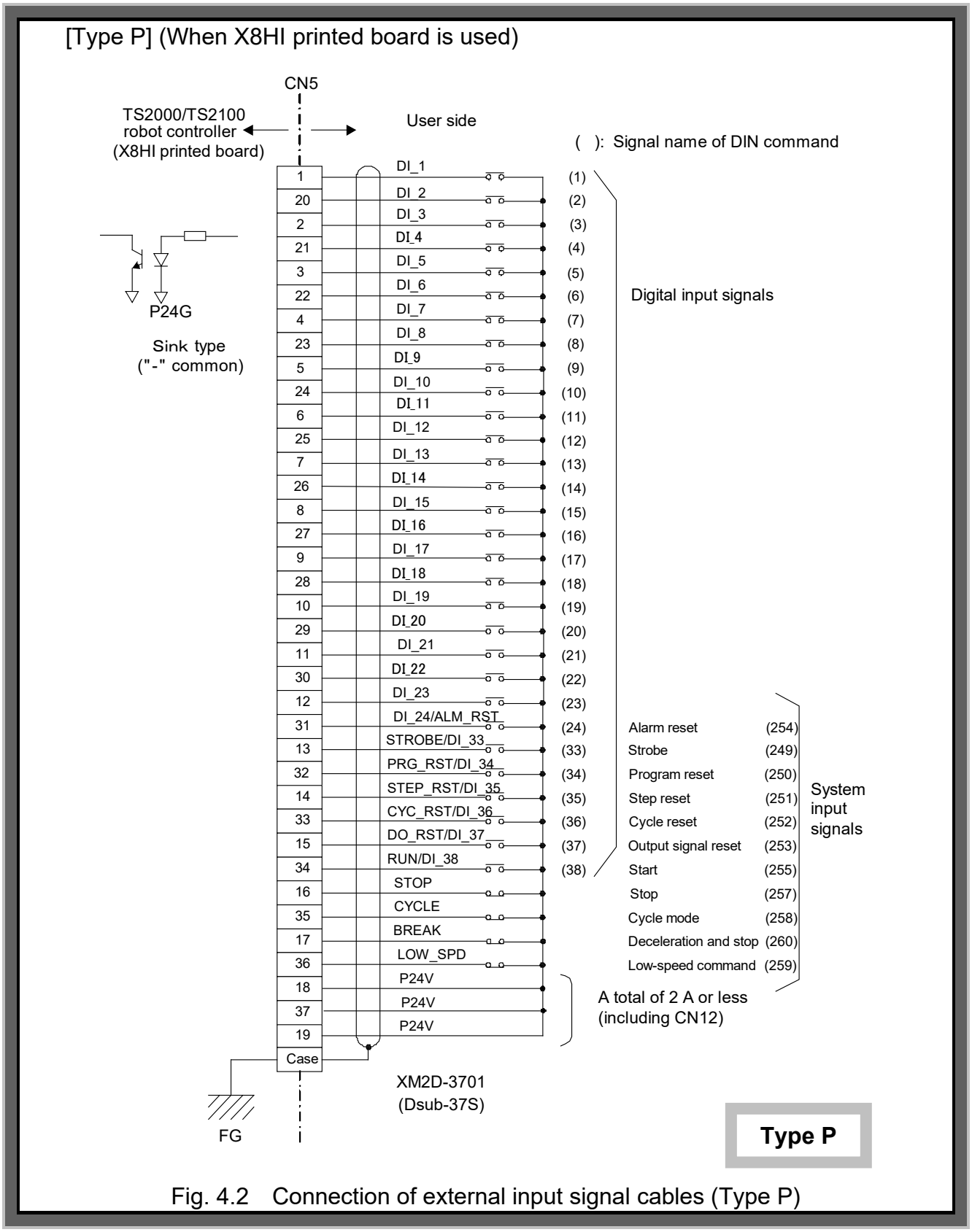


Fig. 4.2 Connection of external input signal cables (Type P)

As shown in Fig. 4.1 and Fig. 4.2 above, the digital input signals are non-voltage contact inputs or open collector inputs. Except for the system input signals of STOP, CYCLE, LOW_SPD, BREAK which are the normal close contact inputs, all others are normal open contact inputs.
For the function, circuit to use etc., of each signal, see Para. 4.4 and 4.5.
Additionally, the external input signals can be controlled by the PLC built in the TS2000/TS2100. For details, see the Simple PLC Function Manual.

4.2 Connecting External Output Signal Cable

To connect the external output signal cable, use the attached connector [XM2A–3701 (plug type connector), XM2S–3711 (connector cover)].
The output common comes in the two (2) types; Type N [X8HN (output sink type) is selected for the I/O printed board and the polarity is the same as in the SR7000 robot] and Type P [X8HI (output source type) is selected for the I/O printed board). After confirming the type of your controller, connect the external output signal cables.

[Type N] (When X8HN printed board is used)

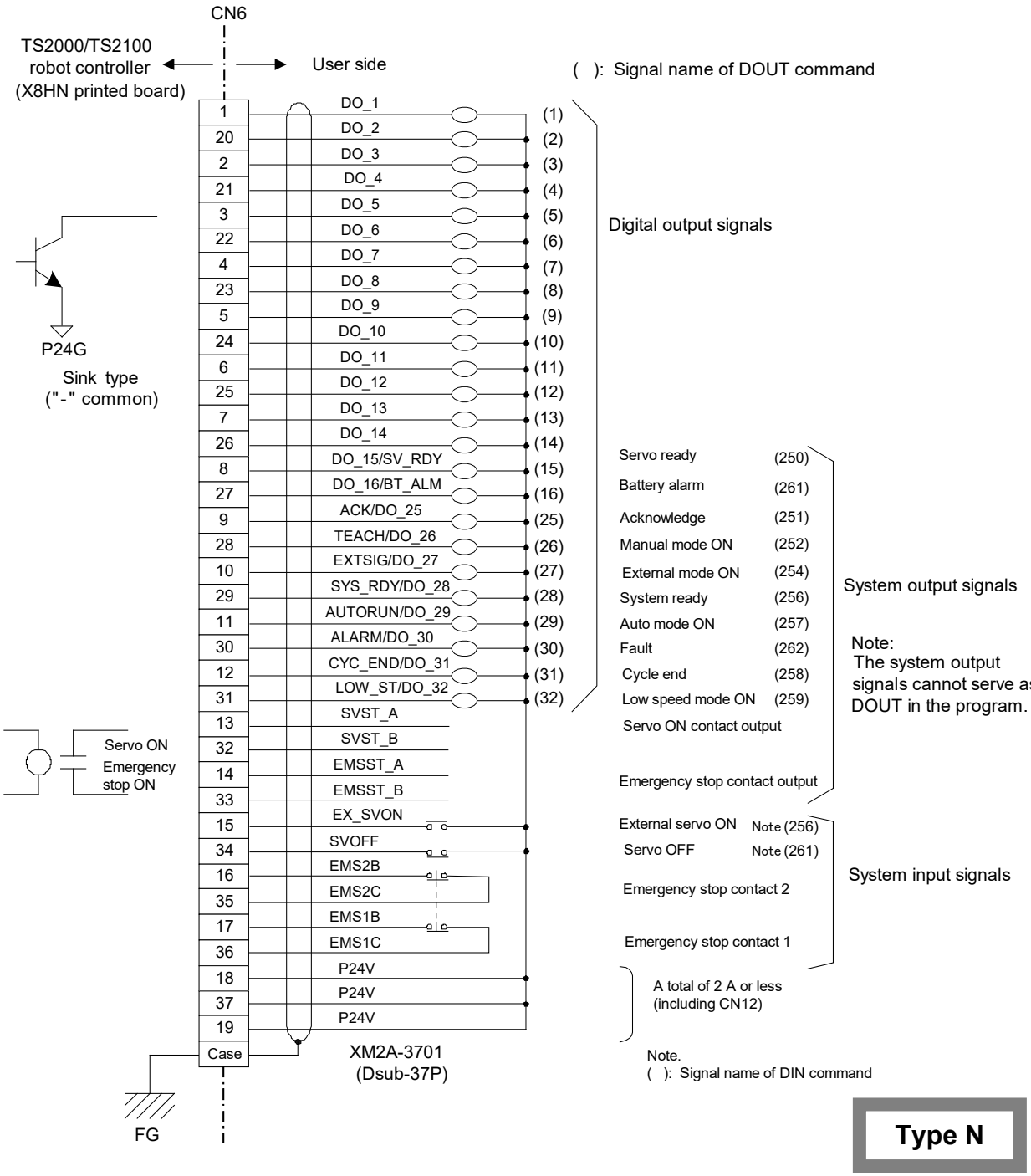


Fig. 4.3 Connection of external output signal cables (Type N)

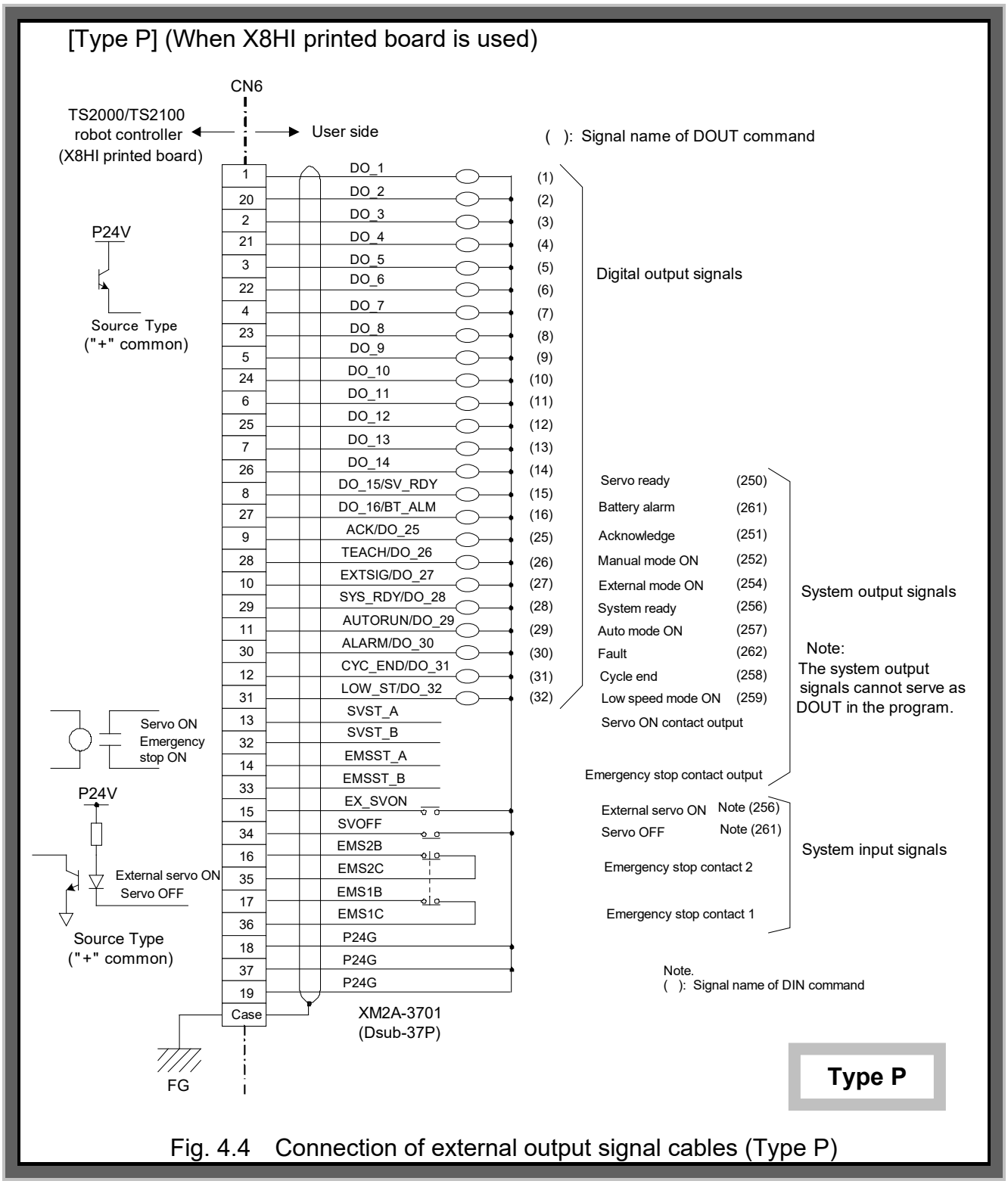


Fig. 4.4 Connection of external output signal cables (Type P)

As shown in Fig. 4.3 and Fig. 4.4 above, all sixteen (16) digital output signals are transistor outputs. Two (2) system output signals (i.e., servo ON contact output and emergency stop contact output) are non-voltage relay contact outputs, and all other system output signals are transistor outputs.
The system input signals of External servo ON, Servo OFF, Emergency stop contact 1, Emergency stop contact 2 (two (2)-contact connection) are non-voltage contact inputs or open collector inputs.
(In the Type P, both External servo ON and Servo OFF are the source type ("+" common), different from the other input signals. Connect them with the external input device, taking careful precautions on the input polarity.)
For the function, circuit to use etc., of each signal, see Para. 4.5, 4.7 and 4.8.
Additionally, the external output signals can be controlled by the PLC built in the TS2000/TS2100. For details, see the Simple PLC Function Manual.

4.3 Connecting External I/O Signal Cable

To connect the external input/output signal cable, use the connector [XM2A–2501 (plug type connector), XM2S–2511 (connector cover)].
This connector is not included in the accessories, which is available from us, however, for an extra price, or which should be provided by the customer.
The I/O common comes in the two (2) types; Type N [X8HN (output sink type) is selected for the I/O printed board and the polarity is the same as in the SR7000 robot] and Type P [X8HI (output source type) is selected for the I/O printed board).
After confirming the type of your controller, connect the external I/O signal cables.

[Type N] (When X8HN printed board is used)

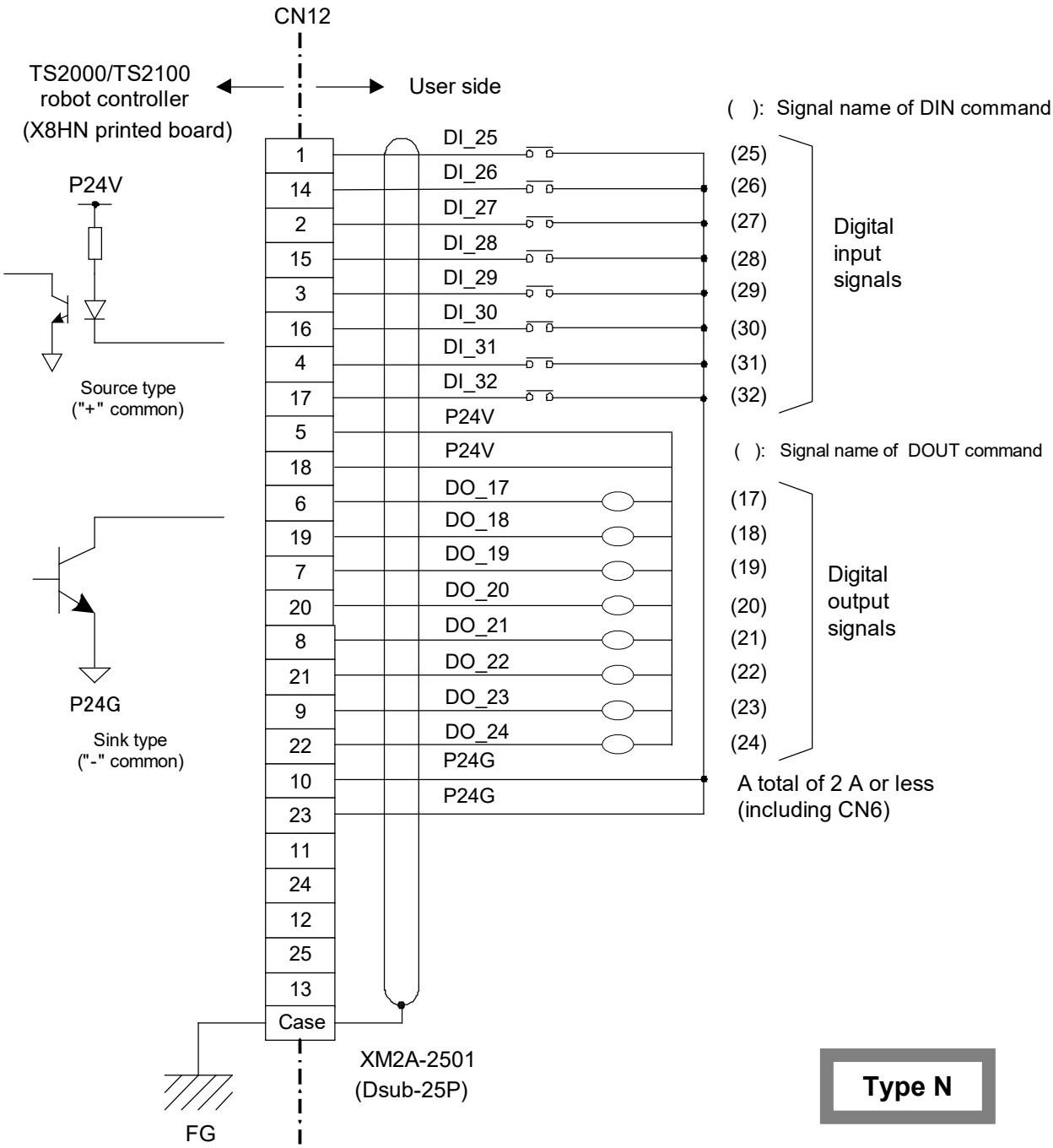


Fig. 4.5 Connection of external I/O signal cables (Type N)

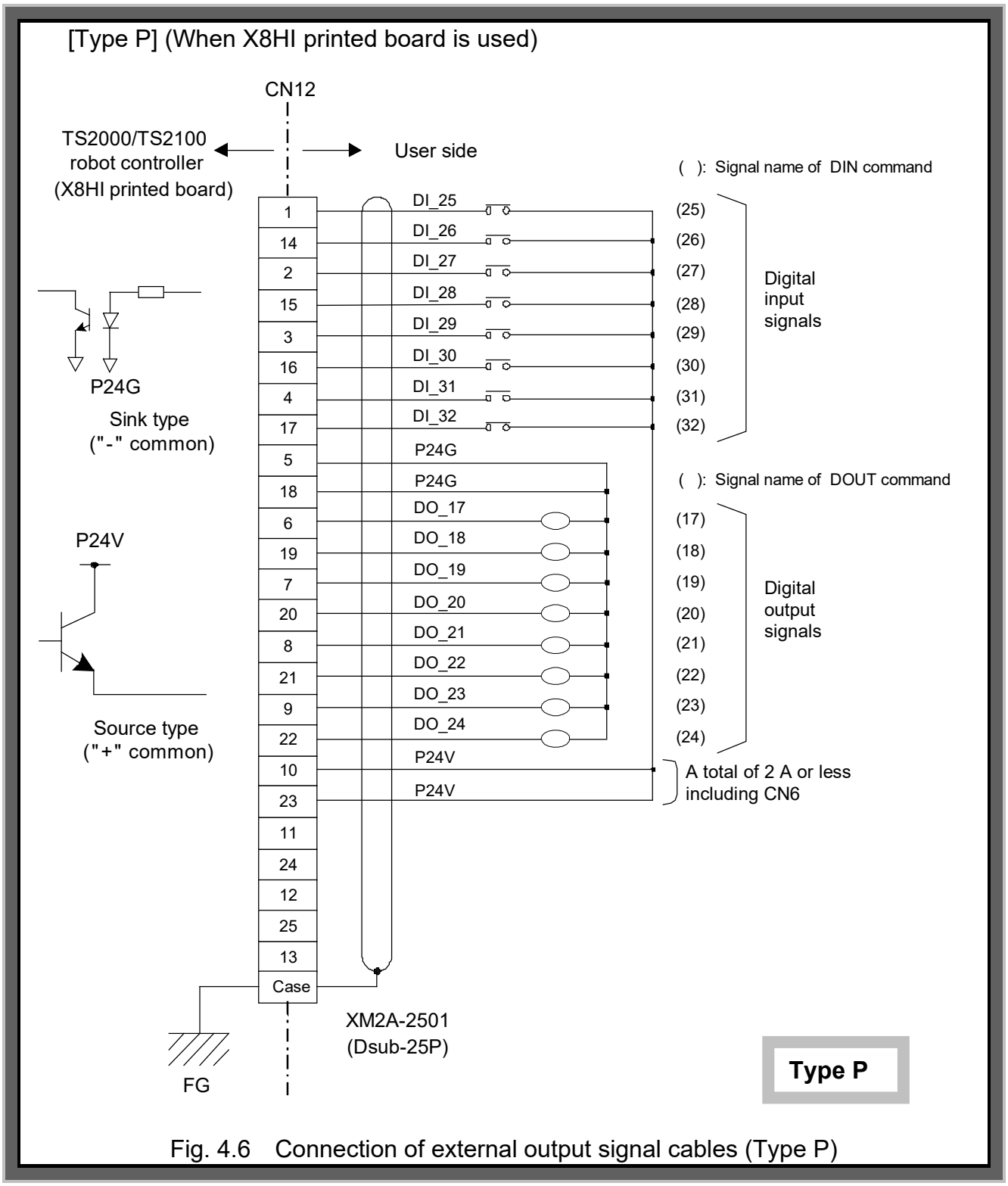


Fig. 4.6 Connection of external output signal cables (Type P)

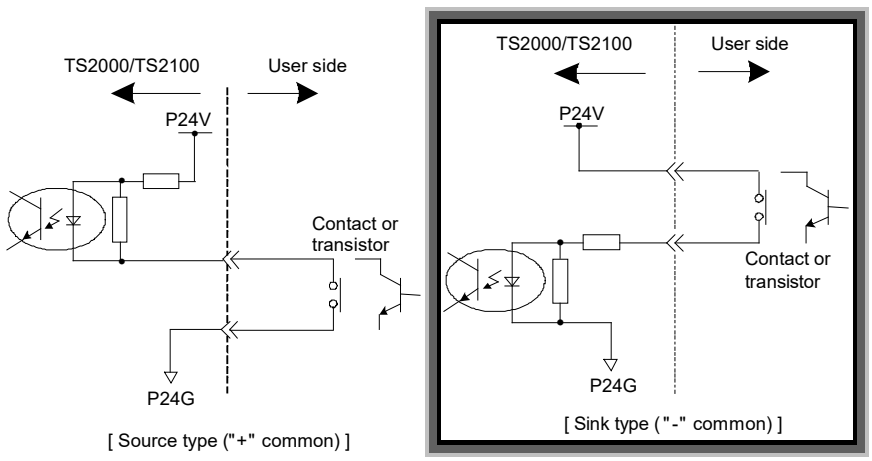
As shown in Fig. 4.5 and 4.6 above, eight (8) digital input signals are non-voltage contact inputs or open collector inputs. Eight (8) digital output signals are transistor outputs.
For the function, circuit to use etc., of each signal, see Para. 4.4 and 4.8.
Additionally, the external input/output signals can be controlled by the PLC built in the TS2000/TS2100. For details, see the Simple PLC Function Manual.

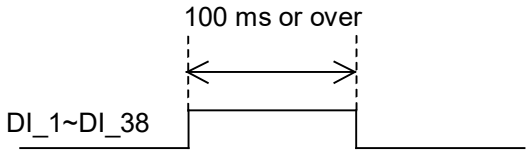
! CAUTION

The COM1, HOST, TCPRG, POD and CN12 connectors of the TS2000/TS2100 robot controller are attached with a connector cap, respectively.

Unless these connectors are used, be sure to attach the connector caps to prevent static electricity and damage.

4.4 Digital Input Signal

Designation	Digital input signal DI_1 ~ DI_32, DI_33 ~ DI_38 (system input signals)
Connector input terminal	Signals DI_1 ~ DI_24 are assigned to CN5–1 ~ 12 pins and 20 ~ 31 pins. (See Fig. 4.1 and 4.2.) DI_24 can be used as signal ALM_RST by changing the user parameter. DI_25 ~ DI_32 are assigned to CN12–1 ~ 4 pins and 14 ~ 17 pins. (See Fig. 4.5 and 4.6.) The system input signals assigned to CN5–13 ~ 15 pins and 32 ~ 34 pins can be used as signals DI_33 ~ DI_38 by changing the user parameter.
Function	Each signal status of DI_1 ~ DI_38 can be identified by the robot program (DIN command) to branch the processing of program. Also, it is possible to perform interruptive processing of each signal (DI_1 ~ DI_38) with change in signal status monitored during the robot operation.
Input type	Non-voltage contact input or transistor open collector input.
Example of circuit (Input circuit structure)	
Signal logic	Input terminal — Signal judgment Open — OFF Short-circuit — ON

Specifications of non-voltage contact and transistor	• Non-voltage contact specifications Contact rating DC24 V, 10 mA or over Circuit current: Approx. 7 mA Minimum current DC 24 V, 1 mA Contact impedance 100 Ω or less • Transistor specifications Withstand voltage between collector and emitter 30 V or over Current between collector and emitter 10 mA or over Circuit current: Approx. 7 mA Leakage current between collector and emitter 100 μA or less
Signal timing	When the pulse type input signals are used, the pulse width should be 100 ms or over. 

* When using the system input signals as the digital input signals:
When "Default" (standard PLC) is specified by user parameter [U11] (I/O mode), you can select whether INPUT23, 24, 33 ~ 38 are the system input signals or digital input signals.

User parameter [U13]
[U13] Select input signal (Default I/O mode only)
{Input 23,24,33,34,35,36,37,38} (0: System 1: User)
= 1 1 0 0 0 0 0 0

Specify "1" for a desired bit underlined above, and appropriate system input signal can serve as the digital input signal. Respective bits signify DI_23, DI_24, STROBE, PRG_RST, STEP_RST, CYC_RST, DO_RST and RUN when viewed from the left, which correspond to digital signals DI_23, 24, 33 ~ 38.

To use only system input signal STROBE as digital input signal DI_33, for instance, change the above bits as shown below.

1 1 1 0 0 0 0 0

To use all system input signals as the digital input signals, specify in the following manner.

1 1 1 1 1 1 1 1

DO NOT specify zero (0; system input) for the bit of INPUT23.
After the above parameter has been changed, save the data, turn the power off and on again. Otherwise, the parameter will not be operative.

0: System input signal
1: Digital input signal

Input signal cable	0: System input	1: Digital input
INPUT23	Reserved (for extending function)	DI_23
INPUT24	ALM_RST	DI_24
INPUT33	STROBE	DI_33
INPUT34	PRG_RST	DI_34
INPUT35	STEP_RST	DI_35
INPUT36	CYC_RST	DI_36
INPUT37	DO_RST	DI_37
INPUT38	RUN	DI_38

4.5 System Input Signal

In addition to a total of thirteen (13) signals which control STOP, CYCLE, etc., of the TS2000/TS2100 robot controller from the external equipment, emergency stop contacts 1 and 2 are also available for the system input signal.
The system input signal is provided with an exclusive input terminal for each function.
Of the above signals, six (6) signals (STROBE, PRG_RST, STEP_RST, CYC_RST, DO_RST, RUN) can be used as the digital input signals (DI_33 ~ DI_38) by changing the user parameter (initial setting).
For the setting procedures, see the descriptions given above.
For ALM_RST, DI_24 is assigned by the initial setting of the user parameter. This signal can be used as ALM_RST by changing the user parameter.
The structure of the system input signal is quite the same as that of the digital input signal. See the descriptions in Para. 4.4. (However, the two (2) signals of emergency stop contact 1 and emergency stop contact 2 differ from the above specifications. See the descriptions on each signal.)

When inputting the system input signal, keep the input status until the output signal corresponding to each input is output to assure each signal input. The relationship between input signal and output signal is stipulated by the timing chart of each signal.
Also, each system input signal becomes valid or invalid by means of the MODE (master mode) selector switch equipped on the control panel. Each signal ON mode is shown in Table 4.1 below.

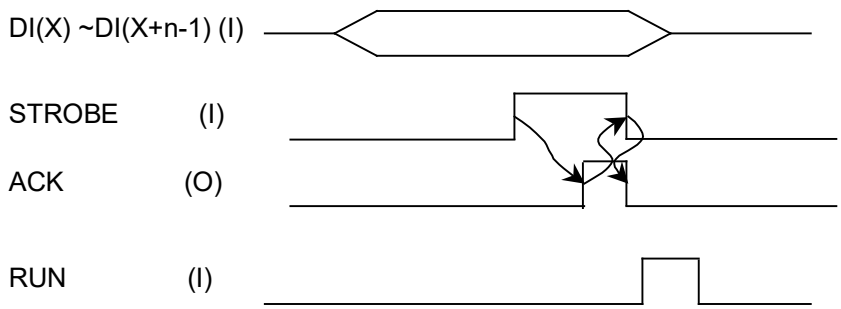
Table 4.1 List of system input signal ON modes

Master mode / Designation	ON mode			
			EXTERNAL	
	TEACHING	INTERNAL	EXT. SIGNAL	EXT. HOST
STROBE (Strobe)			O	
PRG_RST (Program reset)			O	
STEP_RST (Step reset)			O	
CYC_RST (Cycle reset)			O	
DO_RST (Output signal reset)			O	
ALM_RST (Alarm reset)			O	
RUN (Start)			O	
EX_SVON (External servo ON)			O	
STOP (Stop)	O	O	O	O
CYCLE (Cycle operation mode)			O	
LOW_SPD (Low speed command)	O	O	O	O
BREAK (Deceleration and stop)	O	O	O	O
SVOFF (Servo OFF)	O	O	O	O
EMS*B ~ EMS*C (Emergency stop contacts 1 and 2)	O	O	O	O

O : ON mode

Designation	STROBE (Strobe)
Input terminal	CN5–13 pin
Exclusive signal name used in the controller	249
Signal logic	

Signal judgment / Signal terminal	ON	OFF
Open		O
Short-circuit	O	

Function	Used to select an execution program for the TS2000/TS2100 robot controller from the external equipment. The program number selected should use any successive "n" numbers (max. eight (8) numbers) of external digital input signal, which are coded. 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 ← Program Number (max. eight (8) bits) DI(X+n–1) ---------- DI(X) ← External digital input signal * n = 1 ~ 8 For the program file name and register of it to the program number, and assignment of bits to external digital input signals, see the User Parameter Manual. This signal can be used only in the EXT. SIGNAL mode.
Signal timing	 With the start of the STROBE signal, the above digital signals are read to select an appropriate program. After the program has been selected, the ACK signal turns on. Turn on the RUN signal and execute the program.
Cautions	The STROBE signal should not be input together with the PRG_RST, CYC_RST, STEP_RST or DO_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid. If a file other than the current file is selected, the program is reset to step 1 and the values of variables are reset also.

* Program file name and register of it to program number, and assignment of bits to external digital input signals:

  To select an execution file (i.e., program selection), using digital input or extension input signals, assignment of bits to the controller input signals is necessary.

  User parameter [U07]
  [U07] Specify signal for EXTSELECT.
  {Signal No.} (1 –)
  {Bit length} (1 – 8)
  = 1 4

  Bits underlined = (Leading signal name in DIN command) (Bit length).
  "Signal name in DIN command" signifies an input signal number assigned in a program to run the internal sequence. (The signal name is predetermined in the program and cannot be changed by the user.)
  For the digital input signal, "signal name in DIN command" represents a number in parentheses as shown on the right side of the signal name in Fig. 4.1 and 4.2.

  (Leading signal name in DIN command):
  Specify the leading number of input signals to be used. (Allowable range: 1 ~ 32, 101 ~ 164, 301 ~ 364)

  (Bit length):
  Specify the number of signals to be used. (Allowable range 1 ~ 8)

  Set value underlined (ex.) = 1 4
  This signifies that four (4) external digital input signals 1 ~ 4 are used.

Correspondence table
between [U07] set value (example) and program file name

Signal name in DIN command / Program file name (EXTRNSEL. SYS)	4	3	2	1
"PROG1"	0	0	0	0
"PROG2"	0	0	0	1
"PROG3"	0	0	1	0
"PROG4"	0	0	1	1
"PROG5"	0	1	0	0
"PROG6"	0	1	0	1
"PROG7"	0	1	1	0
"PROG8"	0	1	1	1
"PROG9"	1	0	0	0
"PROG10"	1	0	0	1
"PROG11"	1	0	1	0
"PROG12"	1	0	1	1
"PROG13"	1	1	0	0
"PROG14"	1	1	0	1
"PROG15"	1	1	1	0
"PROG16"	1	1	1	1

When bits underlined = 12, two (2) external digital input signals 1 and 2 are used, and the number of program files selected is four (4) from "PROG1" ~ "PROG4".

To register the program file name to the program number, use the EXTRNSEL. SYS file.

Copyright (C) 2001 by SHIBAURA MACHINE CO., LTD.
All rights reserved.
External select file "EXTRNSEL. SYS"
*** [00 - 0F] *****
= "PROG00"
= "PROG01"
= "PROG02"
= "PROG03"

= "PROG04"
= "PROG05"
= "PROG06"
= "PROG07"
= "PROG08"
= "PROG09"
= "PROG0A"
= "PROG0B"
= "PROG0C"
= "PROG0D"
= "PROG0E"
= "PROG0F"

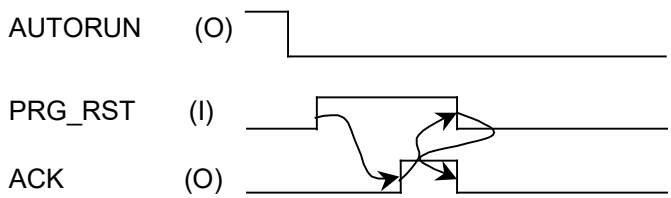
The initial setting is as shown above. Specify a file name you registered beforehand for the underlined of "PROG**".

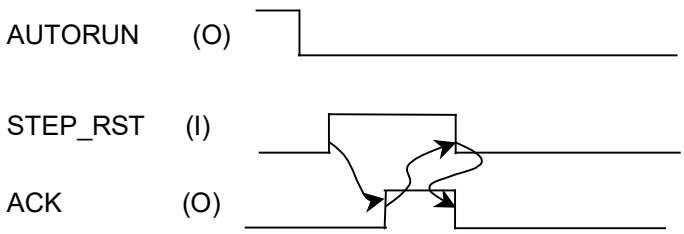
Example:
= "AAA"
= "BBB"
= "CCC"
= "DDD"

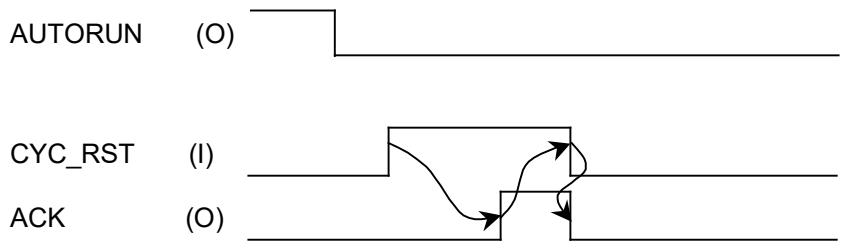
Specify the above program names in advance. (If the following steps are taken without registering the program names, a "Compile Error" occurs.) Then specify zero (0) for both bits 1 and 2 of DIN command, which are set by user parameter [U07] (assume that [U07] = 1 2) and input the STROBE signal. Program "AAA" is automatically selected now.

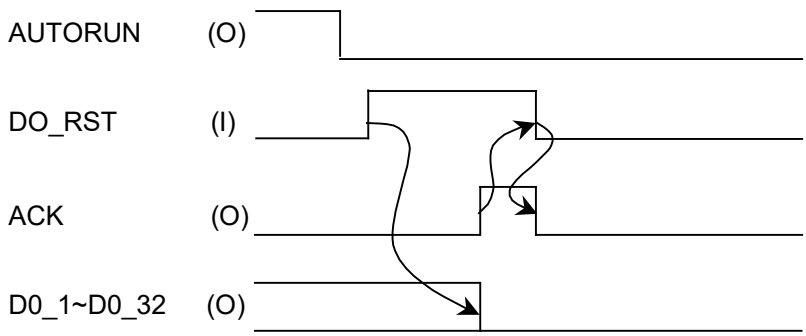
After the above parameter has been changed, save the data, turn the power off and on again. Otherwise, the parameter will not be operative.

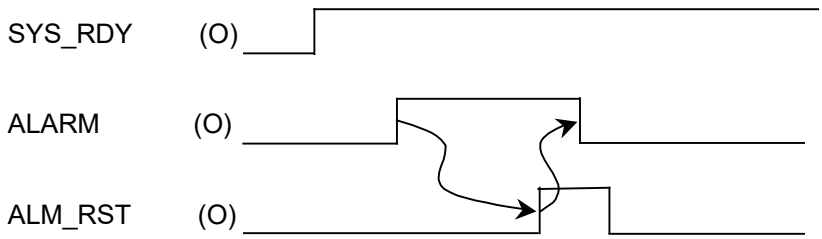
For the EXTRNSEL. SYS file, only the line described as "= File name" is effective and the other lines are regarded as the comment.

<table>
<tr><td>Designation</td><td>PRG_RST (Program reset)</td></tr>
<tr><td>Input terminal</td><td>CN5–32 pin</td></tr>
<tr><td>Exclusive signal name used in the controller</td><td>250</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>Open</td><td></td><td>O</td></tr>
<tr><td>Short-circuit</td><td>O</td><td></td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to reset a currently stopped program to step 1. The value of each variable is also reset to zero (0).
This signal can be used only in the EXT. SIGNAL mode.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>1. The PRG_RST signal should not be input together with the STROBE, CYC_RST, STEP_RST or DO_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid.
2. This signal cannot be used while AUTORUN (automatic operation mode ON) is set ON.</td></tr>
</table>

Designation	STEP_RST (Step reset)
Input terminal	CN5–14 pin
Exclusive signal name used in the controller	251
Signal logic	<table><tr><td>Signal terminal \ Signal judgment</td><td>ON</td><td>OFF</td></tr><tr><td>Open</td><td></td><td>O</td></tr><tr><td>Short-circuit</td><td>O</td><td></td></tr></table>
Function	Used to reset a currently stopped program to step 1. The value of each variable used in the program remains unchanged. This signal can be used only in the EXT. SIGNAL mode.
Signal timing	
Cautions	1. The STEP_RST signal should not be input together with the STROBE, PRG_RST, CYC_RST or DO_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid. 2. This signal cannot be used while AUTORUN (automatic operation mode ON) is set ON.

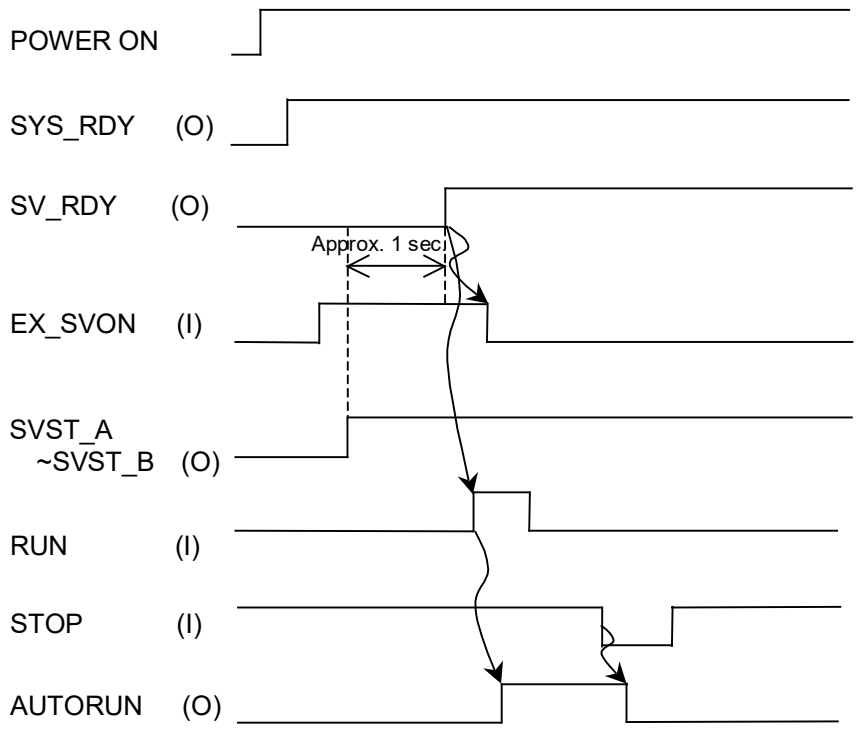
<table>
<tr><td>Designation</td><td>CYC_RST (Cycle reset)</td></tr>
<tr><td>Input terminal</td><td>CN5–33 pin</td></tr>
<tr><td>Exclusive signal name used in the controller</td><td>252</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>Open</td><td></td><td>O</td></tr>
<tr><td>Short-circuit</td><td>O</td><td></td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to reset a currently stopped program to the step labeled "RCYCLE". The value of each variable used in the program remains unchanged.
This signal can be used only in the EXT. SIGNAL mode.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>1. The CYC_RST signal should not be input together with the STROBE, PRG_RST, STEP_RST or DO_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid.
2. This signal cannot be used while AUTORUN (automatic operation mode ON) is set ON.</td></tr>
</table>

<table>
<tr><td>Designation</td><td>DO_RST (Output signal reset)</td></tr>
<tr><td>Input terminal</td><td>CN5–15 pin</td></tr>
<tr><td>Exclusive signal name used in the controller</td><td>253</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>Open</td><td></td><td>O</td></tr>
<tr><td>Short-circuit</td><td>O</td><td></td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to reset digital output signals (DO_1 ~ DO_32) of the TS2000/TS2100 robot controller from the external equipment. (Extension I/O signals DO_101 (133) ~ DO_120 (152) are also reset.)
When reset, all signals of DO_1 ~ DO_32 turn off. When DO_15, DO_16, DO_25 ~ DO_32 are set as the system output signals, however, they are turned on.
This signal can be used only in the EXT. SIGNAL mode.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>1. The DO_RST signal should not be input together with the STROBE, PRG_RST, CYC_RST or STEP_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid. Neither system output signals nor hand output signals are reset.
2. This signal cannot be used while AUTORUN (automatic operation mode ON) is set ON.</td></tr>
</table>

Designation	ALM_RST (Alarm reset)
Input terminal	CN5–31 pin
Exclusive signal name used in the controller	254
Signal logic	<table><tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr><tr><td>Open</td><td></td><td>O</td></tr><tr><td>Short-circuit</td><td>O</td><td></td></tr></table>
Function	Used to cancel an alarm from the external equipment, which occurred while the TS2000/TS2100 robot controller was ready to start. This signal can be used only in the EXT. SIGNAL mode.
Signal timing	
Cautions	If an alarm of the emergency stop level which will not allow the processing of EX_SVON, or EMSST_A ~ EMSST_B is output, alarm reset by the ALM_RST signal is not possible.

Designation	RUN (Start)
Input terminal	CN5–34 pin
Exclusive signal name used in the controller	255
Signal logic	

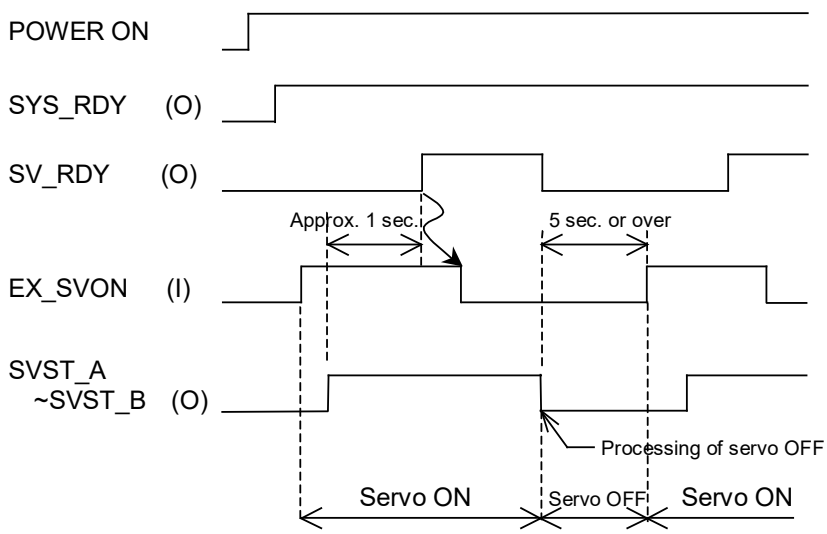
Signal terminal \ Signal judgment	ON	OFF
Open		O
Short-circuit	O	

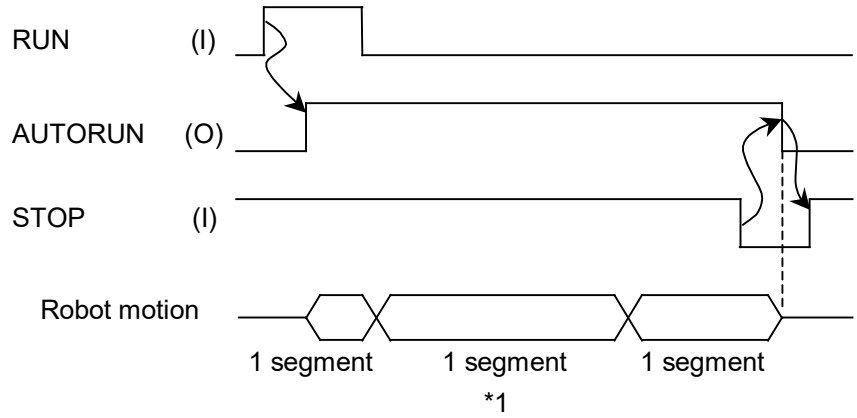
Function	Used to start a program registered in the TS2000/TS2100 robot controller from the external equipment to execute an automatic cycle operation. This signal can be used only in the EXT. SIGNAL mode.
Signal timing	

Cautions	After servo ON, automatic operation starts with the start of the RUN signal. It takes about one (1) second from the input of EX_SVON to the time when the robot is actually ready to work. Set ON the RUN signal only after the SV_RDY signal is ON. Even if the RUN signal is input before the SV_RDY signal is ON, it is neglected and the automatic operation will not be started.

Designation	EX_SVON (External input servo ON)
Input terminal	CN6–15 pin
Exclusive signal name used in the controller	256
Signal logic	

Signal terminal \ Signal judgment	ON	OFF
Open		O
Short-circuit	O	

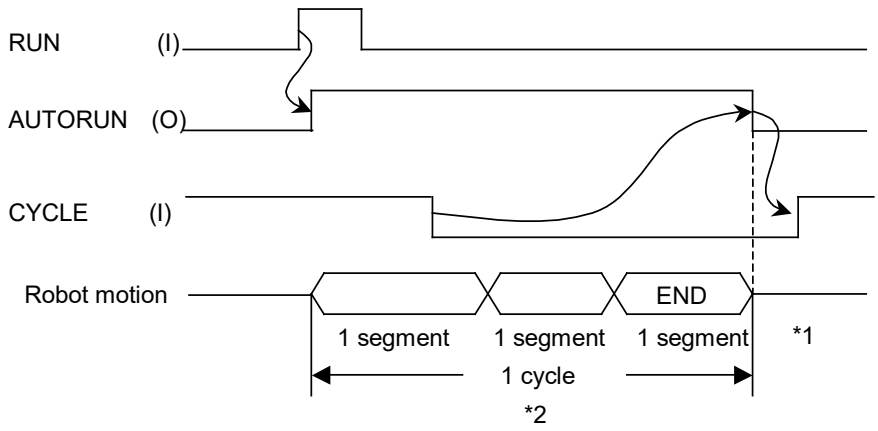
Function	Used to turn on the servo driver main power from the external equipment. Once the servo power is turned on, it is maintained even after this signal turns off. This signal can be used only in the EXT. SIGNAL mode.
Signal timing	
Cautions	1. It takes about one (1) second from the servo ON to the time when the robot is actually ready to work. Therefore, program so that the RUN signal, etc., can turn on only after the SV_RDY signal turns on. 2. To prevent an internal damage, the servo cannot be turned on about 4.5 seconds after it is turned off. To turn the servo on again, wait at least five (5) seconds after the SVST_A ~ SVST_B signal turns off.

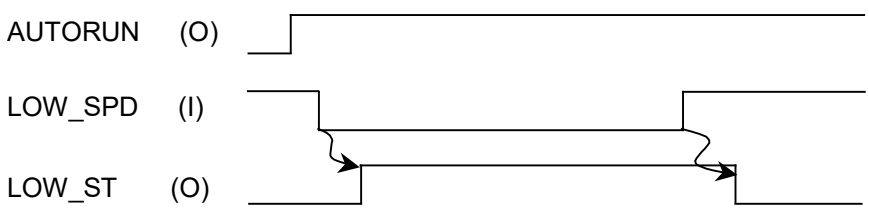
<table>
<tr><td>Designation</td><td>STOP (Stop)</td></tr>
<tr><td>Input terminal</td><td>CN5–16 pin</td></tr>
<tr><td>Exclusive signal name used in the controller</td><td>257</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>Open</td><td>O</td><td></td></tr>
<tr><td>Short-circuit</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to stop executing a program registered in the TS2000/TS2100 robot controller from the external equipment.
When this signal is open, the program stops after the current motion command has been executed.
When this signal is open, the robot cannot be operated.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 

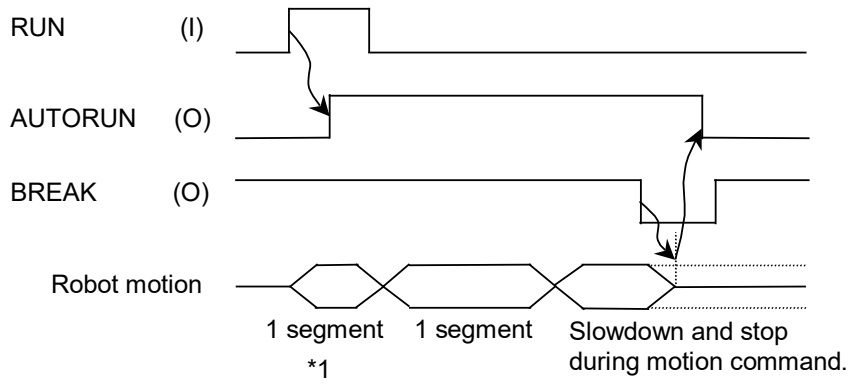
*1 Duration from the start of one motion command to just before the start of next motion command is called "1 segment".</td></tr>
<tr><td>Cautions</td><td>1. When the RUN command is executed after cancel of STOP, the program restarts from the step next to the interrupted step.
2. RUN signal input is ineffective at the input of STOP signal.
3. Unless this signal is used, short-circuit (0 V) input terminal CN5–16.</td></tr>
</table>

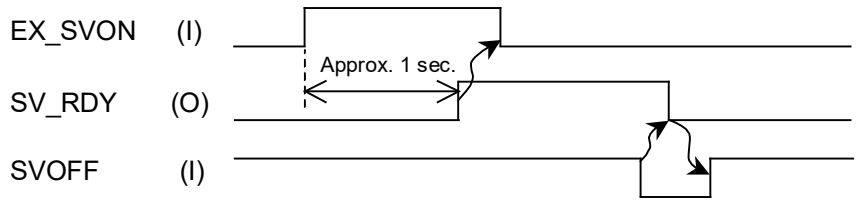
Designation	CYCLE (Cycle operation mode)
Input terminal	CN5–35 pin
Exclusive signal name used in the controller	258
Signal logic	

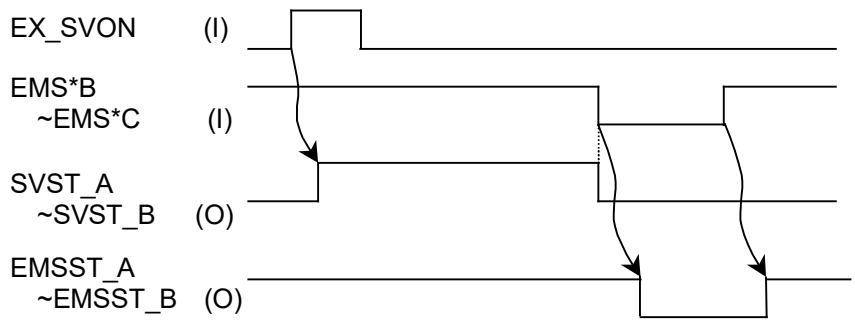
Signal terminal \ Signal judgment	ON	OFF
Open	O	
Short-circuit		O

Function	Used to stop from the external equipment a program registered in the TS2000/TS2100 robot controller after current one (1) cycle operation has been executed during automatic operation. This signal can be used only in the EXT. SIGNAL mode.
Signal timing	 *1 Duration from the start of one motion command to just before the start of next motion command is called "1 segment". *2 Duration from the top of the main program to the END command is called "1 cycle".
Cautions	1. Unless this signal is used, short-circuit (0 V) input terminal CN5–35 as necessary.

<table>
<tr><td>Designation</td><td>LOW_SPD (Low speed command)</td></tr>
<tr><td>Input terminal</td><td>CN5–36 pin</td></tr>
<tr><td>Exclusive signal name used in the controller</td><td>259</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>Open</td><td>O</td><td></td></tr>
<tr><td>Short-circuit</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to cause the robot operation speed to low speed from the external equipment.
The robot operates at a low speed (the low speed command is valid) while this signal is open.
The robot operation speed override in the low speed mode can be set by means of the parameter. (Initial set value: 25 %)
When this signal is short-circuited, the previously set value (initial set value: 100 %) takes effect again.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>1. If the speed override lower than the parameter set value is used, even if the LOW_SPD signal is made valid, the speed override value remains unchanged.
2. Unless this signal is used, short-circuit (0 V) input terminal CN5–36.</td></tr>
</table>

Designation	BREAK (Deceleration and stop)
Input terminal	CN5–17 pin
Exclusive signal name used in the controller	260
Signal logic	<table><tr><th>Signal judgment / Signal terminal</th><th>ON</th><th>OFF</th></tr><tr><td>Open</td><td>O</td><td></td></tr><tr><td>Short-circuit</td><td></td><td>O</td></tr></table>
Function	Used to stop the robot motion from the external equipment. The robot slows down and stops at the same time that this signal is open. After the stop, the robot enters a STOP (RETRY) status. Even if this signal is short-circuited again after the stop of robot motion, the robot will not operate. To restart the robot, short-circuit this signal, then execute the RUN command. If this signal is open, the robot cannot be started. This signal can always be used, irrespective of the master mode selected by means of the MODE switch.
Signal timing	 *1 Duration from the start of one motion command to just before the start of next motion command is called "1 segment".
Cautions	1. If the robot is operating, processing of execution is interrupted and the robot slows down and stops. 2. Unless this signal is used, short-circuit (0 V) input terminal CN5–17.

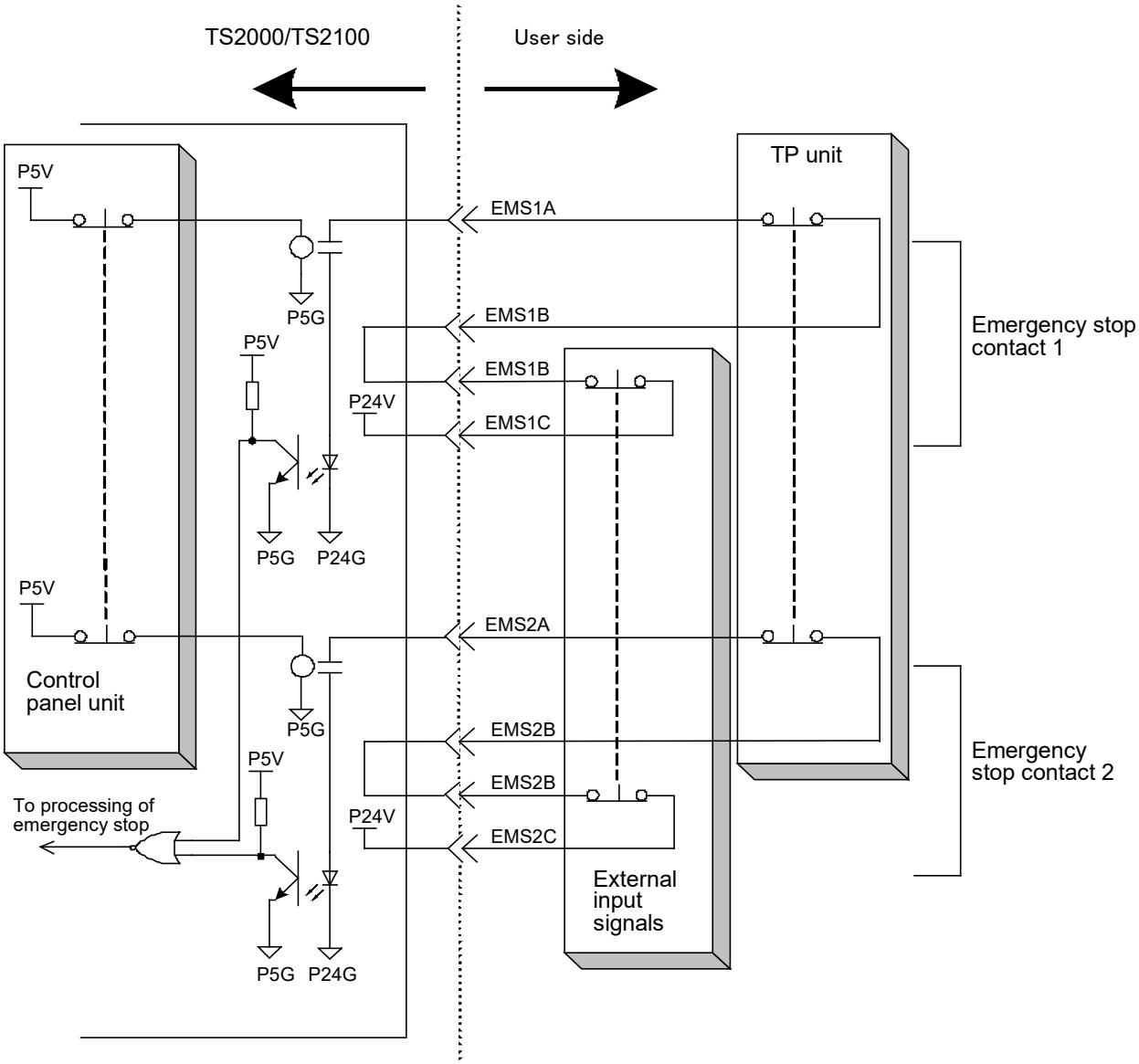
<table>
<tr><td>Designation</td><td>SVOFF (Servo OFF)</td></tr>
<tr><td>Input terminal</td><td>CN6–34 pin</td></tr>
<tr><td>Exclusive signal name used in the controller</td><td>261</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON (Servo OFF)</td><td>OFF (Normal)</td></tr>
<tr><td>Open</td><td>O</td><td></td></tr>
<tr><td>Short-circuit</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to turn off the servo driver main power from the external equipment.
While this signal is open, the servo power is turned off.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>1. While this signal is open, the servo power cannot be turned on in any mode.
2. Unless this signal is used, short-circuit between CN6–18 and CN6–34.</td></tr>
</table>

<table>
<tr><td>Designation</td><td>EMS*B ~ EMS*C (Emergency stop contacts 1 & 2)</td></tr>
<tr><td>Input terminal</td><td>Between CN6–16 and CN6–35 (Emergency stop contact 2)
Between CN6–17 and CN6–36 (Emergency stop contact 1)</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><th>Signal judgment / Signal terminal</th><th>ON (Emergency stop)</th><th>OFF (Normal)</th></tr>
<tr><td>Open</td><td>O</td><td></td></tr>
<tr><td>Short-circuit</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to emergency-stop the robot from the external equipment.
While this signal is open, the processing of robot emergency stop is executed.
Use this signal by connecting a safety device such as external emergency stop switch, photoelectric type sensing safety device and safety mat switch.
When the emergency stop contact is open, system output signals EMSST_A ~ EMSST_B are short-circuited.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
</table>

Cautions	1. While this signal is open, the servo power cannot be turned on in any mode. 2. EMS*A ~ EMS*B are assumed to have two (2) normal close contacts interconnected, which should be turned on and off at the same time. If there is a delay in contact operation, the non-conformity detecting function of hardware works to effect an emergency stop. It is not possible, therefore, to short-circuit the one side and use the other side as the emergency stop. When this happens, the system can be restored only by tuning off and on again the controller power. For the contact structure of the emergency stop switch, see the descriptions on the emergency stop signal line given below. 3. Unless this signal is used, short-circuit between CN6–16 and CN6–35, and between CN6–17 and CN6–36.

* Emergency stop signal line

The connection diagram of the emergency stop switch is shown below.



4.6 Jumper of Safety Measure Signal

Of the system input signal cables, the following signals are used for the safety measures.

System input signal cables	·····	CN5–16	(STOP)
		CN6–34	(SVOFF)
		CN5–17	(BREAK)
		CN6–16, 35	(EMS2B ~ EMS2C)
		CN6–17, 36	(EMS1B ~ EMS1C)

For the connectors provided with the TS2000/TS2100 robot controller, these signals are already jumpered. If these signals are used or changed, perform wiring with the jumper of connector removed. When operating the robot without using system input signals, be sure to connect the attached connectors to the CN5, CN6 connectors on the controller side.
Unless the following signals are used as the system signals, jumper the cables also.

CN5–36	(LOW_SPD)
CN5–35	(CYCLE)

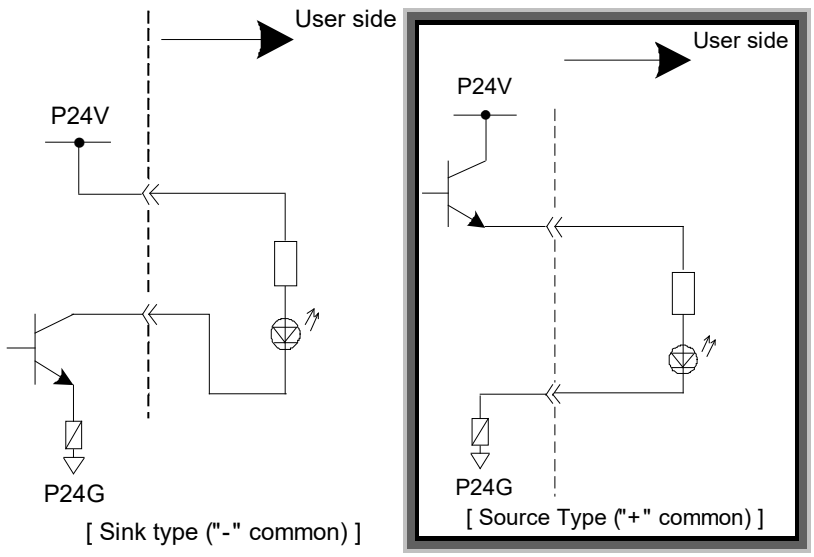
Jumper of connectors

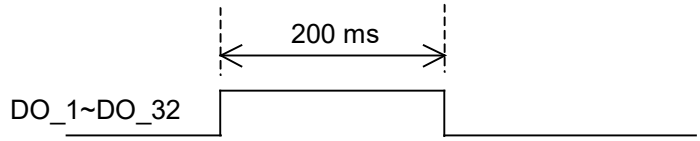
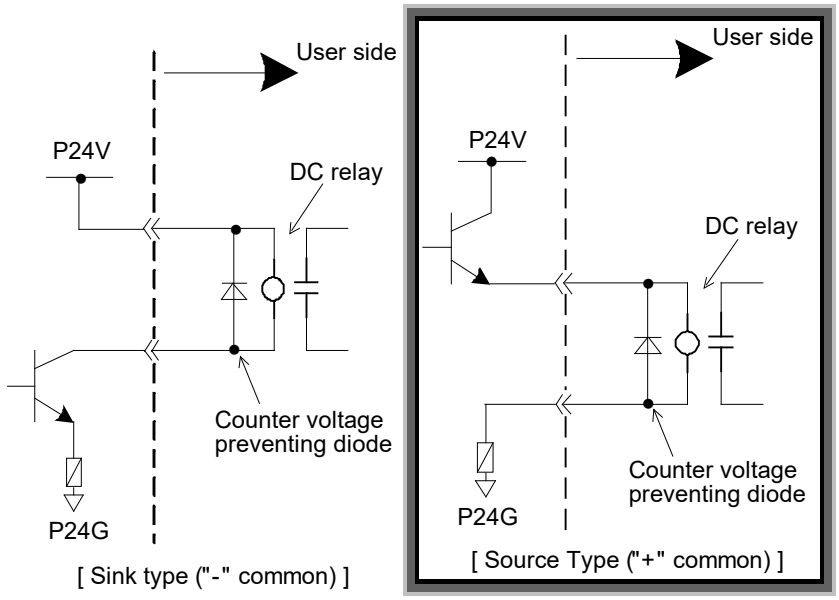
CN5		CN6	
16–18	17–18	18–34	—
(35–37)	(36–19)	16–35	17–36

! CAUTION

1. Unless the signals of SVOFF and emergency stop contacts 1 and 2 are jumpered, the controller servo power cannot be turned on.
2. Unless the CYCLE signal is jumpered, the controller enters the cycle operation mode.
3. Unless the LOW_SPD signal is jumpered, the low speed is selected for the robot speed during automatic operation.
4. Unless the signals of STOP and BREAK are jumpered, automatic operation of the robot is not possible.

4.7 Digital Output Signal

Designation	Digital output signal DO_1 ~ DO_24, DO_25 ~ DO_32 (system output signals)
Connector output terminal	Signals DO_1 ~ DO_16 are assigned to CN6–1 ~ 8 pins and 20 ~ 27 pins. (See Fig. 4.3 and 4.4.) DO–15 and 16 can be used as SV–RDY and BT_ALM by changing the user parameter. DO_17 ~ DO_24 are assigned to CN12–6 ~ 9 pins and 19 ~ 22 pins. (See Fig. 4.5 and 4.6.) The system output signals assigned to CN6–9 ~ 12 pins and 28 ~ 31 pins can be used as signals DO_25 ~ DO_32 by changing the user parameter.
Function	ON/OFF of signals DO_1 ~ DO_32 and pulse output can be performed by the robot program (BCDOUT command and PULOUT command).
Output type	Transistor output
Output circuit structure	
Electric rating	Rated voltage: DC24 V Rated current: 100 mA (max.) **Caution:** If the current which exceeds the rated output current is supplied, the output device may be damaged or the printed board may be burnt. To avoid this, be sure to use within the rated output current.

Signal timing	When performing pulse output by the PULOUT command, the output pulse width should be 200 ms or over. 
Example of circuit	

* When using the system output signals as the digital output signals:
When "Default" (standard PLC) is specified by user parameter [U11] (I/O mode), you can select whether OUTPUT13 ~ 16, 25 ~ 32 are the system output signals or digital output signals.

User parameter [U14]
[U14] Select output signal (Default I/O mode only)
{Output 13 14 15 16} (0: System 1: User)
= 1 1 1 1
{Output 25 26 27 28 29 30 31 32} (0: System 1: User)
= 0 0 0 0 0 0 0 0

Specify "1" for a desired bit underlined above, and appropriate system output signal can serve as the digital output signal. Respective bits on the upper side signify DO_13, DO_14, BT_ALM and SV_RDY when viewed from the left, and respective bits on the lower side represent ACK, TEACH, EXTSIG, SYS_RDY, ALARM, AUTORUN, CYC_END and LOW_ST when seen from the left. They correspond to digital signals DO_13 ~ DO16, DO_25 ~ DO_32, respectively.

To use only system output signal ACK as digital output signal DO_25, for instance, change the above bits as shown below.

1 0 0 0 0 0 0 0

To use all system output signals as the digital output signals, specify in the following manner.

1 1 1 1 1 1 1 1

DO NOT specify zero (0; system output) for the bits of OUTPUT13 and 14. After the above parameter has been changed, save the data, turn the power off and on again. Otherwise, the parameter will not be operative.

0: System output signal
1: Digital output signal

Output signal cable	0: System output	1: Digital output
OUTPUT13	Reserved (for extending function)	DO_13
OUTPUT14	Reserved (for extending function)	DO_14
OUTPUT15	SV_RDY	DO_15
OUTPUT16	BT_ALM	DO_16
OUTPUT25	ACK	DO_25
OUTPUT26	TEACH	DO_26
OUTPUT27	EXTSIG	DO_27
OUTPUT28	SYS_RDY	DO_28
OUTPUT29	AUTORUN	DO_29
OUTPUT30	ALARM	DO_30
OUTPUT31	CYC_END	DO_31
OUTPUT32	LOW_ST	DO_32

4.8 System Output Signal

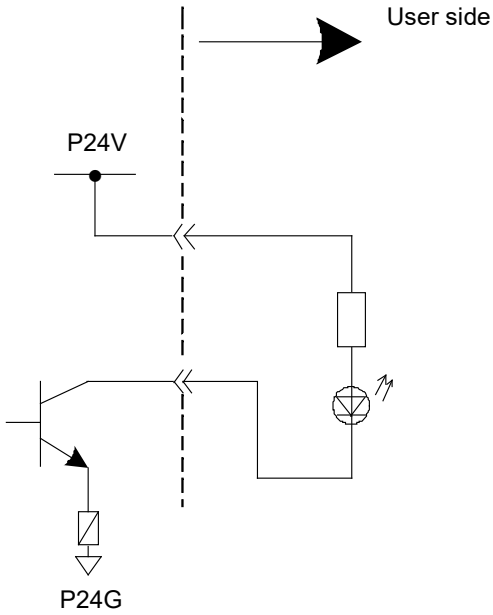
A total of twelve (12) system output signals are available. Of them, ten (10) signals are used to output the run status of the TS2000/TS2100 robot controller and the remaining two (2) signals are output via relay contact.
The system output signal is provided with an exclusive output terminal for each function and can be used as the digital output signal by changing the user parameter (initial setting).
For SV_RDY and BT_ALM, DO_15 and 16 are assigned by the initial setting of the user parameter. They can be used as SV_RDY and BT_ALM by changing the user parameter.
The two (2) relay output signals are SVST_A ~ SVST_B and EMSST_A ~ EMSST_B. The former signal (SVST_A ~ SVST_B) is of a normal open contact type and the latter signal (EMSST_A ~ EMSST_B) is of a normal close contact type. They differ in contact operation at the time of output signal ON/OFF. When using, take careful precautions not to mistake.
The specifications of the system output signals are as follows:

- Output type
  (1) Transistor output (system output signal)
  (2) Relay contact output (SVST_A ~ SVST_B, EMSST_A ~ EMSST_B)

  Note: SVST_A ~ SVST_B : Normal open contact output
  EMSST_A ~ EMSST_B : Normal close contact output

- Output circuit structure
  When Type N is selected (when X8HN printed board is used)

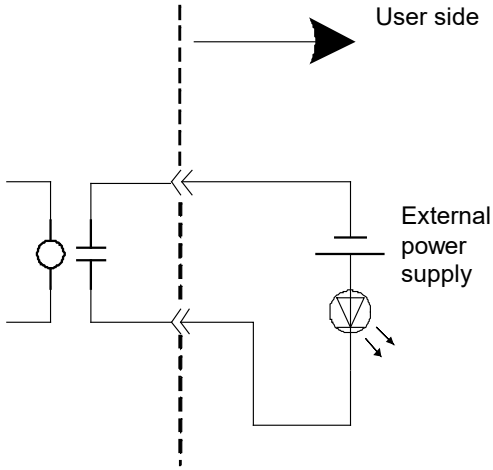
(1) Transistor output



[Sink type ("-" common)]

Type N

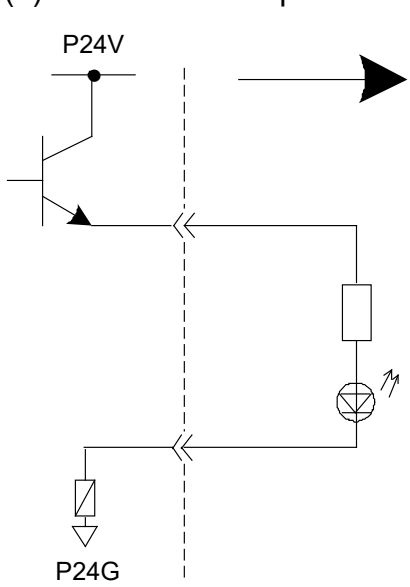
(2) Relay contact output



* The above figure exemplifies a normal open contact output structure.
The servo ON contact output is of a normal open contact type and the emergency stop contact output is of a normal close contact type. Take utmost care when using.

When Type P is selected (when X8HI printed board is used)

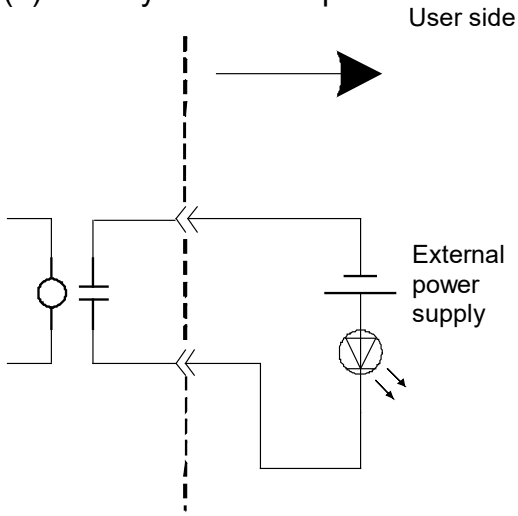
(1) Transistor output



[Source Type ("+" common)]

Type P

(2) Relay contact output

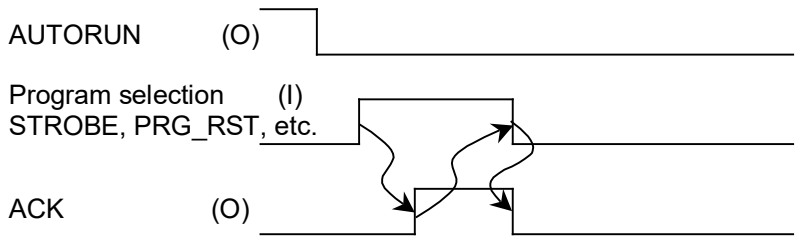


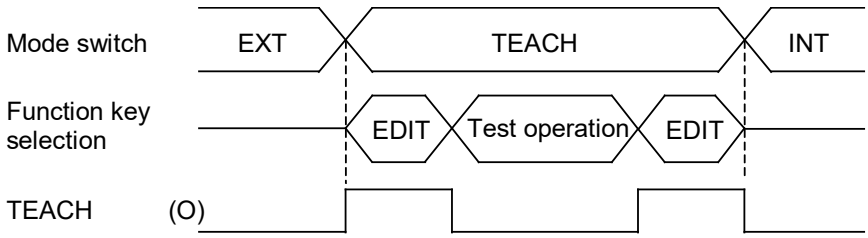
* The above figure exemplifies a normal open contact output structure.
The servo ON contact output is of a normal open contact type and the emergency stop contact output is of a normal close contact type. Take utmost care when using.

- Transistor specification
  Rated voltage: DC 24 V
  Rated current: 100 mA max.

- Relay contact specification
  Maximum rating: AC125 V, 0.5 A / DC60 V, 1.0 A

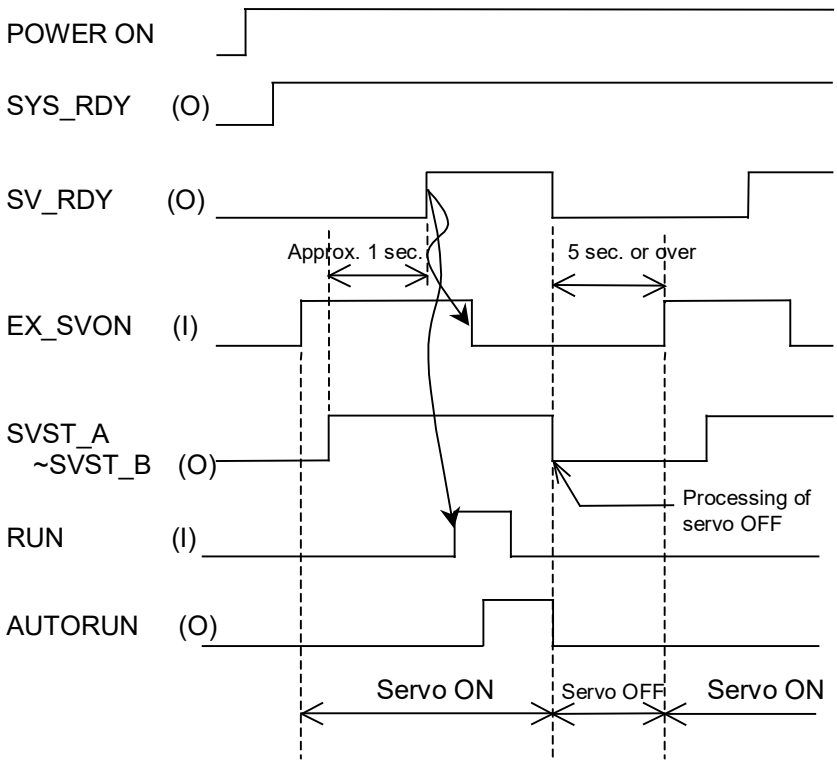
! **CAUTION**	If the current which exceeds the rated output current is supplied, the output device may be damaged or the printed board may be burnt. To avoid this, be sure to use within the rated output current.

<table>
<tr><td>Designation</td><td>ACK (Acknowledge)</td></tr>
<tr><td>Output terminal</td><td>CN6–9 pin</td></tr>
<tr><td>Signal name</td><td>251</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><th>Signal judgment / Output terminal</th><th>ON</th><th>OFF</th></tr>
<tr><td>High</td><td>O</td><td></td></tr>
<tr><td>Low</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>This is a response signal to the STROBE, PRG_RST, STEP_RST, CYC_RST and DO_RST signals. When one of these signals is input, the ACK signal is sent back to inform that appropriate processing has finished.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>If two (2) or more input signals shown above are input at the same time, only the signal which was input first is processed, then the ACK signal is output.</td></tr>
</table>

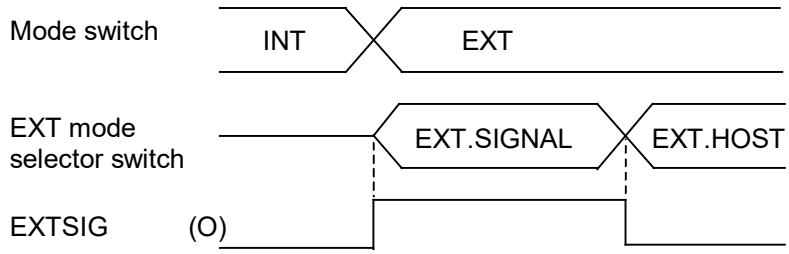
Designation	TEACH (Manual mode ON)
Output terminal	CN6–28 pin
Signal name	252
Signal logic	<table><tr><th>Signal judgment / Output terminal</th><th>ON</th><th>OFF</th></tr><tr><td>High</td><td>O</td><td></td></tr><tr><td>Low</td><td></td><td>O</td></tr></table>
Function	This signal turns on when the MODE switch of the TS2000/TS2100 robot controller is set to "TEACH" and the test operation mode is not selected. During the output of this signal, the robot arm can be guided through the teach pendant.
Signal timing	
Cautions	

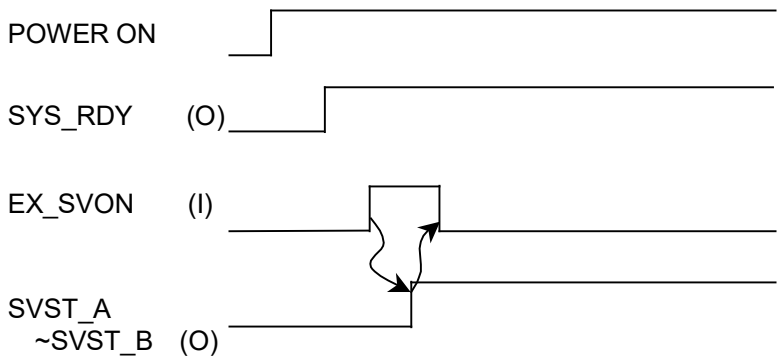
Designation	SV_RDY (Servo ready)
Output terminal	CN6–8 pin
Signal name	250
Signal logic	

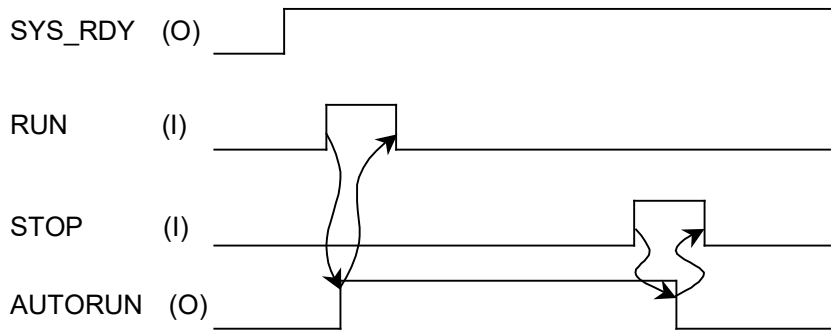
Output terminal \ Signal judgment	ON	OFF
High	O	
Low		O

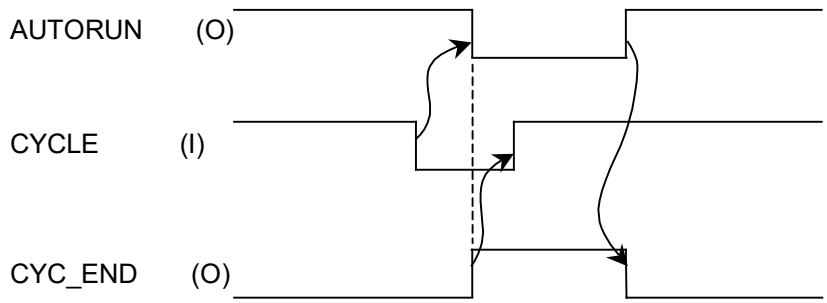
Function	This signal indicates that the servo power of the TS2000/TS2100 robot controller has turned on with the robot ready for starting an operation. This signal can always be used, irrespective of the master mode selected by means of the MODE switch.
Signal timing	

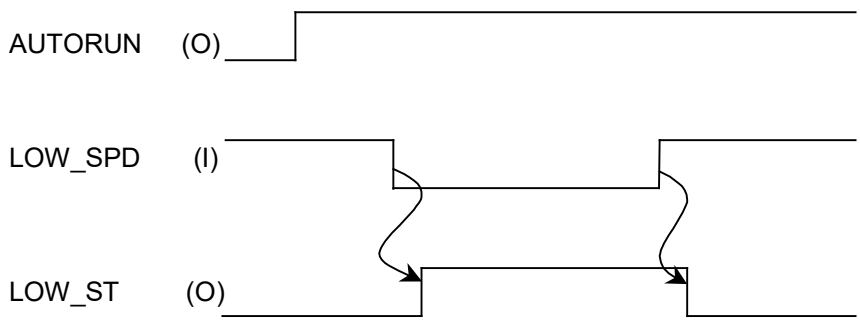
Cautions	1. Program so that the RUN signal can be turned on only after the SV_RDY signal turns on. 2. To prevent an internal damage, the servo cannot be turned on about 4.5 seconds after it is turned off. To turn the servo on again, wait at least five (5) seconds after the SVST_A ~ SVST_B signal turns off.

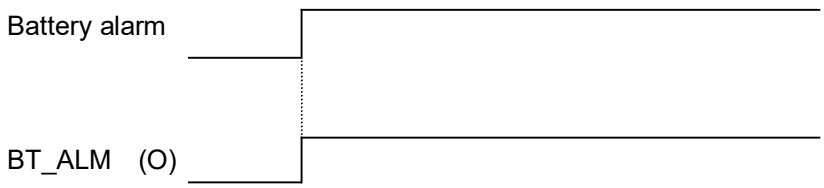
Designation	EXTSIG (External mode ON)
Output terminal	CN6–10 pin
Signal name	254
Signal logic	<table><tr><th>Signal judgment / Output terminal</th><th>ON</th><th>OFF</th></tr><tr><td>High</td><td>O</td><td></td></tr><tr><td>Low</td><td></td><td>O</td></tr></table>
Function	This signal turns on while the TS2000/TS2100 robot controller is in the EXT. SIGNAL mode.
Signal timing	
Cautions	When the EXT mode is selected by means of the MODE switch and the EXT. SIGNAL mode is selected by means of the EXT mode selector switch, all system input signals become operative.

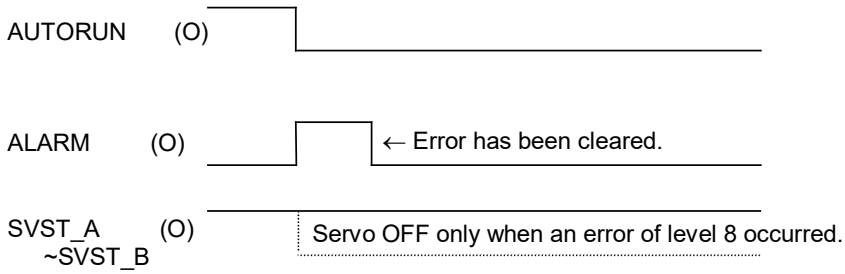
Designation	SYS_RDY (System ready)
Output terminal	CN6–29 pin
Signal name	256
Signal logic	<table><tr><th>Signal judgment / Output terminal</th><th>ON</th><th>OFF</th></tr><tr><td>High</td><td>O</td><td></td></tr><tr><td>Low</td><td></td><td>O</td></tr></table>
Function	This signal turns on when the robot can start after the controller main power is turned on. With this signal, it is possible to confirm that the robot can be operated. This signal can always be used, irrespective of the master mode selected by means of the MODE switch.
Signal timing	
Cautions	

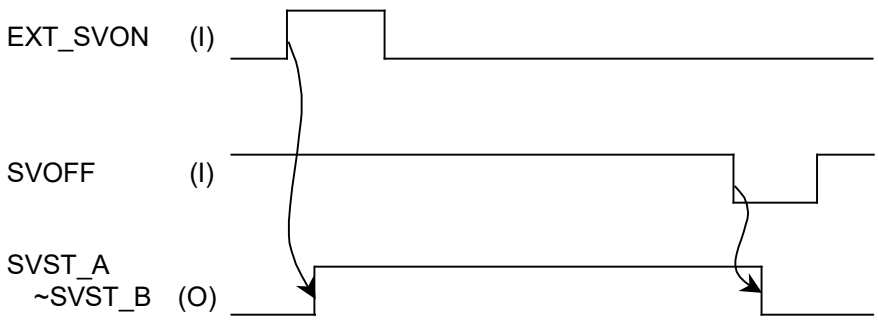
Designation	AUTORUN (Auto mode ON)
Output terminal	CN6–11 pin
Signal name	257
Signal logic	<table><tr><th>Signal judgment / Output terminal</th><th>ON</th><th>OFF</th></tr><tr><td>High</td><td>O</td><td></td></tr><tr><td>Low</td><td></td><td>O</td></tr></table>
Function	This signal turns on when the robot is operating in the automatic operation mode. When the RUN signal is made valid after the SYS_RDY signal is turned on, automatic operation starts with this signal output. This signal remains on as long as the robot is operating in the automatic operation mode. This signal turns on when the INT or EXT (EXT. SIGNAL, EXT. HOST) mode is selected by means of the MODE switch of the TS2000/TS2100 robot controller.
Signal timing	
Cautions	This signal will not turn on while the TEACH mode is selected by means of the MODE switch of the TS2000/TS2100 robot controller.

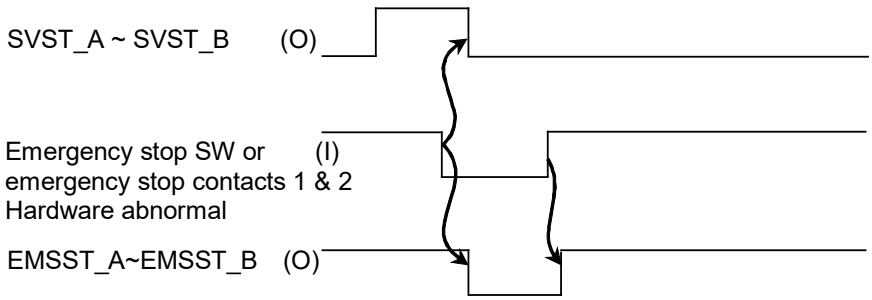
<table>
<tr><td>Designation</td><td>CYC_END (Cycle end)</td></tr>
<tr><td>Output terminal</td><td>CN6–12 pin</td></tr>
<tr><td>Signal name</td><td>258</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment
Output terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>High</td><td>O</td><td></td></tr>
<tr><td>Low</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>This signal turns on after the stop of 1-cycle automatic operation only when the CYCLE signal is made valid in the program execution (RUN) mode and an automatic operation is executed.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>1. This signal turns off when the RUN mode takes effect.
2. This signal turns on also when the cycle operation has been stopped by the STOP, BREAK or ALARM signal.</td></tr>
</table>

Designation	LOW_ST (Low speed mode ON)
Output terminal	CN6–31 pin
Signal name	259
Signal logic	<table><tr><th>Signal judgment / Output terminal</th><th>ON</th><th>OFF</th></tr><tr><td>High</td><td>O</td><td></td></tr><tr><td>Low</td><td></td><td>O</td></tr></table>
Function	This signal turns on while the robot is operating in the low speed mode by the input of system input signal LOW_SPD. This signal can always be used, irrespective of the master mode selected by means of the MODE switch.
Signal timing	
Cautions	

Designation	BT_ALM (Battery alarm)
Output terminal	CN6–27 pin
Signal name	261
Signal logic	<table><tr><th>Signal judgment / Output terminal</th><th>ON</th><th>OFF</th></tr><tr><td>High</td><td>O</td><td></td></tr><tr><td>Low</td><td></td><td>O</td></tr></table>
Function	This signal turns on if a battery alarm has occurred in the robot or robot controller. The battery alarm detects all encoders of axis 1 to axis 5 and battery level in the main control board (X8HC). This signal can always be used, irrespective of the master mode selected by means of the MODE switch.
Signal timing	
Cautions	If the battery alarm has occurred, replace the battery immediately by identifying a battery whose voltage has dropped and referring to the Maintenance Manual.

Designation	ALARM (Alarm)
Output terminal	CN6–30 pin
Signal name	262
Signal logic	<table><tr><th>Signal judgment / Output terminal</th><th>ON</th><th>OFF</th></tr><tr><td>High</td><td>O</td><td></td></tr><tr><td>Low</td><td></td><td>O</td></tr></table>
Function	This signal turns on if an error of level 2, 4 or 8 has occurred in the robot or robot controller. This signal is kept ON during error detection and turns off after the error has been cleared. For details on the errors, see the Operator's Manual. This signal can always be used, irrespective of the master mode selected by means of the MODE switch.
Signal timing	
Cautions	

<table>
<tr><td>Designation</td><td>SVST_A ~ SVST_B (Servo ON contact output)</td></tr>
<tr><td>Output terminal</td><td>Between CN6–13 pin and CN6–32 pin (contact output)</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><th>Signal judgment / Output terminal</th><th>Open</th><th>Short-circuit</th></tr>
<tr><td>OFF</td><td>O</td><td></td></tr>
<tr><td>ON</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>While the servo power is turned on, the output terminals are short-circuited.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>The signal is of a non-voltage contact output type. The logic is a normal open contact output. The contact capacity is AC 125 V (max.), 0.5 A (max.) / DC 60 V (max.), 1 A (max.).
It takes about one (1) second from the servo ON to the time when the robot is actually ready to work.</td></tr>
</table>

<table>
<tr><td>Designation</td><td>EMSST_A ~ EMSST_B (Emergency stop contact output)</td></tr>
<tr><td>Output terminal</td><td>Between CN6–14 pin and CN6–33 pin (contact output)</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><th>Signal judgment / Output terminal</th><th>Open</th><th>Short-circuit</th></tr>
<tr><td>OFF</td><td></td><td>O</td></tr>
<tr><td>ON</td><td>O</td><td></td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>While the EMERGENCY stop pushbutton switch provided on the control panel or teach pendant is pressed ON or system input signals "Emergency stop contacts 1 & 2" are ON or when the hardware abnormal(*), the output terminals of this signal are open.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>The signal is of a non-voltage contact output type. The logic is a normal close contact output. The contact capacity is AC 125 V (max.), 0.5 A (max.) / DC 60 V (max.), 1 A (max.).
(*) hardware abnormal
・Servo Amp error(X8HS,X8HW)
・IPM error
・PW Over Volt
・Servo Power Volt Low
・Volt Overheat
・DC24 Regulator error
・EI/O Fuse Broken
・Bus timeout error
・Instantaneous power failure
・AC Alarm</td></tr>
</table>

4.9 Fabricating External I/O Signal Cable

When fabricating an external I/O signal cable, observe the following matters.

a. Connectors CN5 and CN6 for the TS2000/TS2100 robot controller should be either the connectors attached to the controller or equivalent connectors.
For CN12, prepare the following connector.
The equivalent of other makers (D-SUB connectors) can also be used.

Connector name	Type of connector	Maker
CN5	XM2D–3701 (socket type connector) XM2S–3711 (connector cover)	OMRON
CN6	XM2A–3701 (pin type connector) XM2S–3711 (connector cover)	OMRON
CN12	XM2A–2501 (pin type connector) XM2S–2511 (connector cover)	OMRON

b. Select the cables which meet the following specifications.

- Core wires : 0.18 mm^2 ~ 0.32 mm^2 twisted wires
- Cable outer diameter : Max. 10.5 mm (CN5, CN6), 9 mm (CN12)
- Shield : Batch shield
- Cable length : 30 m or less

NEVER use cables which do not conform to the specifications. Otherwise, short-circuit or cable melting due to overheated cable core may be caused. Additionally, the robot may work improperly due to noise.

c. The CN5, CN6 and CN12 connectors are of a solder cup type which joints cables by soldering. For how to joint cables to connectors and how to shield cables, see Section 12.

4.10 Attaching and Detaching External I/O Signal Cable

Before attaching or detaching an external I/O signal cable to or from the TS2000/TS2100 robot controller, be sure to turn off the POWER switch equipped on the rear side of the controller, or the main power in the user's control panel.

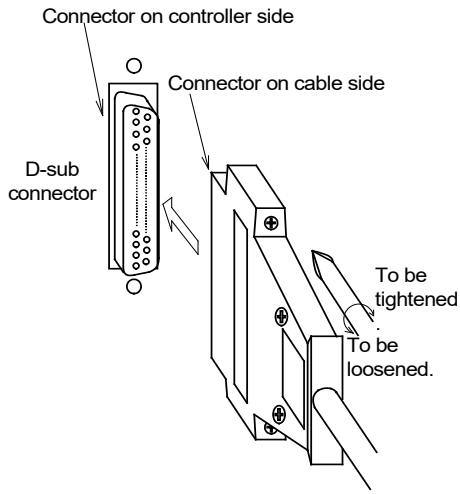


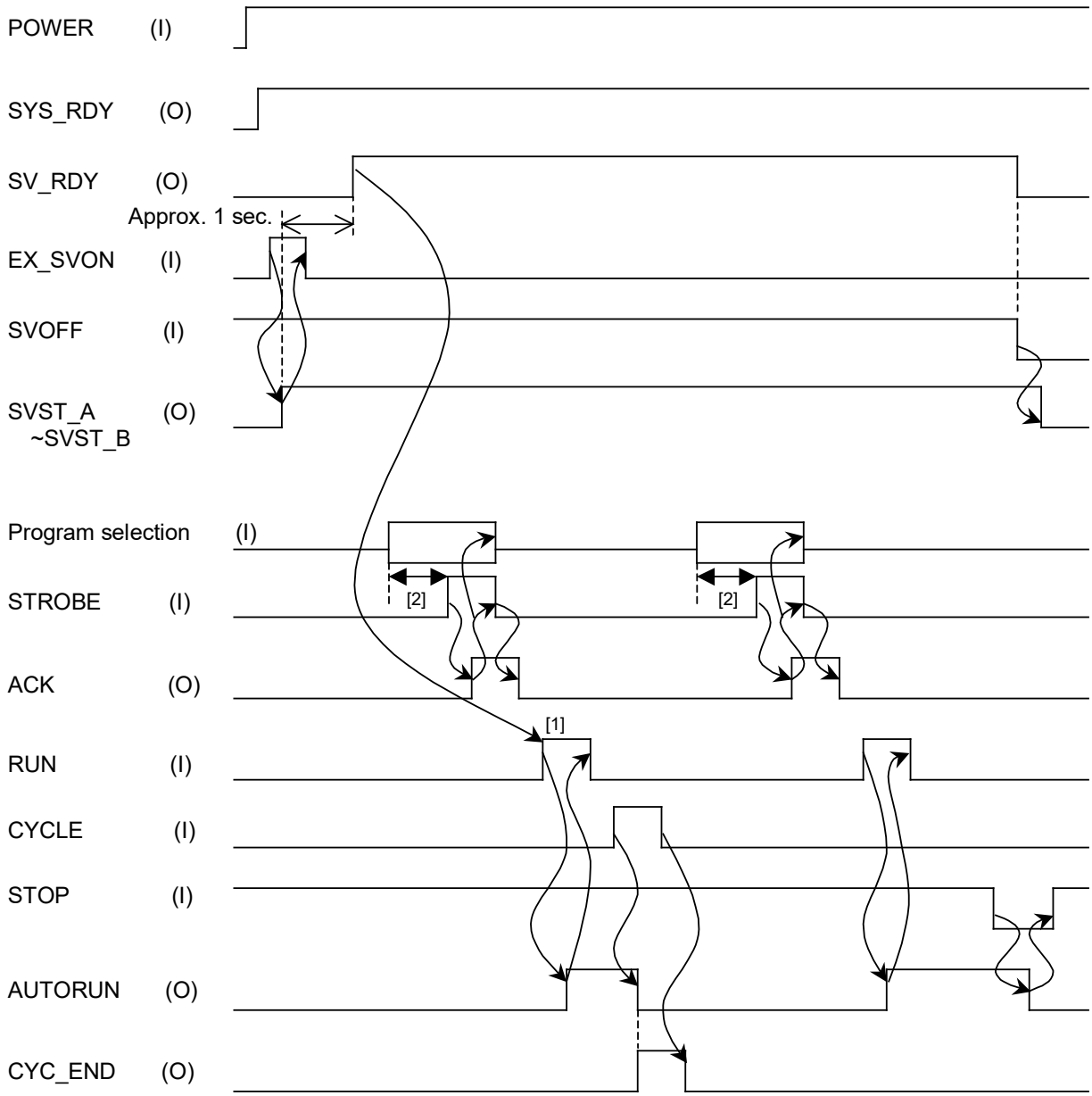
Fig. 4.7 Attaching and detaching external I/O signal cable

To connect an external I/O signal cable, completely insert the cable side connector into the controller side connector and tighten the lock screws on both sides of the connector with a screwdriver, as shown in Fig. 4.7.
To disconnect the cable, loosen the lock screws, then pull out the connector. At this time, DO NOT pull the cable instead of the connector. Otherwise, an excessive force will be exerted on the cable, resulting in damage of the cable.

! **CAUTION**	When attaching or detaching the cable, be sure to hold the connector rather than the cable.

4.11 Example of Controller Operation, Using External Signals

(1) Normal operation sequence

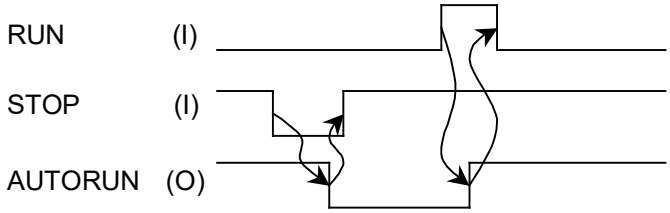


[1] Set ON the RUN signal only after the SV_RDY signal is ON.

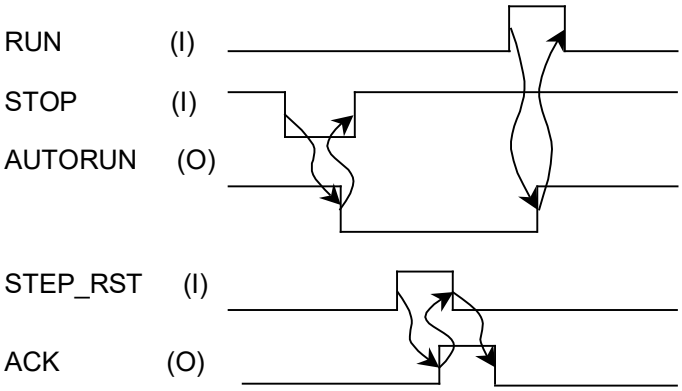
[2] Include a sufficient delay time before the program select signal starts between the program select signal input and STROBE signal input. (200 msec or so)

(2) Example of restarting the stopped robot

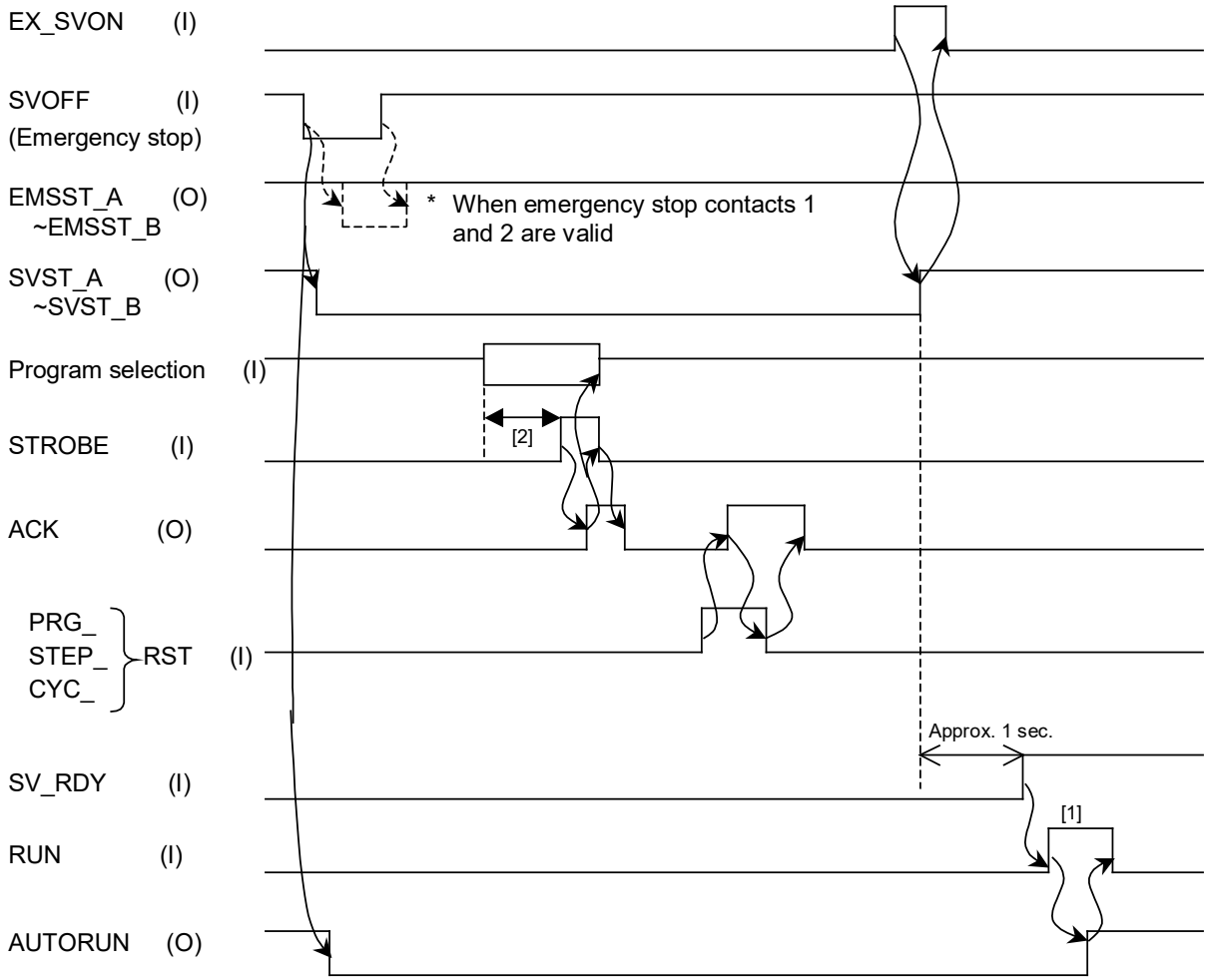
Restart after stop (to resume the operation)



Restart after stop (to start the program from the top)



(3) Example of restarting the robot after servo OFF (emergency stop)
Restart after servo OFF (emergency stop)



[1] Set ON the RUN signal only after the SV_RDY signal is ON.

[2] Include a sufficient delay time before the program select signal starts between the program select signal input and STROBE signal input. (200 msec or so)

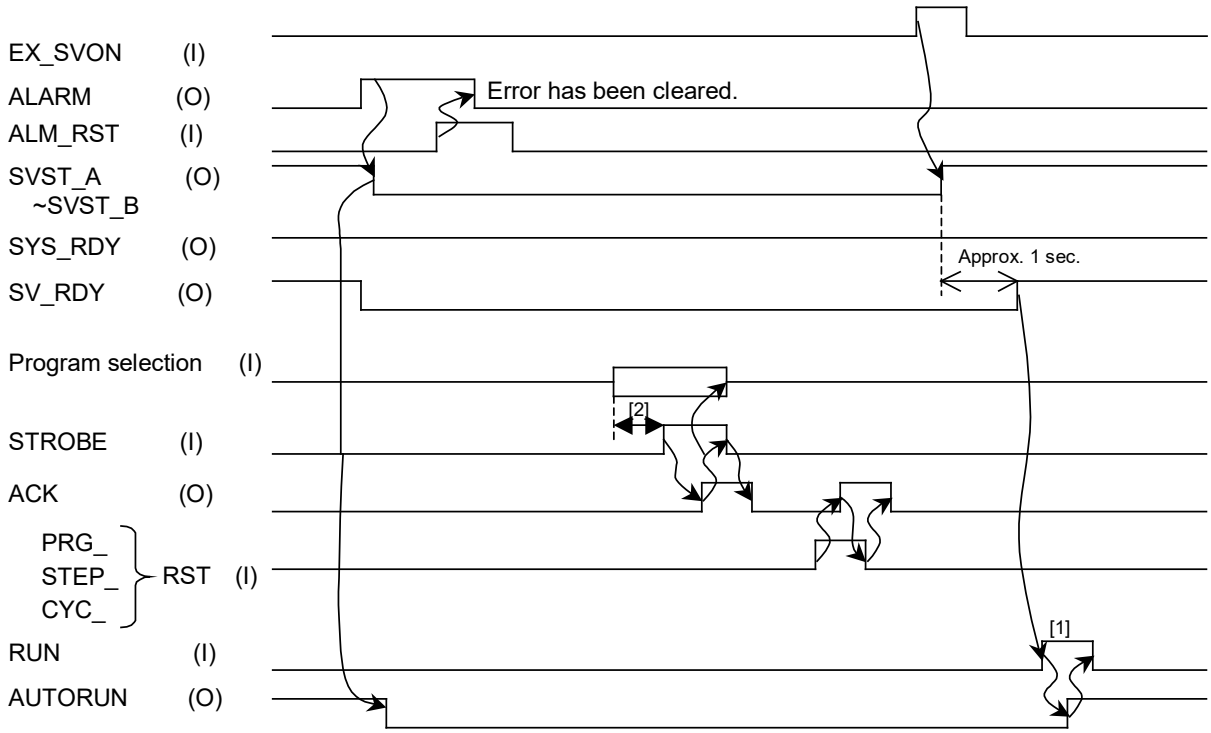
Note 1: Signals EMSST_A ~ EMSST_B (emergency stop contact output) turn on when the emergency stop contacts 1 and 2 become valid.

Note 2: Program selection, PRG_RST, STEP_RST and CYC_RST can be skipped unless necessary.

Note 3: Program selection, PRG_RST, STEP_RST and CYC_RST can be executed at any timing unless the auto operation mode is selected.

(4) Example of restarting the robot after error correction

To resume the automatic operation after an error has occurred (i.e., processing of automatic operation after error correction):



[1] Set ON the RUN signal only after the SV_RDY signal is ON.

[2] Include a sufficient delay time before the program select signal starts between the program select signal input and STROBE signal input. (200 msec or so)

Note 1: To continue the automatic operation, program selection, PRG_RST, STEP_RST and CYC_RST are necessary.

Note 2: The servo power may not be turned off with some type of error.

Note 3: Program selection, PRG_RST, STEP_RST and CYC_RST can be executed at any timing unless the auto operation mode is selected.

5. Connecting Serial Signal Cable

5.1 Connecting Serial I/O Signal Cables COM1, HOST, TCPRG, POD and COM2 (Option)

The TS2000/TS2100 robot controller is provided with four (4) channels of RS232C serial I/O signals COM1, HOST, TCPRG and POD, and with one (1) channel of optional signal COM2. For the connection, see the figure below.
It is recommended to use commercially available cables.
Connection of the RS232C may differ with the communicating device. Before use, therefore, confirm the signals of the communicating device and the connection of commercially available cables.

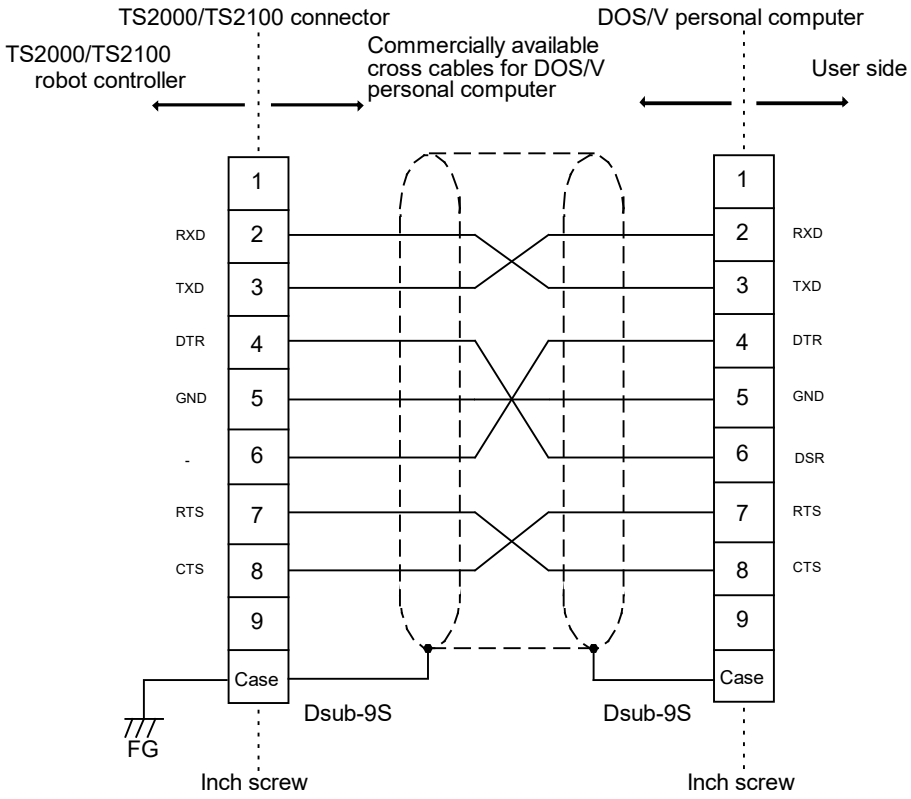


Fig. 5.1 Connection of serial I/O signals COM1, HOST, TCPRG, POD, COM2 (option)

5.2 Attaching and Detaching Serial I/O Signal Cables COM1, HOST, TCPRG, POD and COM2 (Option)

The procedures for attaching and detaching serial I/O signal cables COM1, HOST, TCPRG, POD and COM2 (option) are the same as in Para. 4.10. For details, see Para. 4.10.

! **CAUTION**	The COM1, HOST, TCPRG, POD and CN12 connectors of the TS2000/TS2100 robot controller are attached with a connector cap, respectively. Unless these connectors are used, be sure to attach the connector caps to prevent static electricity and damage.

6. Connecting TP (Teach Pendant) Cable

Connecting TP cable

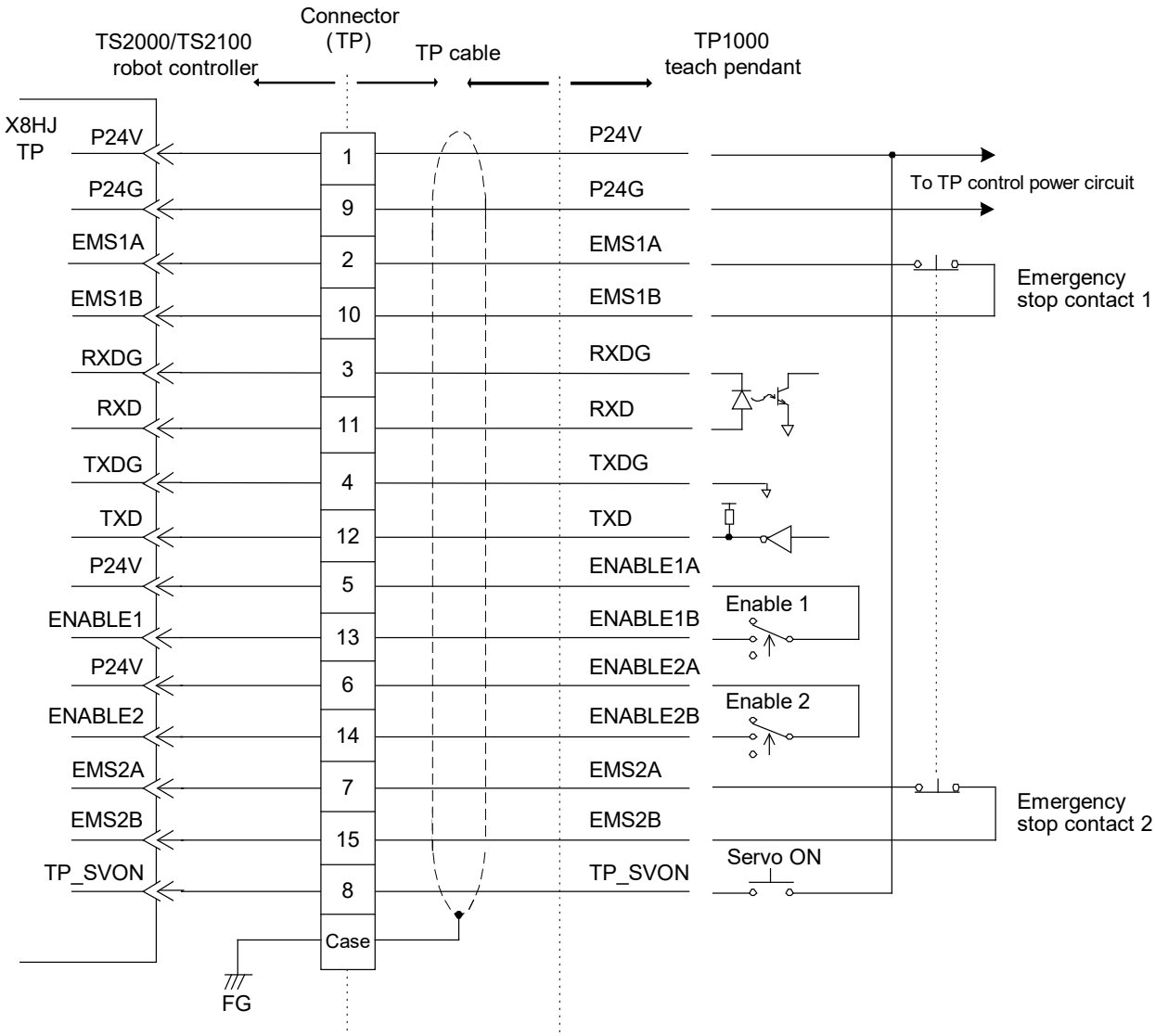


Fig. 6.1 Connection of TP1000 teach pendant I/O signals

The TP cable is a serial I/O signal cable used only for the TP1000 teach pendant (option). One side of the TP cable is secured to the teach pendant and the other side has a D-SUB 15-pin connector.

Unless the teach pendant is connected, insert the attached dummy connector to connector TP on the front panel of the controller.

When connecting the teach pendant, remove this dummy connector first. (The attaching and detaching procedures are the same as in Para. 4.10. For details, see Para. 4.10.)

The dummy connector can be removed while the POWER switch is set ON or OFF. If the dummy connector is removed while the servo power is turned on, however, the servo power is turned off automatically. (While the dummy connector or teach pendant is not connected, safety measures are effected to prevent the servo power from actuating.)
The TP disconnect switch is provided on the lower left side of the TP connector (when seen from the controller front side). When the dummy connector or teach pendant connector is disconnected while pressing this switch, the controller servo power remains on. Change the connector while pressing the TP disconnect switch continuously.

If the TP disconnect switch is kept pressed for long hours, however, the safety device works to turn off the servo power (i.e., emergency stop contact output is ON). The time for attaching or detaching the connector should be ninety (90) seconds or less.

! **CAUTION**	Unless the TP cable or dummy connector is connected with the TP connector of the TS2000/TS2100 robot controller, the servo power cannot be turned on.

! **CAUTION**	NEVER keep pressing the TP disconnect switch intentionally.

As shown in Fig. 6.1, the teach pendant I/O signals include the I/O signals for data transmission between the teach pendant and controller, signals for identifying servo ON, emergency stop and enable input from the teach pedant, and P24V and P24G signals for supplying the power to the teach pendant.

Data transmission between the teach pendant and TS2000/TS2100 robot controller is performed in the current loop system.
These I/O signals are used exclusively for the TP1000 teach pendant, and cannot be used for any other equipment.
While the teach pendant is disconnected, processing of emergency stop is effective and the servo power cannot be turned on. Unless the teach pendant is used, use the attached jumper connector (D-SUB 15 pin).

7. Connecting EXT–I/O Cable

7.1 Connecting EXT–I/O Cable

The EXT–I/O cable is used to connect the TR48DIOCN/TR48DIOC module (option) or SHIBAURA MACHINE's designated remote I/O module (option) serving as the slave station.
To supply the controller internal power to the TR48DIOCN/TR48DIOC module, the EXT–I/O cable is connected with P24V (24 V) and P24G (0 V). (The maximum allowable current capacity of the remote I/O module is 1.2 A in total.)
If the current exceeds the maximum current capacity of 3 A (including the external I/O) when the internal power supply is used, use an external power supply that can supply both 24 V and 0 V.
NEVER use the internal power supply together with the external power supply. Otherwise, the power supply may be damaged.

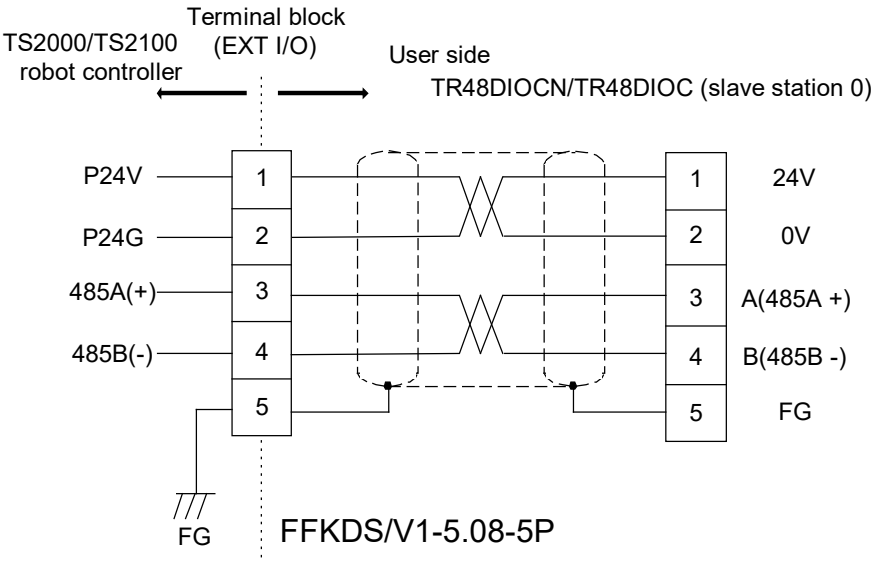


Fig. 7.1 Connection of RS485 cable

a. Use the cable attached to the TR48DIOCN/TR48DIOC module, or the cable which satisfies the following specifications.

- Core wires : 0.3 mm^2 ~ 0.75 mm^2 twisted wires
- Twisted pair : Required.
- Cable outer diameter : Max. 7 mm
- Shield : Batch braided shield
- Cable length : 10 m or less in total

(When the external power is supplied to the remote I/O module and only the RS485 communication cable is extended, the total cable length can be 400 m or less.)

- Characteristic impedance : 100 Ω
- Others : For RS485 250 kbps data communication.

Recommended twisted pair cable with shield
TKVVBS 0.3 sq. × 2 P (made by Tachii Densen)

NEVER use a cable which does not conform to the specifications. Otherwise, short-circuit or cable melting due to overheated cable core may be caused. Additionally, the robot may work improperly due to noise.

b. When connecting two (2) TR48DIOCN/TR48DIOC modules for I/O extension, be sure to branch from the terminal block of slave station 0, as shown in Fig. 7.2. (For the I/O extension, up to two (2) TR48DIOCN/TR48DIOC modules can be connected.)

For details on the TR48DIOCN/TR48DIOC module, see the Simple PLC Function Manual.

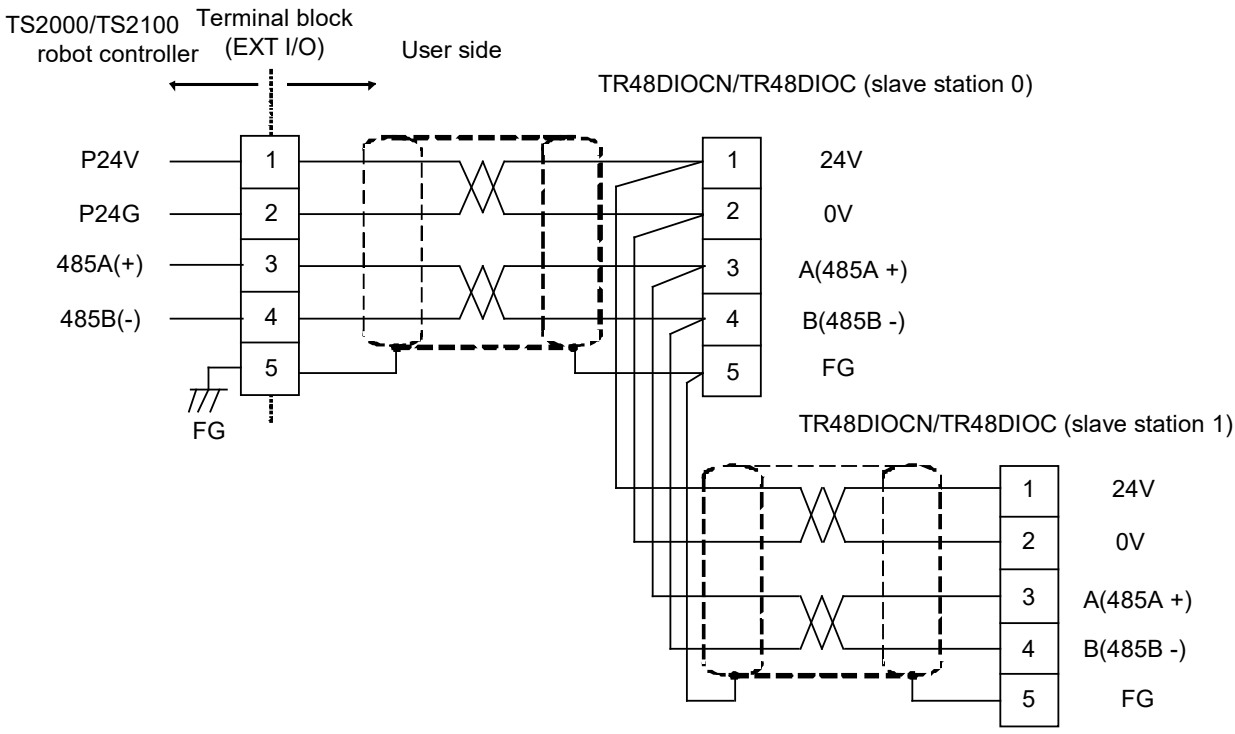


Fig. 7.2 Connection when relaying EXT–I/O cable

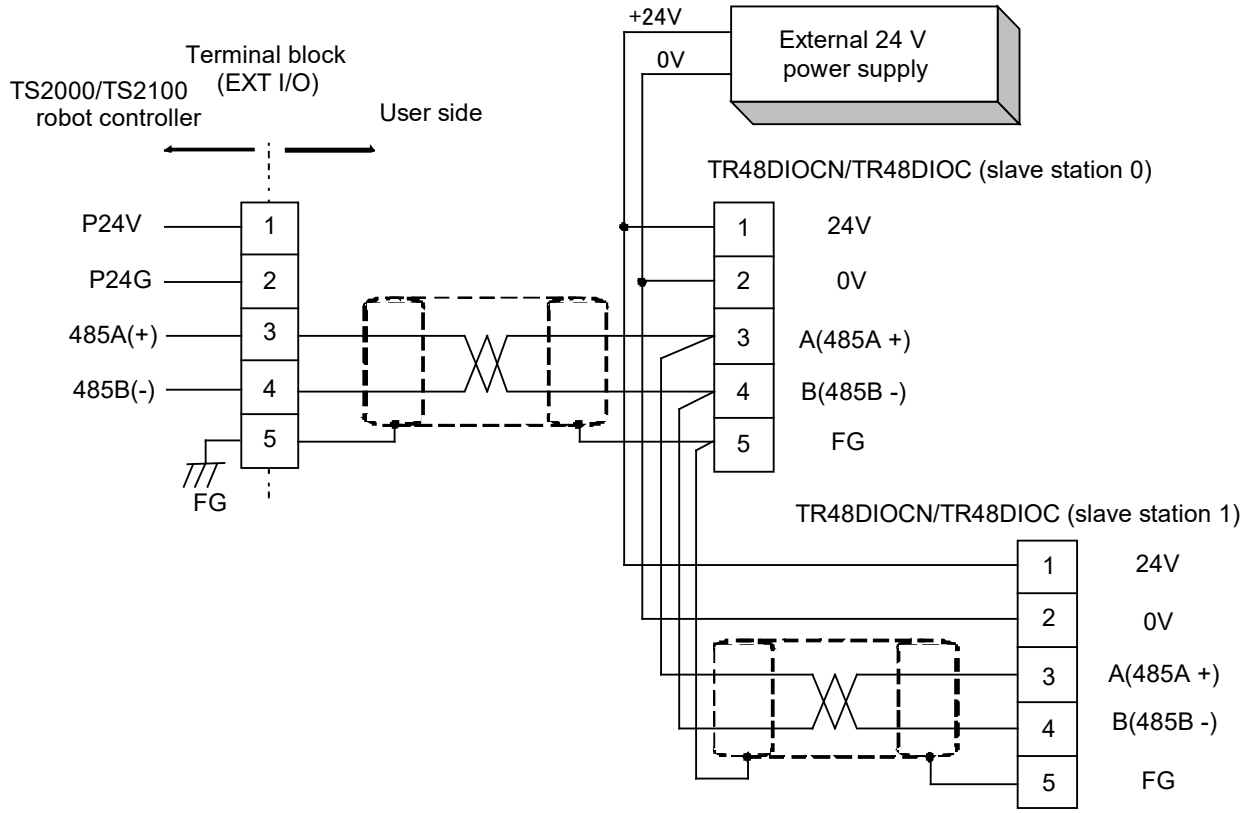


Fig. 7.3 Connection of EXT–I/O cable when external power supply is used

! **CAUTION**	The maximum current capacity of the internal 24 V power supply of the TS2000/TS2100 robot controller, which can be fed to the external equipment, is 3 A. Therefore, the total current used for the external I/O, hand I/O and remote I/O modules should be 3 A or less. If the current used exceeds the rated value, the internal 24 V power supply will be damaged or the fuse will be blown out. To avoid this, be sure to use within the rated capacity.

7.2 EXT–I/O Communication

If the number of digital I/Os is insufficient for the system, up to two (2) TR48DIOCN/TR48DIOC modules [option; twenty-eight (28) inputs and twenty (20) outputs] can be connected as the I/O function extension unit. When the TS2000/TS2100 robot controller is assigned to the master station and the TR48DIOCN/TR48DIOC module to the slave station, inputs and outputs can be controlled via the RS485 serial communication.
In addition to the TR48DIOCN/TR48DIOC module, a remote I/O module designated by SHIBAURA MACHINE, which can serve as the slave station, can be used also.

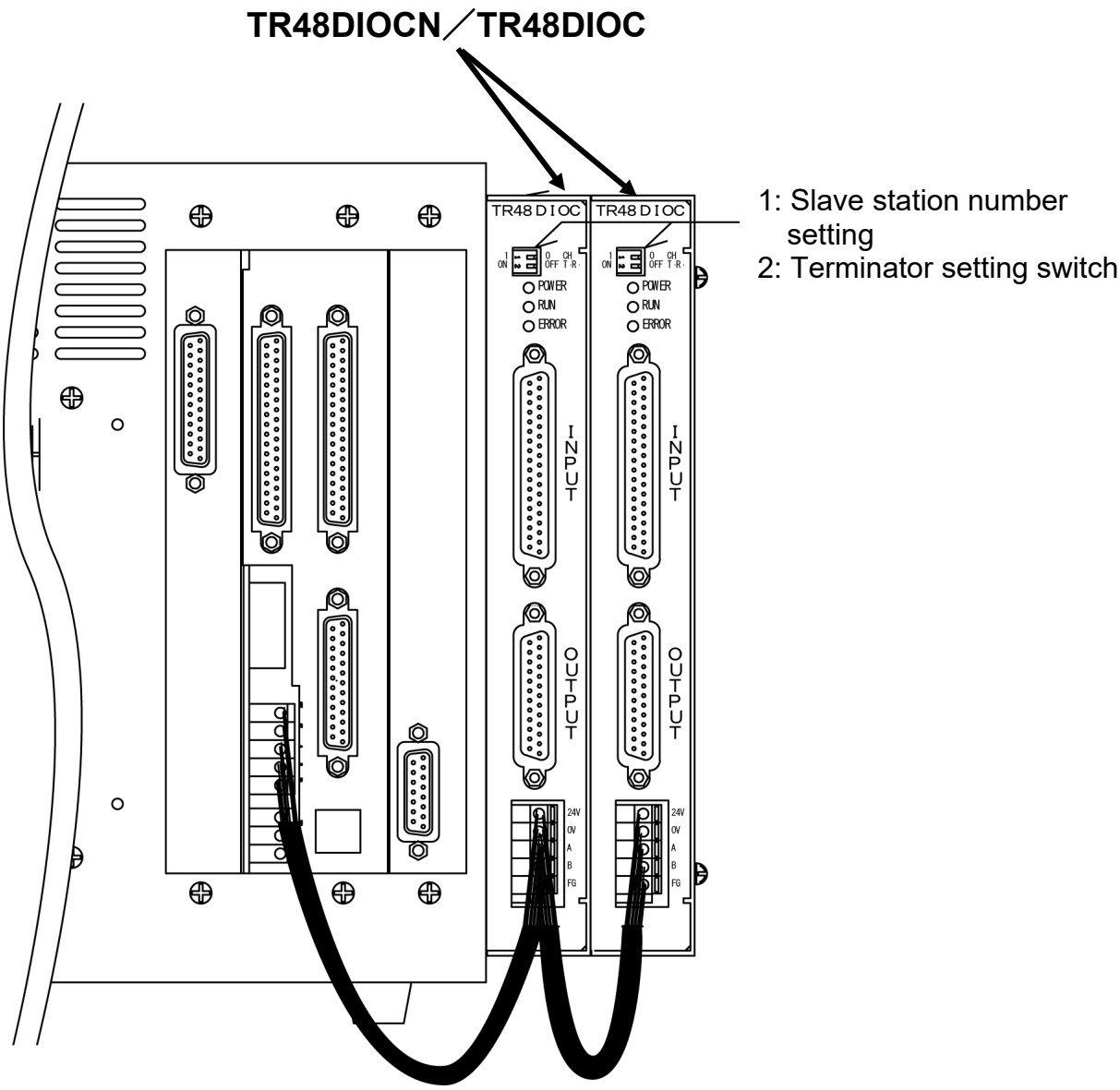


Fig. 7.4 Connection of TR48DIOCN/TR48DIOC module

- User parameter setting procedures

  * To designate the slave station number of the TS2000/TS2100, observe the following steps. Under [U12] of the USER.PAR (user parameter) file, you can find the following parameter setting related to the extension input and output.

    [U12] Extend I/O setting
    {Use/Not Use} (0: Not Use, 1: Use)
    {Not Use}
    {Not Use}
    = 0 28 20 → Corresponds to station 0.
    = 0 28 20 → Corresponds to station 1.

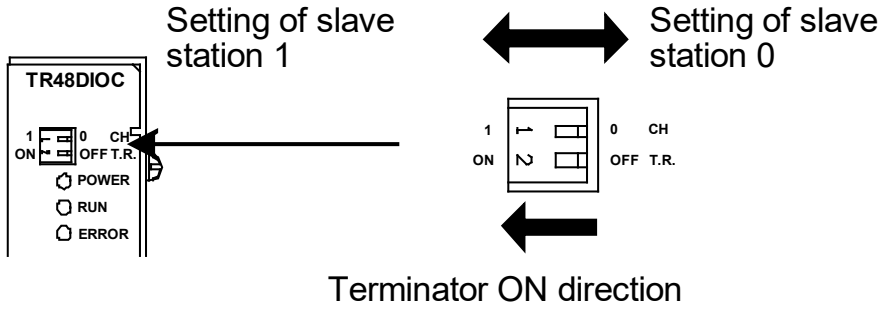
    If you wish to use only station 0 (i.e., one (1) TR48DIOCN/TR48DIOC module), for instance, change the underlined bit corresponding to station 0 to "1".
    = 1 28 20
    = 0 28 20

    After saving the parameter file, turn off the controller power and on again. Then the above parameter becomes effective.

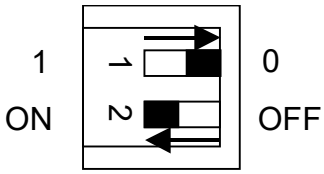
    If you wish to use both stations 0 and 1 (i.e., two (2) TR48DIOCN/TR48DIOC modules), specify as follows and perform the same operation as above to make the parameter valid.
    = 1 28 20
    = 1 28 20

- Station number and terminator setting

  The switches (2 pins) provided on the front upper side of the TR48DIOCN/TR48DIOC module represent pin 1 (upper) for slave station number setting and pin 2 (lower) for terminator setting.



Ex.1 When adding one (1) TR48DIOCN/TR48DIOC



Ex.2 When adding two (2) TR48DIOCN/TR48DIOC



The slave station setting is already described above. Specify the slave station of TR48DIOCN/TR48DIOC to be used, according to the station number set in USER. PAR.

For the terminator, when only one (1) TR48DIOCN/TR48DIOC module is used, set ON the terminator setting switch equipped on the module.
When two (2) TR48DIOCN/TR48DIOC modules are used, see Fig. 7.4 for example. As the TR48DIOCN/TR48DIOC module on the extreme right side as viewed from the controller rear side is the terminal station in terms of cable wiring, set ON the terminator setting switch equipped on this module alone. The terminator of the left TR48DIOCN/TR48DIOC module should remain OFF.

When using the TR48DIOCN/TR48DIOC module, careful precautions should be taken on the following matters.

[1] Make sure that the slave station number set in "USER.PAR" coincides with the station number setting of the TR48DIOCN/TR48DIOC module.

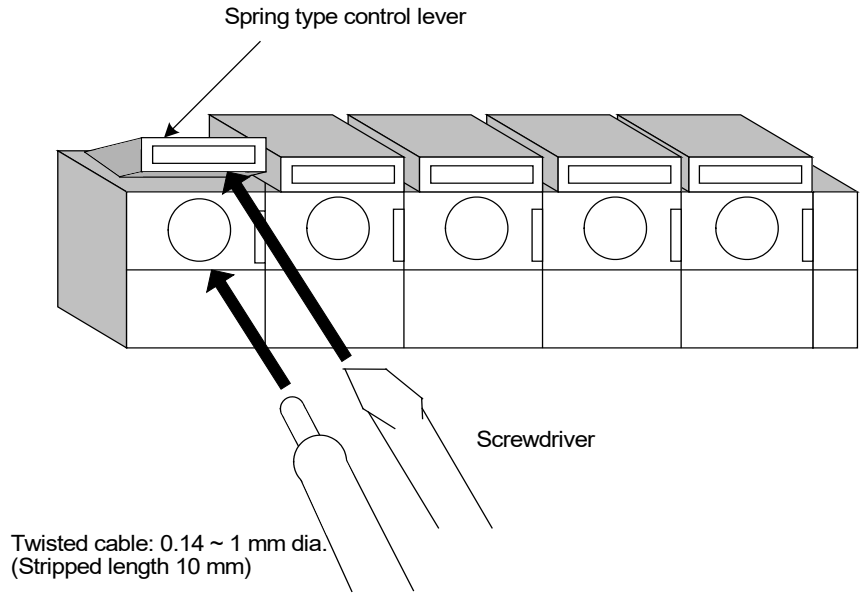
[2] When the TR48DIOCN/TR48DIOC module or modules are connected, set ON the terminator setting switch equipped on the TR48DIOCN/TR48DIOC module which is located on the extreme right side in terms of cable wiring when seen from the controller.

Be sure to execute Items [1] and [2] above, irrespective of the presence or absence of external power supply. Otherwise, the system may not function normally or go wrong.

With the above setting, make sure that both the POWER and RUN LEDs on each TR48DIOCN/TR48DIOC module are illuminated.

7.3 Attaching and Detaching EXT–I/O Cable

For the EXT–I/O cable, use the cable attached to the TR48DIOCN/TR48DIOC module, or a cable which meets the specifications.
Before attaching or detaching the EXT–I/O cable to or from the TS2000/TS2100 robot controller, be sure to turn off the POWER switch equipped on the rear side of the controller, or the main power in the user's control panel.



While holding the control lever with a screwdriver, push the twisted cable, then remove the lever from the control lever.

Fig. 7.5 Attaching and detaching EXT–I/O cable

As shown in Fig. 7.5 above, when connecting the EXT–I/O cable, keep holding the spring type control lever above the connector with a screwdriver and completely push the attached cable. Remove the screwdriver from the control lever then. (Lightly pull the cable and make sure that the cable is connected completely and will not come off. DO NOT pull the cable by force, however. If an unusually large force is exerted on the cable, the cable may be damaged.)
To detach the cable, hold the control lever with the screwdriver, draw out the cable and remove the screwdriver.

8. Connecting Extension I/O Signal Cable (Option)

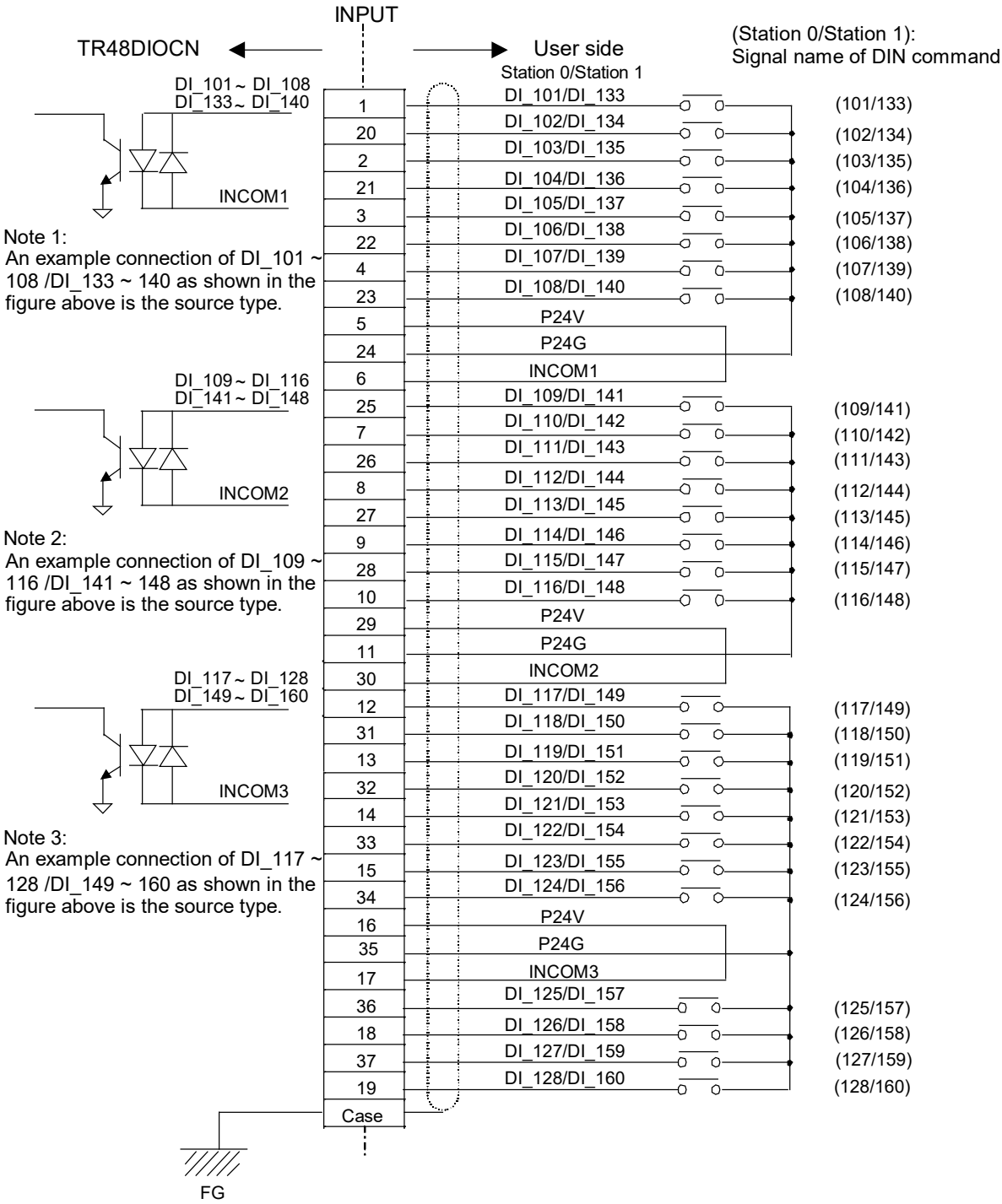
The TR48DIOCN and TR48DIOC modules are available to serve as the extension I/O modules exclusive to the TS2000/TS2100 robot controller.
They are provided with twenty-eight (28) inputs and twenty (20) outputs, respectively.
For the TS2000/TS2100 robot controller, up to two (2) stations can be extended.
The output specifications (source type or sink type) differ between the TR48DIOCN and TR48DIOC modules.
After confirming the type of your extension I/O module, connect the extension I/O signal cables.

8.1 TR48DIOCN

The output specifications of the TR48DIOCN are the same as those of the SR7000 (i.e., sink type).
For the input specifications, a bidirectional photocoupler is used in the input circuit, and it is possible to change over the source type or sink type by the selection of INCOM*.

8.1.1 Connecting Extension Input Signal Cable

To connect the extension input signal cable, use the connector attached to the TR48DIOCN module [XM2A–3701 (plug type connector), XM2S–3711 (connector cover)]. Connect the inputs of the TR48DIOCN (DI_101 ~ 128 (station 0) and DI_133 ~ 160 (station 1) to connector INPUT provided on the module surface.

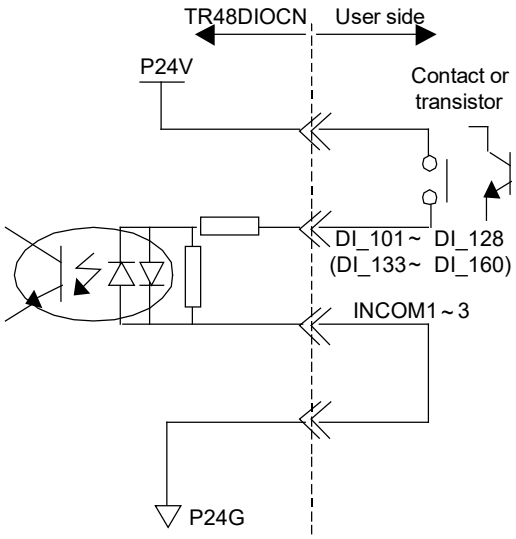

INPUT
TR48DIOCN
User side
Station 0/Station 1
(Station 0/Station 1):
Signal name of DIN command
DI_101 ~ DI_108
DI_133 ~ DI_140
INCOM1
Note 1:
An example connection of DI_101 ~ 108 /DI_133 ~ 140 as shown in the figure above is the source type.
DI_109 ~ DI_116
DI_141 ~ DI_148
INCOM2
Note 2:
An example connection of DI_109 ~ 116 /DI_141 ~ 148 as shown in the figure above is the source type.
DI_117 ~ DI_128
DI_149 ~ DI_160
INCOM3
Note 3:
An example connection of DI_117 ~ 128 /DI_149 ~ 160 as shown in the figure above is the source type.
1 DI_101/DI_133 (101/133)
20 DI_102/DI_134 (102/134)
2 DI_103/DI_135 (103/135)
21 DI_104/DI_136 (104/136)
3 DI_105/DI_137 (105/137)
22 DI_106/DI_138 (106/138)
4 DI_107/DI_139 (107/139)
23 DI_108/DI_140 (108/140)
5 P24V
24 P24G
6 INCOM1
25 DI_109/DI_141 (109/141)
7 DI_110/DI_142 (110/142)
26 DI_111/DI_143 (111/143)
8 DI_112/DI_144 (112/144)
27 DI_113/DI_145 (113/145)
9 DI_114/DI_146 (114/146)
28 DI_115/DI_147 (115/147)
10 DI_116/DI_148 (116/148)
29 P24V
11 P24G
30 INCOM2
12 DI_117/DI_149 (117/149)
31 DI_118/DI_150 (118/150)
13 DI_119/DI_151 (119/151)
32 DI_120/DI_152 (120/152)
14 DI_121/DI_153 (121/153)
33 DI_122/DI_154 (122/154)
15 DI_123/DI_155 (123/155)
34 DI_124/DI_156 (124/156)
16 P24V
35 P24G
17 INCOM3
36 DI_125/DI_157 (125/157)
18 DI_126/DI_158 (126/158)
37 DI_127/DI_159 (127/159)
19 DI_128/DI_160 (128/160)
Case
FG

The specifications of the extension input signal are as follows:

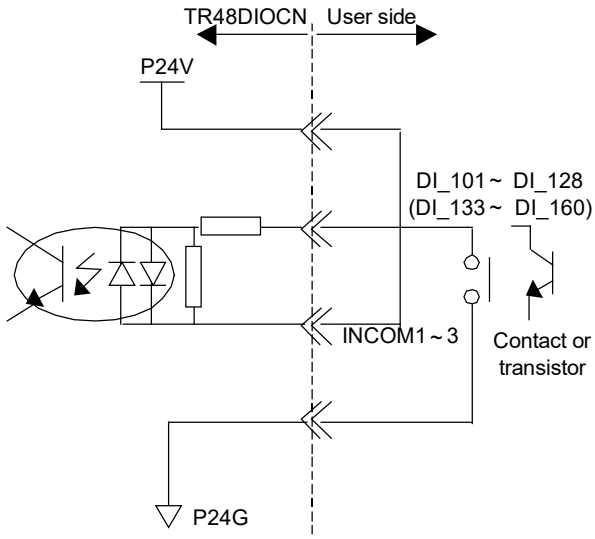
- Type of input:
  Non-voltage contact input or transistor open collector input

- Example of application circuit and structure of input circuit

Sink type ("–" common)



Source type ("+" common)

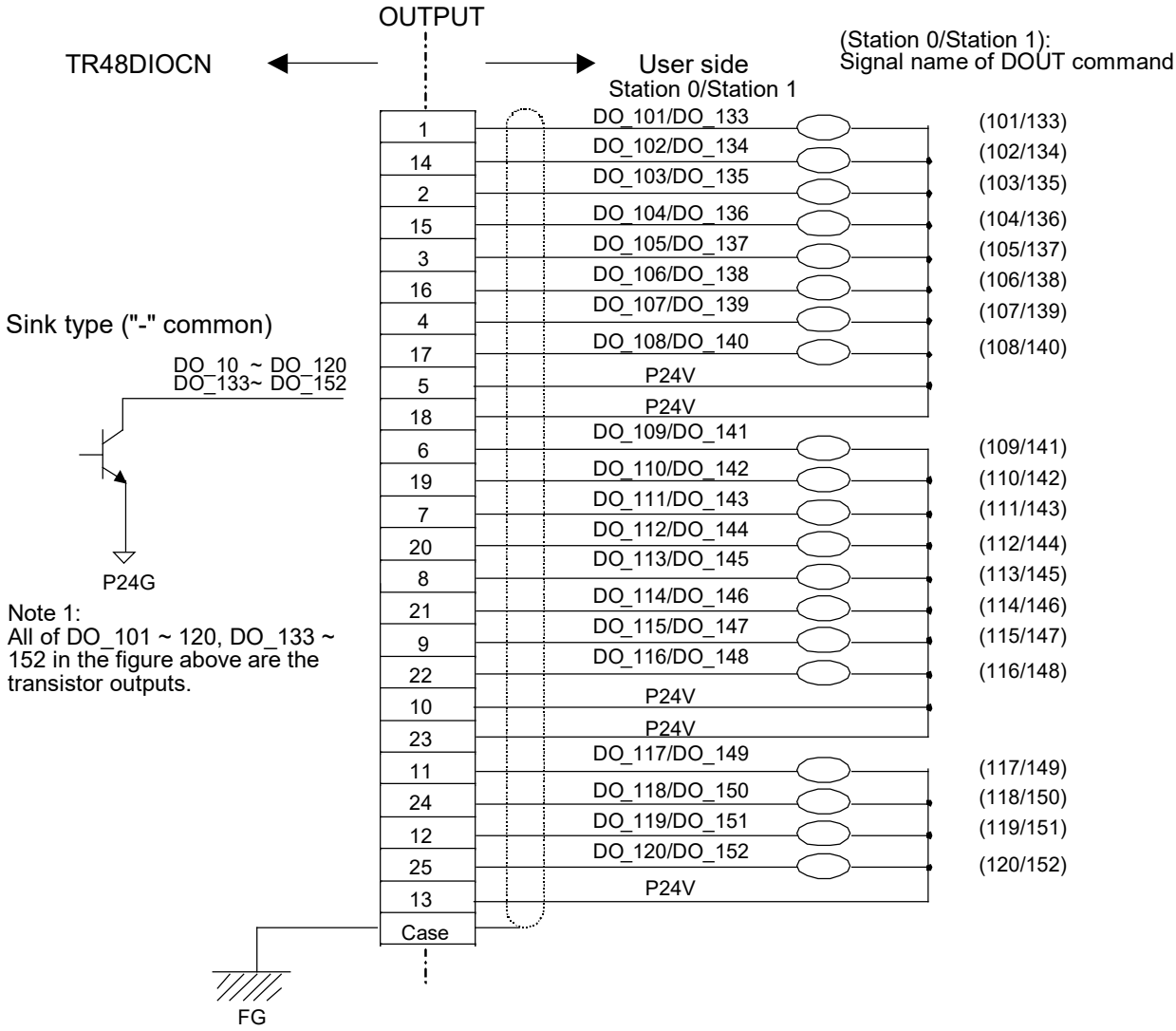


- Non-voltage contact/transistor specifications

Non-voltage contact specifications		Transistor specifications
Contact rating	DC24 V, 10 mA or over	Voltage between collector and emitter 30 V or over
Circuit current	Approx. 7 mA	Current between collector and emitter 10 mA
Minimum current	DC24 V, 1 mA	Circuit current Approx. 7 mA
Connected impedance	100 Ω or less	Leak current between collector and emitter 100 μA

8.1.2 Connecting Extension Output Signal Cable

To connect the extension output signal cable, use the connector attached to the TR48DIOCN module [XM2A–2501 (plug type connector), XM2S–2511 (connector cover)]. Connect the outputs of the TR48DIOCN (DO_101 ~ 120 (station 0) and DO_133 ~ 152 (station 1) to connector OUTPUT provided on the module surface.



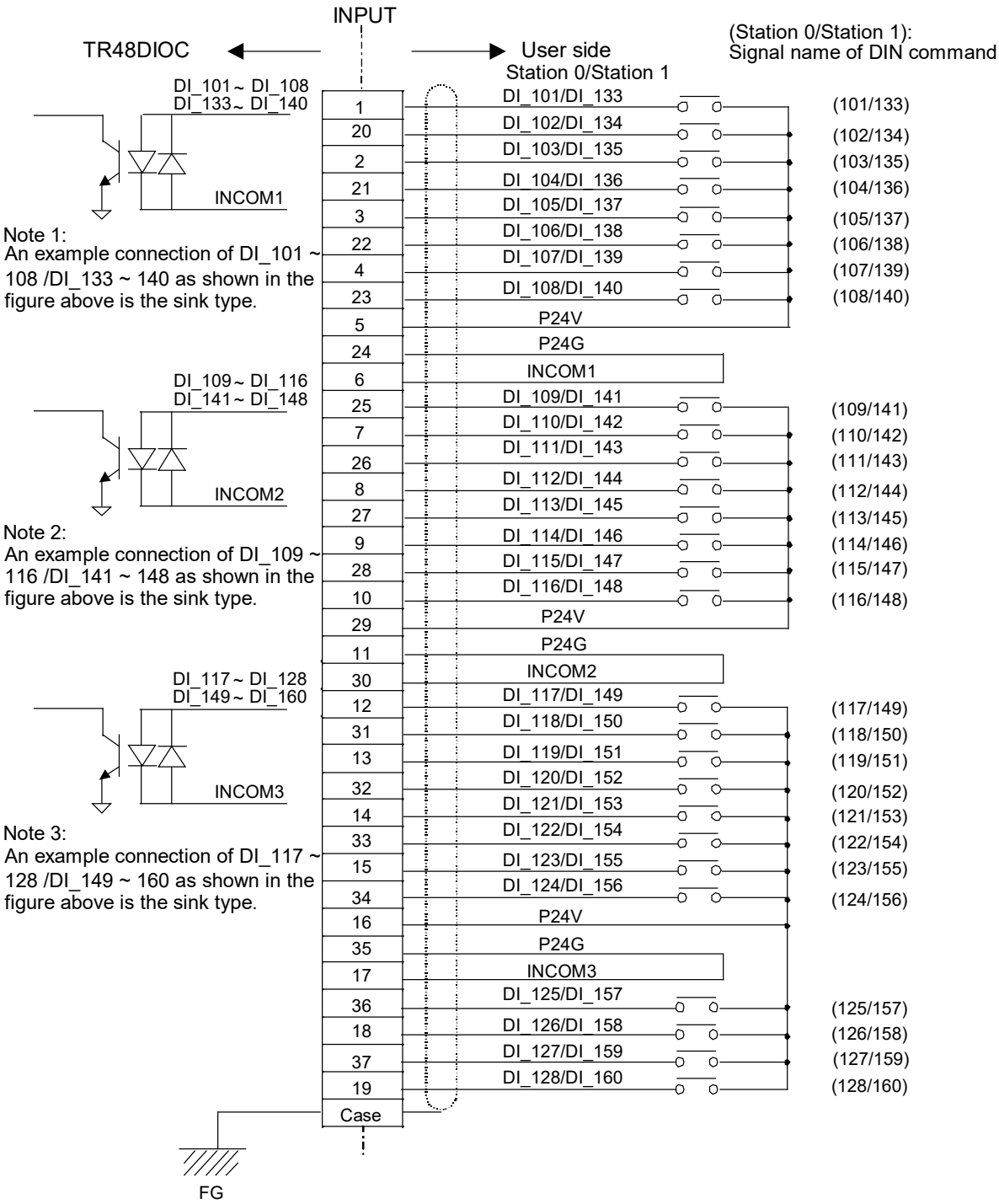
The specifications of the extension output signal are the same as those of the digital output signal (i.e., sink type) which are described in Para. 4.7.

8.2 TR48DIOC

The output specifications of the TR48DIOC differ from those of the SR7000 (i.e., source type). The polarity is the same as the output polarity of I/O printed board X8HI of this controller.
For the input specifications, a bidirectional photocoupler is used in the input circuit, and it is possible to change over the source type or sink type by the selection of INCOM*.

8.2.1 Connecting Extension Input Signal Cable

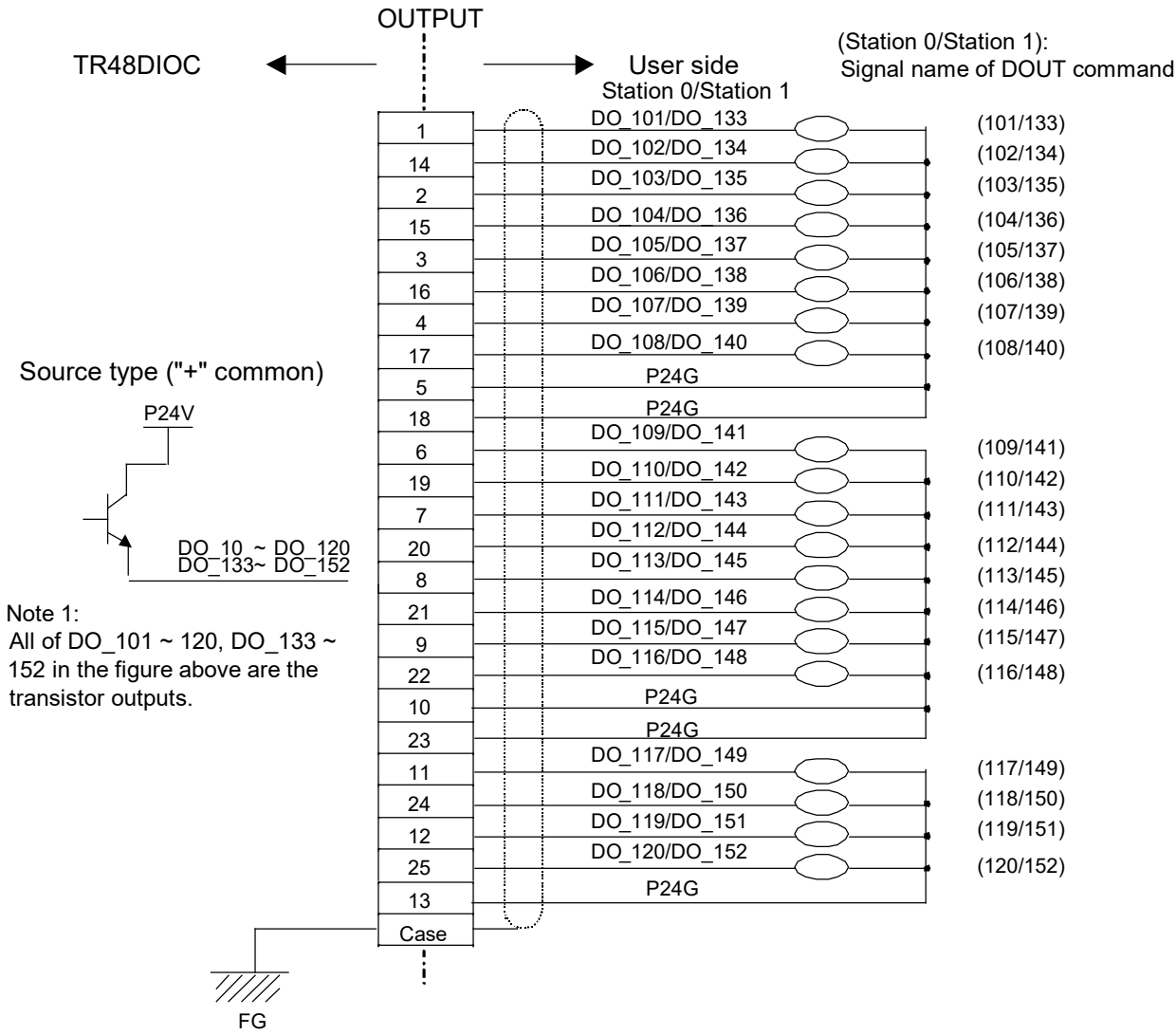
To connect the extension input signal cable, use the connector attached to the TR48DIOC module [XM2A–3701 (plug type connector), XM2S–3711 (connector cover)]. Connect the inputs of the TR48DIOC (DI_101 ~ 128 (station 0) and DI_133 ~ 160 (station 1) to connector INPUT provided on the module surface.



The specifications of the TR48DIOC extension input signal are the same as those of the TR48DIOCN extension input signal. For details, see Para. 8.1.1.

8.2.2 Connecting Extension Output Signal Cable

To connect the extension output signal cable, use the connector attached to the TR48DIOC module [XM2A–2501 (plug type connector), XM2S–2511 (connector cover)]. Connect the outputs of the TR48DIOC (DO_101 ~ 120 (station 0) and DO_133 ~ 152 (station 1) to connector OUTPUT provided on the module surface.



The specifications of the extension output signal are the same as those of the digital output signal (i.e., source type) which are described in Para. 4.7.

- Electric rating

Electric rating	Cautions
Rated voltage DC24 V Rated current 100 mA (max.)	Rated current of the built-in power supply is 3 A. (The total current of the hand, external output and extension output modules should be 3 A or less.) If the current which exceeds the rated output current is supplied, the output device may be damaged or the printed board may be burnt. To avoid this, be sure to use within the rated output current. NEVER use an external power supply. Otherwise, the system will malfunction.

8.3 Fabricating Extension I/O Signal Cable

When fabricating the extension I/O signal cable, see Para. 4.9.

8.4 Attaching and Detaching Extension I/O Signal Cable

For attaching and detaching the extension I/O signal cable, see Para. 4.10.

9. Connecting High-Speed Input Signal Cable (Option)

The X8HL printed board is available as the optional module which is exclusively used for the TS2000/TS2100 robot controller.
This X8HL printed board is provided with eight (8) high-speed inputs. Though a filter of 0.22 ms (calculated value) is normally provided for the inputs, only a filter of 2.2 μs (calculated value) is provided for the high-speed inputs. To prevent mis-operation, input a signal without chattering and less noise.
For the input specifications, a bidirectional photocoupler is used in the input circuit, and it is possible to change over the source type or sink type by the selection of COM.
To connect the high-speed input signal cable, use the connector [XM2D–1501 (socket type connector), XM2S–1511 (connector cover)] attached to the X8HL printed board.

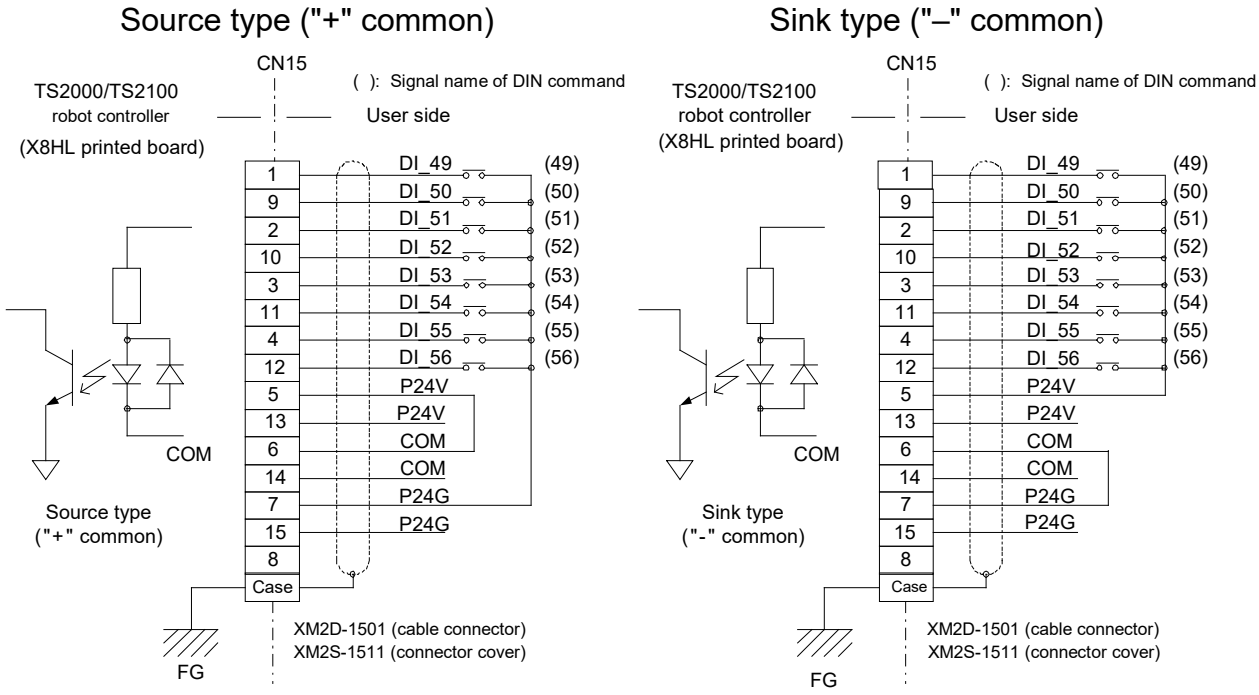


Fig. 9.1 Connection of high-speed input signal cable

The specifications of the high-speed input signal are the same as in the TR48DIOCN/TR48DIOC modules. For details, see Para. 8.1.1.
When this input signal is used as the conveyor synchronous start trigger input function, it cannot serve as the high-speed input function.

9.1 Fabricating High-Speed Input Signal Cable

When fabricating the high-speed input signal cable, see Para. 4.9.

9.2 Attaching and Detaching High-Speed Input Signal Cable

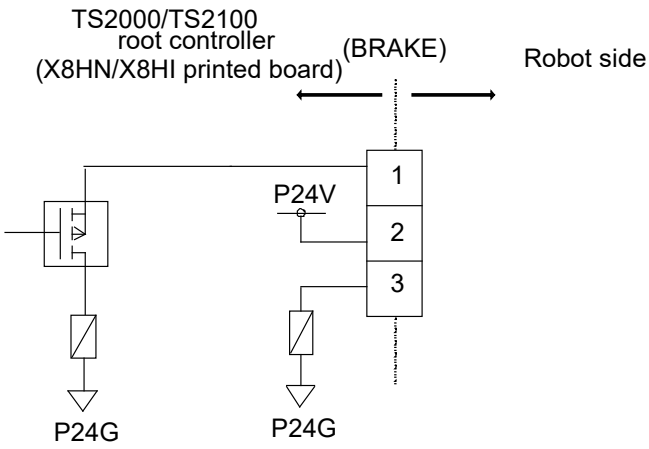
For attaching and detaching the high-speed input signal cable, see Para. 4.10.

10. Robot Control Signal Cable "BRAKE"

This cable is used to turn on and off the robot motor brake, apart from the CN4.
It is also possible to turn on and off the robot motor brake, irrespective of the control of the controller.

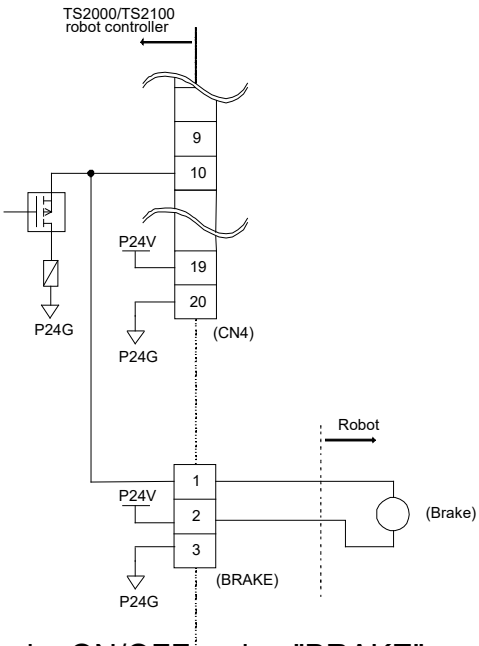
[Connector and contact on the external user side]

- Type of connector: 1–1318120–3, made by Tyco Electronics
- Type of contact: 1318107–1, made by Tyco Electronics

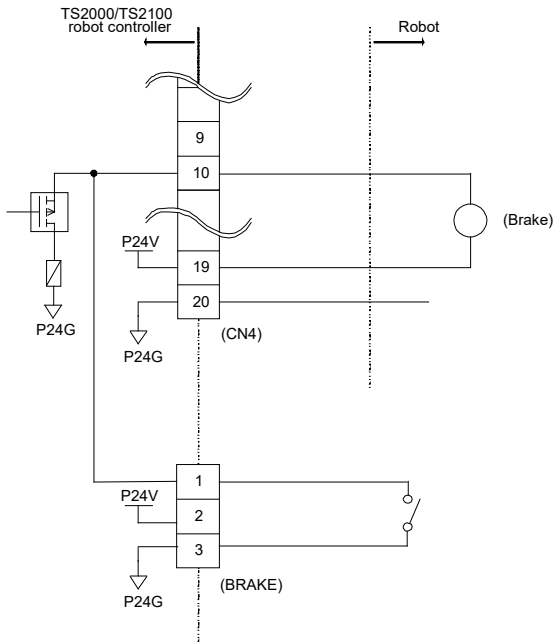


Sink type ("-" common)

[Example of application circuit]



Motor brake ON/OFF, using "BRAKE" connector



Motor brake ON/OFF, irrespective of control on controller side

11. Connecting Digital Output Power Select Cable

The power of P24 V used for the thirty-two (32) digital outputs is supplied from terminal block TB2 equipped on the rear side of the controller. Normally, terminal block TB2 1–2 pin is jumpered (a jumper wire is attached), and the power of P24V in the controller is used.
When using an external power supply, connect 24V and GND to the 2-pin and 3-pin of terminal block TB2, respectively.
When using an internal power supply whose maximum current capacity exceeds 3 A (including the hand I/O and EXT–I/O), use an external power supply.
NEVER use the internal power supply together with the external power supply. Otherwise, the power supply may be damaged.
For connecting the cable to and from terminal block TB2, see Para. 7.3.

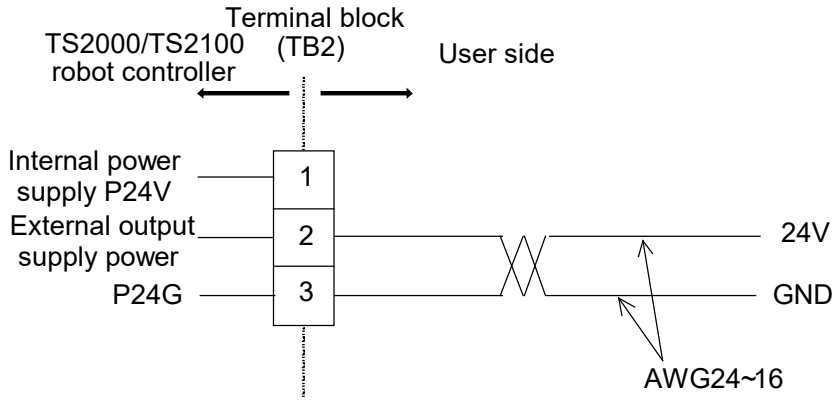


Fig. 11.1 When using external power supply

The cables designed for terminal block TB2 are AWG24 ~ 16.
Select the best one according to the user's system specifications (current capacity).

! **CAUTION**	The maximum current capacity of the internal 24 V power supply of the TS2000/TS2100 robot controller, which can be fed to the external equipment, is 3 A. Therefore, the total current used for the external I/O, hand I/O and EXT–I/O (remote I/O) modules should be 3 A or less. If the current used exceeds the rated value, the life of the power supply will be shortened extremely, internal 24 V power supply will be damaged or the fuse will be blown out. To avoid this, be sure to use within the rated capacity.

12. Appendixes

12.1 System Signal Table

Name	I/O	Function	Signal judgment
STROBE	I	Selects an execution program file from the external equipment.	ON: Short-circuit OFF: Open
PRG_RST	I	Resets a program currently stopped to step 1. The value of each variable is also reset to "0".	ON: Short-circuit OFF: Open
STEP_RST	I	Resets a program currently stopped to step 1. The value of each variable remains unchanged.	ON: Short-circuit OFF: Open
CYC_RST	I	Resets a program currently stopped to the step labeled "RCYCLE". The value of each variable remains unchanged.	ON: Short-circuit OFF: Open
DO_RST	I	Batch-resets the digital output signals (DOUT1 ~ 32, DOUT101 ~ 120, DOUT133 ~ 152).	ON: Short-circuit OFF: Open
ALM_RST	I	Cancels an alarm which has occurred while the robot controller was able to work.	ON: Short-circuit OFF: Open
RUN	I	Starts a program currently stopped to execute an automatic operation.	ON: Short-circuit OFF: Open
EX_SVON	I	Turns on the servo power.	ON: Short-circuit OFF: Open
STOP	I	Stops a program during automatic operation.	ON: Open OFF: Short-circuit
CYCLE	I	Stops a program currently executed in automatic operation after one (1) cycle has finished.	ON: Open OFF: Short-circuit
LOW_SPD	I	Reduces the robot operating speed in automatic operation.	ON: Open OFF: Short-circuit
BREAK	I	Slows down and stops the ongoing robot operation.	ON: Open OFF: Short-circuit
SVOFF	I	Turns off the servo power.	ON: Open OFF: Short-circuit

Name	I/O	Function	Signal judgment
EMS*B ~ EMS*C	I	Emergency-stops the robot.	ON: Open OFF: Short-circuit
ACK	O	Serves as a response signal to input signals STROBE, PRG_RST, STEP_RST, CYC_RST and DO_RST.	ON: High OFF: Low
TEACH	O	This signal is output when the robot arm can be guided through the teach pendant.	ON: High OFF: Low
SV_RDY	O	This signal is output when the servo power is turned on and the robot is ready to start.	ON: High OFF: Low
EXTSIG	O	This signal is output when the EXT. SIGNAL mode is selected by means of the MODE switch.	ON: High OFF: Low
SYS_RDY	O	System ready signal	ON: High OFF: Low
AUTORUN	O	This signal is output while the program is executed in automatic operation.	ON: High OFF: Low
CYC_END	O	This is the output signal for verifying that the program has stopped by the CYCLE signal input.	ON: High OFF: Low
LOW_ST	O	This signal is output while the robot is operating at a low speed by the input of low speed command.	ON: High OFF: Low
BT_ALM	O	Battery voltage level error output for backup power supply.	ON: High OFF: Low
ALARM	O	Controller fault output.	ON: High OFF: Low
SVST_A ~ SVST_B	O	Servo power ON contact output.	ON: Contact close OFF: Contact open Non-voltage contact

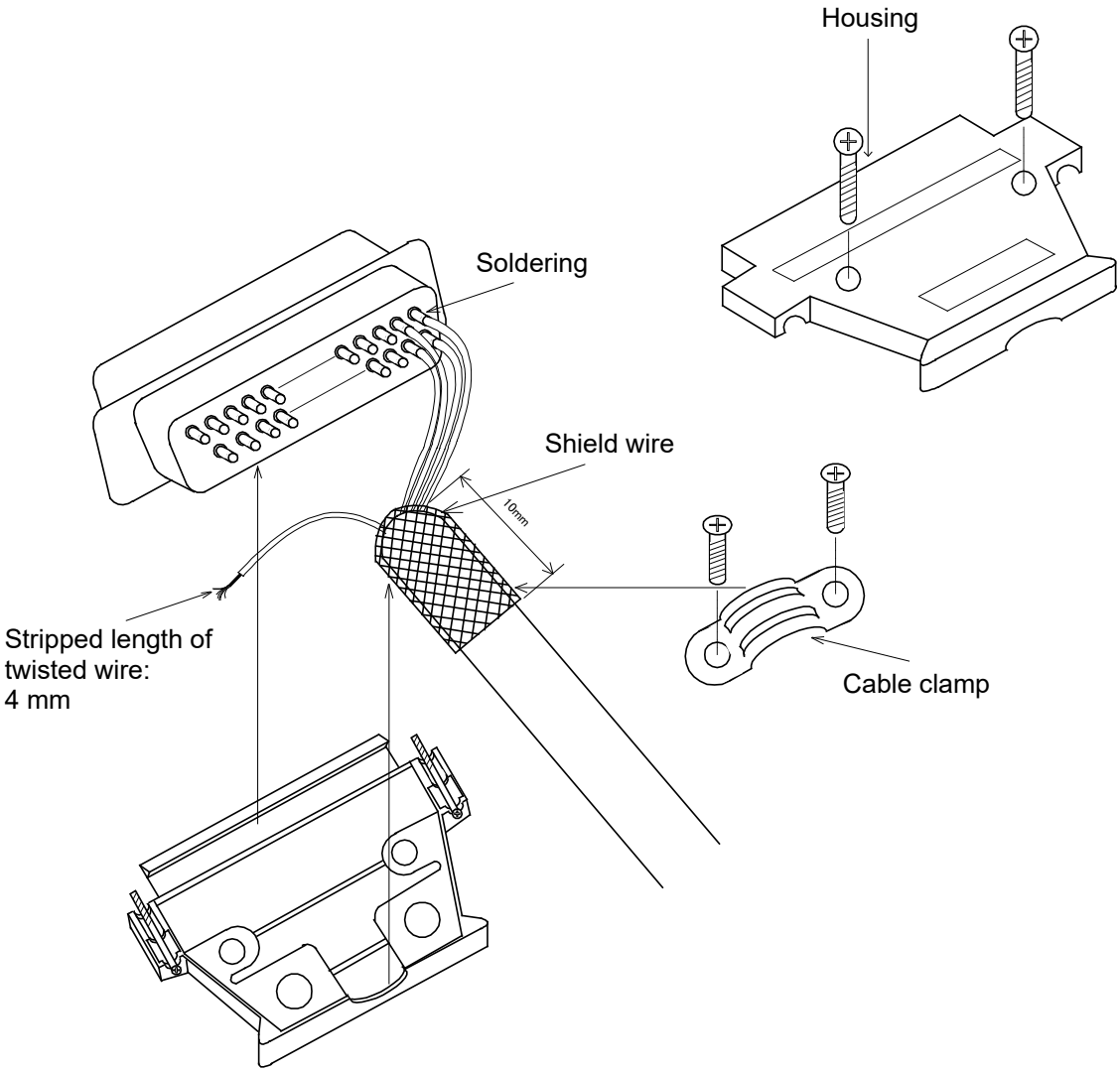
Name	I/O	Function	Signal judgment
EMSST_A ~ EMSST_B	O	Used to detect the EMERGENCY stop switch equipped on the control panel or teach pendant, and system input signal of emergency stop.	ON: Contact open OFF: Contact close Non-voltage contact

I : Input signal

O : Output signal

12.2 Fabricating Cable Using D-SUB Connector

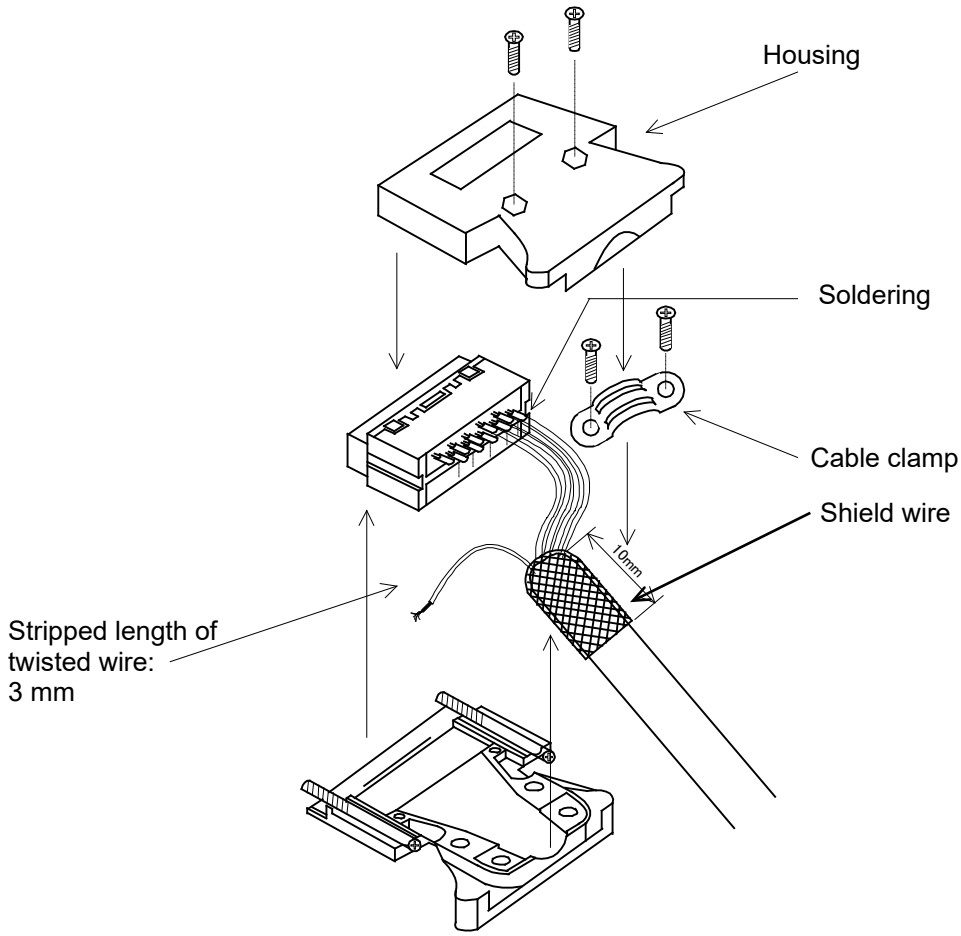
The cables to TS2000/TS2100 external I/O signal connectors CN5, CN6 and CN12 should be fabricated as shown below.



Note: The shield wire of the cable using a D-SUB connector is connected to the ground of the TS2000/TS2100 robot controller through the housing. Completely connect the shield wire to the housing using the cable clamp.

12.3 Fabricating Cable Using Half-Pitch Connector

A cable to connector CN4 of the TS2000/TS2100 robot control signal (controller side) should be fabricated as shown below.



Note: The shield wire of the robot control signal cable is connected to the ground of the TS2000/TS2100 robot controller through the housing. Completely connect the shield wire to the housing using the cable clamp.
Also, it is recommended to attach a protective cover such as thermal contraction tube to the soldered part to prevent a short-circuit.